Development and Design of a Multi-Modal Finger Exercising Device for Stroke Rehabilitation

Software, Testing in Healthy Hands Before Patient Trials, and an Extension to Parkinson's Disease and Dyslexia

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Abstract

Recovering finger control after a stroke requires practice, but patients often do not get enough therapy time, and home-based games rarely check the measurements they record. The author wrote the measurement software for the Curtin finger rehabilitation device: ten game modes on one engine and four force sensors, including Force Pilot for Parkinson's disease and Syllables for dyslexia. The software was checked in three ways that no other finger rehabilitation device in a four-database search combines: its measurements against independent references, its games against known effects in healthy adults, and each press timed from when force first started to rise. These checks found a force conversion error that made readings 4.5 times too high and an audio delay of 87 ms assumed to be 40 ms. On Curtin's EEG sequence task, all four healthy adults slowed by 91 ms on average when the sequence turned random, showing they had learned it. Presses were counted a median 40 to 55 ms after force began. That delay held within one 5 ms sample when the sequence turned random, but its shortening with practice made up 8 to 30 percent of three participants' speed-up. All 1,859 and 1,794 EEG event markers matched the logs within one sample, and a volunteer ran a session alone, saving all six blocks played. The device is ready for a patient pilot, with a normative study alongside it.

Nomenclature

Term	Meaning
Asynchrony	Press time minus beat time in Rhythm, in ms; negative when the press is early
BPM	Beats per minute, the pace of cued trials
Challenge point	The difficulty at which a learner learns best, stretched but not overwhelmed (Section 2.2)
Check codes	R for Reaction, P for Muscle Memory, C for Chords, Rh for Rhythm, M for Mirror, F for Force Pilot, B for Buzz Hunt, E for Echo, A for Adaptive and S for Syllables, numbered as in Table 44; a trailing i or p marks an exploratory row
Count	The moment a press is registered; in the study, as the smoothed force crossed the press threshold
CRN	Correct-response negativity, the smaller EEG dip after a correct press
dz	Cohen's effect size for paired data, the mean difference over its standard deviation
EEG	Electroencephalography
Enslaving	Force that appears in the fingers meant to stay still when one finger presses (Section 2.5)
ER	Enslaving ratio, the gap-scaled peak of the fingers that should stay still over that of the pressed ones
ERN	Error-related negativity, the EEG dip after a wrong press
ERP	Event-related potential, the EEG averaged over many repeats of an event; its peaks are named by sign and order
FCz, Cz, CPz, Pz, Oz, O1, O2, C3	Scalp electrode sites of the 10-20 system, from front to back; z marks the midline, odd numbers the left side
Foreperiod	The wait before the cue in Reaction
Gap	A finger's span from its resting level to its light calibration press, in counts; the thresholds sit inside it
Half-area latency	The time by which an ERP wave has built half of its area, a steady measure of its timing
Holm correction	A step-down adjustment of p values across several tests
I2C	Inter-integrated circuit, the two-wire bus the force sensors use
ICA	Independent component analysis, which separates eye signals from the EEG
ICC(2,1)	Intraclass correlation, two-way random effects, absolute agreement, single measure
Individuation	Moving one finger without the others (Section 2.5)
Lag	Time from the start of a push to its count; the count lag of Section 3.3.2 runs from the raw force's crossing to the count
Learning score	The sequence task's post-test median RT minus block 8's, in ms (Section 3.7)
MAE	Mean absolute error of force tracking, percent of the finger's maximum
Maximum, in Force Pilot	A finger's comfortable maximum: the median peak of three presses as hard as is comfortable, taken before the ladder; not a maximal voluntary contraction
MDC95	Minimal detectable change at 95 percent confidence
MDS-UPDRS	Movement Disorder Society Unified Parkinson's Disease Rating Scale
Mu, beta, theta	EEG rhythms of about 8 to 12, 13 to 30 and 4 to 7 Hz; mu and beta over the motor cortex fall during movement
N1 and N2 waves	Negative ERP peaks about 100 and 200 ms after a stimulus
Normative study	The planned session in eight or more healthy people that gives each measure its normal range (Section 6.3)
O codes, such as OA1	The objectives of Table 4: OA to OD from the progress report, OV and OE added in Semester 2
P1 and P3 waves	Positive ERP peaks about 100 and 300 ms after a stimulus

Term	Meaning
Pass	One run through a session's modes; pass 2 repeats four of them after a rest, for reliability
Pe	Error positivity, the positive EEG wave 200 to 400 ms after a wrong press
PLA	Polylactic acid, the plant-based plastic the chassis is printed in
Planned rule	The sequence task's scoring, fixed before the data: correct responses under 1000 ms, without each block's first trial (Section 3.7)
Press check	Each counted press timed against the start of its push on the force stream (Section 4.3.2)
Press threshold	*T_on* of Equation 2, the level the smoothed reading must pass to count a press; the release threshold, *T_off*, ends it (Section 3.3.2)
Push	A finger's rise in force that ends in a count; its start is found on the raw stream (Section 3.8.1)
Rank-biserial correlation	The effect size of a signed-rank test, from −1 to 1
RAS	Rhythmic auditory stimulation
REML	Restricted maximum likelihood, the usual fit for a mixed model
Reversal	An A-B-A run of targets in the sequence task, slower to answer (Section 2.6)
RT	Reaction time, from the cue (flip or buzz) to the logged count
SEM	Standard error of measurement
Session	One person's run through the study battery, 15 to 60 minutes (Section 3.6)
SQ1 to SQ5	The engineering questions of Table 3
SR codes, such as SR3	The system requirements of Table 9
SRT	Serial reaction time task
T1 to T5	The five core measures whose reliability is estimated (Appendix K)
W1, W3 to W5	The within-block checks of Appendix K; W2 was dropped with Mirror

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Chapter 1. Introduction

1.1 Finger rehabilitation after stroke

This thesis builds on Curtin University's finger rehabilitation device by turning its games into a system that can measure finger performance and checking those measurements in healthy adults before testing with patients. Around 12.2 million people experience a stroke worldwide each year [1], with an estimated 41,100 strokes occurring in Australia during 2023 [2]. Regaining fine finger control is one of the more difficult parts of recovery. Among patients whose arm is initially flaccid after a stroke, only around 38 percent regain some dexterity within six months [3]. Index or middle finger extension at three weeks also strongly predicts grasping ability at 13 weeks [4]. Everyday activities such as buttoning a shirt or holding a pen require control of individual fingers, rather than simply squeezing with the whole hand.

The amount of practice needed for recovery is often difficult to achieve through clinic appointments alone. Lang et al. [5] observed 312 therapy sessions after stroke and found that sessions involving functional arm practice averaged only 32 repetitions. In comparison, a study focused on high-repetition training achieved 322 repetitions per session [6]. Home rehabilitation tools such as MusicGlove [7], spring-loaded gloves, tablet apps and switch games can make extra practice easier, but often provide limited information for clinicians to review. More advanced systems such as the FINGER robot [8] and PRIDE palmar robot [9] can measure finger movement, with FINGER also recording some finger force. However, these devices are mainly used in research laboratories and specialist clinics. This project aims to provide something between these two types of systems: a device suitable for regular use that also records the force and timing of every finger press.

1.2 Earlier Curtin theses and the gap

Earlier versions of the Curtin device could record some performance data, but none had fully validated its measurements (Table 1). This thesis begins that work. The original device was developed by Lim [10] using an Arduino Uno, force-sensing resistors and a Python interface. Palmer [11], Lew [12] and Saini [13] later moved the system to an Arduino Nano and added a test screen, audio feedback and an early

rhythm mode. Their training modes could calculate maximum force, reaction time and finger enslaving. In 2025 the project was split between four students [14]-[17], who worked on the chassis, buttons, sensors and software. Although the hardware and software improved, the final design did not include closed-loop control or performance tracking for every trial. EEG integration was also outside the project's scope. During 2026, the author developed and integrated the software while the electronics team worked on the circuit boards, printed circuit board (PCB) and firmware. Table 1 summarises the development of the device over the different project years.

Table 1: What each version of the Curtin device added.

Version	Hardware	Software and data	Measurements validated
2023 [10]	Arduino Uno, force-sensing resistors	Python interface recording reaction times	No
2024 [11]-[13]	Arduino Nano, tabletop chassis	Test screen with audio feedback; training modes measuring maximum force, reaction time and finger enslaving; early rhythm mode	No
2025 [14]-[17]	Chassis, input buttons and characterised SingleTact force sensors	Software base with serial communication protocol and fixed press thresholds	No; sensors bench-tested and learning observed in six adults
2026, this thesis	Circuit boards, PCB and firmware from electronics team; adjustable chassis from Semester 1	Ten game modes, hand calibration, adaptive difficulty, force and timing recorded for each press, EEG event markers, automated analysis and modes for Parkinson's disease and dyslexia	Press timing checked against force, sequence learning reproduced in four participants and markers checked in two EEG recordings (Chapter 4)

Recording reaction time or peak force for every press is useful only when the timing and force values have been checked against independent references.

1.3 Project scope

Semester 1 focused on the four main areas outlined in the progress report [18]. These were an adaptive difficulty engine based on the challenge-point approach (A), a music-based rhythm mode (B), support for two hand devices running together (C), and automatic device detection and setup (D). In Semester 2, most of the author's work focused on developing and testing the software across four areas (Table 2). The platform grew from four game modes to ten, with six designed around effects already reported in healthy adults (Section 3.3.4). The patient trials proposed in the progress report could not be completed within the remaining project time. Testing with

healthy participants was also needed first to understand the device's measurements before using it with patients. Instead, the device was tested in healthy adults and prepared with the software and analysis tools needed for a larger normative study. The author also developed an EEG event-marker system and tested it at Curtin's EEG laboratory. Two additional game modes extended the platform beyond stroke rehabilitation. Force Pilot measures force characteristics reported to differ between healthy adults and people with Parkinson's disease, while Syllables uses the finger sensors in a phonics game designed for readers with dyslexia across different age groups.

Table 2: The four parts of the Semester 2 work, including their deliverables and research focus. The Semester 1 work is reviewed in Chapter 5.

Part	Deliverable	Research focus
Platform	Ten game modes, calibration for each participant, force and timing recorded for every press	Established tasks ranging from reaction time to force tracking [19], [20]
Verification	Independent checks of constants and delays, healthy participant testing, study kit and analysis notebook with a check for each mode	Measurement validity and reliability [21], [22]
EEG	Event markers sent on the display frame and the laboratory's sequence task reproduced trial by trial	Accurate timing for event-related potentials [23]
Beyond stroke	Force Pilot measures relevant to Parkinson's disease and Syllables with age profiles and spoken feedback	Force release in Parkinson's disease [24] and phonics teaching [25]

These conditions affect different abilities. Stroke can reduce finger control, independence, force, speed and sensation, which are measured by all modes except Syllables. Parkinson's disease affected around 6.1 million people worldwide in 2016 [26] and can cause slower movements that become smaller with repetition, along with difficulty releasing force. These changes can be measured through Force Pilot and Rhythm. Dyslexia affects around 7 percent of primary school children [27] and can make reading slower and make it harder to connect written letters with speech sounds, which Syllables is designed to practise. The project also involved work across different areas of engineering and research. The EEG component was developed with Welber Marinovic and Curtin's EEG laboratory. The software reproduces the laboratory's PsychoPy serial reaction time task trial by trial, allowing the device to be used during EEG recordings. The laboratory has also published research using a related task [28], [29]. Before recording, the laboratory requested that the event-marker timing be checked and that each counted press be compared

with the start of the finger's push. Using the laboratory's onset detection settings, this check found a delay of 40 to 55 ms between the beginning of the push and the registered press (Section 4.3.2).

The force analysis also builds on Rayyan Raiz's bench testing using the same sensors. Since the independent sessions were designed to run without a researcher present, participant privacy, safety and the option to stop at any time were considered in the study design (Section 3.9). The main Semester 2 milestones were the publication of the study design on 2 September, EEG recordings on 8 and 22 October, and participant testing between 15 and 23 October. Appendix E compares these with the original plan: of the fifteen planned tasks, five were completed, four were replaced with different checks, two were partly completed, two were not carried out and two were transferred to the hardware team.

1.4 Project question and objectives

The Semester 1 progress report examined whether adaptive difficulty, rhythm cues, two-device operation and automatic setup could all work together in one rehabilitation device that records performance on every trial [18]. During Semester 2, the focus shifted towards checking whether those measurements could be trusted. Before collecting patient data, the device's force and timing measurements needed to be tested and compared with independent references. This led to the main research question:

Can a multi-modal finger rehabilitation platform be engineered to record trustworthy measures of finger function, verified against independent references and checked in healthy hands, so that it is ready for a patient pilot and normative testing?

Five engineering questions were developed from this, with a specific requirement for what would count as a successful result (Table 3).

The five questions, SQ1 to SQ5, are used to answer the main research question. The objectives in Table 4 also allow the work to be compared with the original progress report, while Table 9 lists the system requirements. All twelve objectives from the progress report were kept, with their final outcomes discussed in Chapter 5. Additional objectives for verification (OV) and extending the device to other conditions (OE) were introduced during Semester 2. Two original objectives, chassis

adjustment across hand sizes (OA3) and the wrist and forearm angles (OD3), depend on hardware work that was deferred when the project became more software-focused. The two-board objectives OC2 and OC3 were tested using simulated devices. These four objectives are separate from the main research question.

Table 3: Engineering questions, criteria for success and where they are answered.

Question	What is being tested	Requirement for success	Section
SQ1 Timing	Are presses and events recorded at the correct time?	Constants and delays match independent references; each study press logged within 50 ms of crossing its threshold	4.2, 4.3.2
SQ2 Function	Can the device reproduce a known effect in healthy adults?	Every participant shows the published learning effect using the planned scoring method; additional analyses identified as post-data	4.3
SQ3 EEG	Are the event markers accurate enough for EEG?	Every marker recorded in order and within one sample of the game log	4.4
SQ4 Readiness	Can someone complete an assessment without assistance?	Session data saved without a researcher present and all result tables produced by the notebook	4.1, 4.6.2
SQ5 Extension	Can the device support another condition through software alone?	Measurements based on the relevant research [30]-[33], a planted 5 percent release error recovered within 0.5 percent of maximum, healthy blocks measured and all age profiles voiced	3.3.5, 3.3.6, 4.7

Table 4: Project objectives. OA to OD come from the progress report, with OV and OE added during Semester 2. The 18 objectives are reviewed in Table 23: five met, three partly met, eight tested in simulation and two deferred.

ID	Objective
OA1	Adaptive challenge-point controller keeps each finger's success rate between 65 and 80 percent across a 32-trial block
OA2	Peak force recorded for every press, in newtons when calibrated
OA3	Adjustable chassis for hands between the 5th and 95th percentiles
OB1	Rhythm mode uses a beatmap engine to score presses against the beat
OB2	Each block records lag-1 entrainment and beat-offset statistics
OB3	Configurable timing windows
OC1	Two devices automatically detected and assigned to each hand, with manual override
OC2	Mirror mode only counts a trial when both hands press together
OC3	Bilateral asymmetry and correlation between hands calculated
OD1	Setup from connection to first sample takes less than 2 s
OD2	Startup delay recorded for every session
OD3	Chassis supports 10 to 20 degrees of wrist extension and approximately 30 degrees of forearm pronation
OV1	Normative study kit including protocol, de-identified import and analysis notebook
OV2	Predefined check for every mode, including a known effect that can fail for six of the eight study modes
OV3	Within-session reliability analysis for the main modes
OV4	Agreement between an uncalibrated sensor and calibrated sensors
OV5	An assessment lasting 15 minutes to one hour that can be completed independently
OV6	EEG event-marker timing and laboratory sequence task

ID	Objective
OE1	Force Pilot calculates Parkinson's-related force measurements from the raw data
OE2	Syllables supports readers aged 6 to over 60, with spoken feedback for every word and syllable

1.5 Contributions and uses of the work

The literature search described in Section 2.14 and Appendix J did not identify another finger rehabilitation device that combined all three checks used in this project: comparing measurements and timing against independent references, testing games against known effects in healthy adults, and measuring each press from the beginning of its force rise. The device uses these methods across ten game modes running through the same software engine. The main contributions of this thesis are:

1. A shared software engine for ten finger rehabilitation games, each based on an established task. The system records force and timing for every press using the same data format, with automated tests covering 87 percent of the software lines.
2. A measurement system with force constants and timing delays checked against independent references. Each counted press was also compared with the beginning of its force rise. The median delay was 40 to 55 ms, and the reduction in this delay with practice accounted for 8 to 30 percent of the improvement in speed for three participants. The device also reproduced Curtin's EEG laboratory sequence-learning effect in four healthy adults, who became 91 ms slower on average when the sequence changed to random.
3. A three-part testing method that could be used for other force-based game devices (Figure 22). This involves checking each measurement against an independent reference, testing each game against an effect reported in healthy adults and timing each press from when force first begins to rise.
4. EEG event markers sent on the display frame, along with a trial-by-trial reproduction of the laboratory's sequence task. These were tested during two EEG recordings, with every marker matching the game log within one sample.
5. A study kit prepared for normative testing, including predefined checks for every game mode, within-session reliability analysis and a notebook that automatically produces the result tables. One volunteer completed a session independently, finishing six of the eight games in 13.8 minutes of play.

6. Two additional game modes for conditions beyond stroke. Force Pilot measures force-release characteristics associated with Parkinson's disease and recovered a planted 5 percent release error to within 0.5 percent of maximum using synthetic signals. Syllables provides a phonics game for readers aged 6 to over 60.

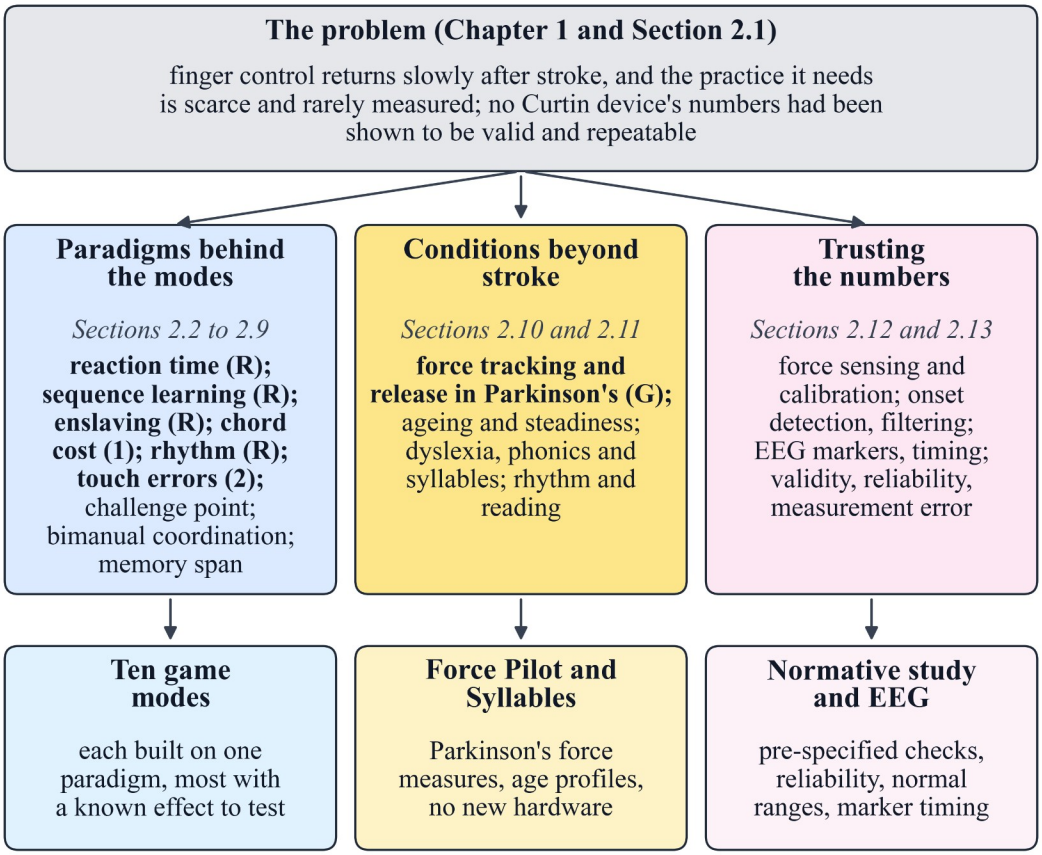
The software can also be used by future project teams. New conditions can be supported by adding game modes to the existing engine without replacing the logging system or measurement checks, as demonstrated with Force Pilot and Syllables. The EEG laboratory can use the games during recordings, with event markers sent when stimuli appear on the screen. Every completed session produces a record of individual presses. These records will be useful for the planned normative study and stroke patient pilot, and could later allow clinicians to review performance remotely during home-based rehabilitation (Section 6.3).

1.6 Thesis outline

Chapter 2 reviews the research behind the different game modes and the methods used to check whether the device's measurements are reliable and valid. Chapter 3 explains the system design and how it was tested. Chapter 4 presents the engineering verification results, sequence-learning task, EEG recordings and testing of the device's own games. Chapter 5 summarises the findings, answers the research question in Table 22 and reviews the progress of each objective in Table 23. Chapter 6 discusses the remaining work and priorities for future projects. The appendices provide further detail on software development (A), study design changes (B), EEG sessions (C), faults found before data collection (D), project progress against the original plan (E), Force Pilot and Syllables (F to H), design decisions (I), the literature search (J), study analysis methods (K), patient pilot consent documents (L) and participant feedback (M).

Chapter 2. Literature Review

Six of the eight modes in the normative study were built to reproduce an effect the literature reports in healthy people, and each effect became a pre-specified check. The review is therefore organised around the modes, as Figure 1 maps: what recovery needs and what devices record, the paradigm and healthy values behind each mode, the two conditions beyond stroke, and the analysis methods. Each mode's main sources were read in full, and the notes are kept with the device search of Appendix J in the repository [34]. Sections 2.2 and 2.3 extend the review in the progress report [18]; its hardware literature, on hand anthropometry and resting posture, belongs to the Semester 1 chassis and is not repeated.



Bold: findings a known-effect check rests on, tagged by their evidence: R a review or several studies, 2 two studies, 1 one study, G one group's (Section 2.14)

Figure 1: How the literature reviewed in Chapter 2 feeds the platform, the game modes and the normative study.

2.1 Recovery after stroke and the measurement gap

A rehabilitation device's value depends on how much function survivors regain through usual therapy and whether the device adds the practice that recovery needs.

Veerbeek et al. [35] reviewed 467 randomised physical therapy trials and found strong evidence favouring intensive, high-repetition, task-specific training. Constraint-induced movement therapy helped whatever its intensity [36], repetitive task training helped modestly [37], and the American stroke guidelines reflect both [38]. More therapy time gives more improvement, perhaps with diminishing returns per extra hour [39], yet arm sessions average 32 functional repetitions when they include any, and half include none [5]. Clinical scales also miss finer deficits. In 80 people after a first stroke, force control was impaired even when the Fugl-Meyer assessment rated them mildly impaired, and two of three force measures improved over six months [40]. A device that records force and timing for every press can therefore follow deficits the usual tests rate as mild.

Between the home tools and the laboratory robots of Section 1.1 sit force-tracking devices (Table 5). Training on a grip force tracking system improved grip force control in 8 of 10 stroke patients, though without a control group [20]. In 45 randomised survivors, fingertip force training with three-dimensional feedback gave larger gains on the Action Research Arm Test, 3.5 against 0.8 points [41]. In another trial every group improved on the trained task and none on the Wolf Motor Function Test [42]. The FINGER robot sets the challenge and measures individuation during a music game [8]; 30 survivors who trained in it for nine sessions improved on hand function and impairment, with high or low assistance [43]. Individuation trained on a virtual keypad, with an actuated glove assisting, improved both impairment and task measures where matched occupational therapy did not [44]. The commercial GripAble grip trainer agreed with a Jamar dynamometer at an ICC of 0.91 but read 69 percent of the Jamar's force, so the two cannot be swapped within one patient's record [45]. Taken together, these trials show gains on the trained task with mixed transfer to function.

Three devices come closer to this one. Two PubMed searches on 15 October 2026, repeated in Europe PMC, IEEE Xplore and OpenAlex, found them on screening the abstracts (Appendix J). AMBER's finger-press measures of strength, reaction time and memory retested at ICCs of 0.56 to 0.90 in 36 healthy adults [46], and an individuation device's index retested at 0.48 to 0.80 in flexion [47]. A fingertip force device's index held together at R of 0.97 or more and tracked manual ability in children with cerebral palsy, though in the preferred hand only [48]. Each shows how

well its measure repeats or hangs together, evidence this project's normative study must also give. Their retest values put the normative study's bar for a press measure at about 0.5 to 0.9. None reports what its timing rests on, checks a training game against an effect published for healthy hands, or carries EEG markers, the gaps Table 8 sets out.

Table 5: Classes of hand rehabilitation device and what each records.

Class	Examples	What it records	Setting	Evidence
Home game or glove	MusicGlove [7]	Limited data, no finger force	Home	Feasible; little data for a clinician
Commercial grip trainer	GripAble [45]	Grip force, not per finger	Home or clinic	Reliable against a dynamometer
Virtual reality games	EVREST [49]	Limited data, no finger force	Supervised clinic	Safe and feasible
Finger robot	FINGER [8]	Movement; force on some notes	Laboratory	Challenge set during a music game; gains in 30 survivors [43]
Palmar robot	PRIDE [9]	Finger position; the four fingers' force as one, in bench tests only	Laboratory or clinic	Bench tests in six healthy adults
Force tracking	[20], [41], [42]	Grip or finger force on every trial	Laboratory	Feasible; mixed transfer to function
Finger training glove	Actuated virtual keypad [44]	Finger movement on every trial	Laboratory	Pilot gains in fine motor control
Motor and cognitive device	AMBER [46]	Each finger's press; strength, reaction time and memory	Portable	Retest ICC 0.56 to 0.90 in 36 healthy adults
Finger individuation device	[47]	Force under each finger, in flexion and extension	Laboratory or clinic	Retest ICC 0.48 to 0.80 in flexion
Fingertip force device	[48]	Force at every fingertip in three dimensions	Laboratory	Individuation tracked manual ability in children with cerebral palsy
This project	Chapter 3	Force per press and at 200 Hz, every trial in one format	Built for clinic and home; used in a laboratory	Sequence learning in four people; one session

The gap is a device meant for everyday use that still records finger-level force and timing for every press, across several recognised paradigms, with measurements shown to be valid and repeatable. No earlier Curtin device met the last part, and it is the main focus of this thesis.

2.2 Motor learning and the challenge point

The adaptive difficulty in the software follows the challenge-point hypothesis of Guadagnoli and Lee [50]: people learn most when a task stretches them without

overwhelming them, and the right difficulty rises with skill. The software adjusts the pace of cued trials and the fingers in play as performance changes. Wilson et al. [51] derived an optimal success rate of about 85 percent, for binary classification, while Guadagnoli and Lee hold that no single rate suits every learner. The platform holds a 65 to 80 percent band, set in Semester 1 a little below Wilson's optimum; no source sets it. Adaptive procedures from psychophysics give a check on it: Levitt's transformed up-down staircases settle at 70.7 percent correct for a two-down, one-up rule and 79.4 percent for three-down, one-up [52], so the band covers both common targets.

Several findings shape how practice data are read. Gains during practice are performance, and a delayed retention test is the better guide to what is kept [53]. Feedback on every trial helps performance while it is present but can weaken learning [54]. Improvement straight after a rest is a passing effect of the rest, not offline learning [55], [56], so a second pass after a rest starts on a rested hand, which the reliability design relies on. Spreading practice over days helped a timed finger sequence where more practice in one session did not [57], and whether sleep adds to it is disputed [58]-[60]. In one session, only a gain in performance can show. Retention needs a second day, and transfer to daily hand use needs a trial.

2.3 Rhythmic auditory stimulation and timing

Most research on rhythmic auditory stimulation (RAS) is in gait [61], [62], with little work on the fingers. Music-supported therapy used instruments in rehabilitation but did not measure timing against the beat [63]; the Rhythm mode records the timing of every press against the song. The synchronisation literature says what a healthy timing record looks like. People tapping to a steady beat press slightly before it, by tens of milliseconds, the negative mean asynchrony [64]. People with Parkinson's disease tapped about 39 ms after the beat at 116 beats a minute, no differently from controls [65], and tapping to music shows less lead than tapping to tones [66]. Offsets this small are measurable only after subtracting the system's own sound delay, which ranged from under 1 ms to about 119 ms across experiment software [67]. The progress report's planned comparison of a fixed cadence with RAS needed patient trials and was not run, so Rhythm is used as a timing measurement (Table 35).

The timing literature gives the analysis its model. A model that splits tapping variability into a central timekeeper and motor delays predicts that successive intervals correlate negatively [68], while correcting only part of each error leaves successive asynchronies positively correlated [64], [69]. Notes meant to sound together are spread by 30 to 50 ms in ensemble playing [70], and the melody lead in piano chords almost vanishes when measured at the fingers [71]. The presses of a healthy chord should therefore land within a few tens of milliseconds of each other.

2.4 Bimanual coordination

Mirror, built in Semester 1, counts a trial when the same finger on both hands presses within a short window. The evidence for bilateral training is contested. Training both hands together may engage both hemispheres [72], and metronome-paced bilateral arm training helped in an uncontrolled pilot [73]. But Cochrane authors [74] disputed a meta-analysis's large pooled effect [75], and the Cochrane review found no benefit over usual care [76]. Mirror therapy [77], [78] helped across 62 studies, though not against the same movements made with both limbs in view [79], which is what the mode delivers. This makes the mode a measurement, with the prediction that healthy hands start together [80]. It needs two boards, so the study left it out.

2.5 Finger individuation and enslaving

Moving one finger without the others is called individuation, and stroke impairs it long term, most in the middle, ring and little fingers [81]. Healthy fingers are not fully independent either. When one finger presses, force appears on the others, an effect Zatsiorsky, Li and Latash called enslaving; it was largest between neighbouring fingers and reached up to 67.5 percent of a finger's maximum at full effort [82]. The Chords mode takes four checks from this result and those below: low enslaving, more misses on harder chords, the ring finger most enslaved and a time cost per extra finger. The ring finger is the least independent [83], and while two fingers held a quarter of their maximum the other two carried about 8 to 10 percent of the total force [84]. Pressing several fingers at once also takes longer [85], but the cost per finger the check uses rests on one study. Verwey [86] measured one, two and three-key chords at 510, 632 and 762 ms of mean key time, about 126 ms per extra finger. Those times average one-hand and two-hand players after 240 practice trials per chord, so an unpractised first session on one hand should cost more.

Stroke damages individuation in a specific way: with the motor cortex or its corticospinal tract injured, the finger muscles fire less selectively, and the loss goes with poorer hand function [87]. In 54 patients followed for a year, strength and individuation recovered together up to about 60 percent of premorbid strength. After that, individuation needed a process of its own [88], so finger control needs its own training target. Each finger produces less when more press together and adding a finger soon stops raising the total [89], and a 60 s maximal effort cut four-finger force by about 43 percent [90]. Individuation is plastic, but the feedback matters: practice aimed only at force accuracy increased enslaving, while the same practice with feedback on independence reduced it [91]. The Chords mode therefore scores the fingers that should stay still as well as the ones that press, and watches for fatigue within a session.

2.6 Sequence learning

The serial reaction time task measures how people learn a sequence of movements without being told there is one. Nissen and Bullemer [92] had one group respond to a repeating 10-item sequence over four positions and another to random order, and the sequence group became faster and more accurate. The usual measure is now within each person: random material swapped in after sequence blocks slows responses, and that rebound is the specific measure of learning [93], [94]. Second-order conditional sequences stop people learning finger frequencies instead of the order [95], and awareness can be checked afterwards by scoring recalled triplets [96]. Destrebecqz et al.'s [96] transfer costs, 91 and 111 ms in groups of 21 and 17, came after twelve training blocks of 96 trials. Muscle Memory's probes follow 108 and 144 trained trials, about a tenth of that practice. A proportional share would be about 10 ms, and learning is fastest early in practice [97], so 10 to 30 ms is expected there; the laboratory's task, at 800 trained trials, comes closer to the published practice. Random blocks must be built with care, since they hold more reversals (A-B-A runs), which are slower, and so can add to the learning measure [98]. The Curtin laboratory's own work reported an advantage for a 500 ms interval and for location-informative tones under explicit instructions [28]. Whether the timing between trials changes what is learned is disputed [93], [99].

What the task measures is largely the link between stimuli and responses [93], and with no time to prepare, people express sequence knowledge in their reaction times

that they can neither recognise nor withhold [100]. The effect is reliable for a group, but an individual's learning score retests poorly, at about $r = 0.28$ [101], replication has been difficult [102], and how long it is retained is disputed [103], [104]. Two findings matter for patients. Implicit learning is preserved in the less-affected hand after stroke [105]. After middle cerebral artery or basal ganglia stroke, telling people about the sequence disrupted their implicit learning, though it helped controls [106], [107], so the platform never tells players during play that a sequence exists.

Learning also shows in the EEG. The change in the error-related negativity from early to late blocks tracked sequence learning better than its average did [108], the kind of within-task change the event markers make measurable. Stimulus-locked averages show learning too. A stimulus that breaks a learned sequence draws a larger N2 [109], and the P3 shrinks as targets become predictable [110]. Motor alpha and beta power [111] and fronto-central theta [112] are higher on random trials. Section 4.4 tests the N2, P3 and beta, and Table 31 reports theta.

2.7 Reaction time and the foreperiod

Reaction needs a choice task whose healthy behaviour is known and a timing a player cannot anticipate. The first is settled. Choice reaction time rises with the logarithm of the number of options and with the information the stimulus carries [113], [114]. But in one study a vibration on the finger that must respond flattened that rise [115]. The second is a design choice. A player can time a fixed wait before the stimulus, while an exponential distribution of waits keeps the chance of the stimulus constant and gives no such cue [19]. What does not carry over between devices is the absolute value, which depends on the hardware and the cue [11], [116]; ordinary computers can add tens of milliseconds [67]. Reaction times are therefore reported as this device's own numbers, by the median of their right-skewed distribution [117], and a short block is stable across sessions [118]. After stroke both hands respond more slowly [119], so the unaffected hand is a comparison within the patient rather than a healthy baseline.

2.8 Touch localisation and sensory retraining

Buzz Hunt needs two things from the literature: the errors healthy fingers make, against which a patient's can be read, and a pulse the fingers feel clearly. Two studies describe the first. Near threshold, healthy people mistake a touch for a neighbouring

finger more often than a distant one [120], and after median or ulnar nerve repair the errors also jump between digits [121], the pattern a finger confusion matrix records. The second rests on one study, in which pulses of about 50 to 200 ms from a phone-type vibration motor were rated best [122]. Training is promising but not settled. Many stroke survivors lose sensation in the hand, and in the SENSe trial ten hours of graded discrimination training improved it more than exposure alone, with gains lasting six months [123]. The evidence elsewhere is mixed: home re-education improved localisation after nerve repair over three years [124] and Braille training reduced focal hand dystonia [125], but sensory relearning after carpal tunnel decompression missed its primary outcome [126]. Buzz Hunt is therefore a measurement for these groups and a candidate training task, and it makes no treatment claim.

2.9 Working memory span

Echo measures how long a sequence a player can hold and repeat; no participant played it (Section 4.1), so this section gives only the basis of its design. Its healthy values are settled for the classic task but not for a finger game. Adults average 6.2 blocks on the Corsi task [127], [128]. For the four-button game Simon, the value of about seven items in college students comes from one study [129]. How a span is given and scored changes the result. Corsi results varied so much with administration and scoring that 25 years of them were hard to compare [130], though a tablet version reproduced the spans of the physical test [131]. Scoring matters as much: a mean span over many trials retests better than the maximum [132], [133], and partial credit or edit distance beats all-or-nothing scoring [134], [135]. Because of this, every Echo block records its settings and is read by the mean span of its two games, with the edit distance beside it. What is disputed is the kind of learning a repeat brings: a sequence repeated every third trial is recalled better [136], [137], but that learning starts abruptly, with awareness of the repeat [138], so Echo does not claim it as implicit.

2.10 Force tracking, Parkinson's disease and ageing

The findings below set Force Pilot's design, its checks and its moderate to good predicted reliability (Table 43). Force tracking asks a person to follow a moving target with their force. After stroke, grip force varies more, with extra slow

modulation near 0.2 Hz [139], and deficits are largest when releasing force [140]. Repeating one wave lets waveform-specific learning be measured, and in stylus tracking the correlation between cursor and target detected it more sensitively than the usual error measure [141]. That is why Force Pilot keeps one fixed wave, Storm, beside a new wave with the same statistics, Uncharted.

Parkinson's disease is the second condition the platform was extended to. One of its main motor signs, bradykinesia, is slowness with movements that shrink and fade as they repeat [142], rated by eye on the MDS-UPDRS [143]. An instrumented force task gives continuous numbers instead: tracking error, lag and tremor power together predicted the total UPDRS score within about 3.5 points, though from 36 predictors on 26 people [31]. The most repeated finding at light force is a problem with letting go. People with Parkinson's disease released grip force less accurately than they built it on slow sine waves [24], and on ramps less accurately than controls [144]. Young adults released at least as accurately as they pressed, older adults less accurately [30]; in early disease the pinch grip relaxed more slowly, at 43.2 against 57.4 percent of maximum per second, but rose no slower [145]. Force Pilot therefore compares release error with press error on its symmetric waves, where a young adult should release as accurately as they press. The evidence is not all one way, since Tobin et al. found both rise and release slowed [146] and the sine study's release deficit showed in the time spent near the target but not in the error measure [24]. That study's young adults also tracked a precision grip with about 4 s of preview, a different task from a single finger on a sensor. In a randomised trial, a tablet game played for twelve weeks made tapping faster but less accurate [147], so a game should reward accuracy at a set speed.

The remaining findings come from single studies, so they set Force Pilot's design, not its checks. The target stays visible, since patients' movements faded more than controls' only without visual feedback, and then only on medication [148]. Regularity is read as sample entropy [149] because force tremor was more regular than controls' and its regularity tracked motor severity [150]. Single-finger presses also leaked more force into the other fingers [151], an enslaving that medication left unchanged [33].

Ageing sits between health and disease. Held steady at 15 percent of maximum, an index finger's force varies by about 1.4 percent of the held force in young adults

[152] against 3.4 percent in controls aged about 63 [146]. A meta-analysis put the link between hand steadiness and task performance in older adults at $r = 0.58$, mostly on pegboard tests [153]. A young cohort gives only the floor of these measures, not the reference for older patients. A repeating target can be tracked with almost no lag even in Parkinson's disease [154], so lag is read on unpredictable waves, as on the unpreviewed target of Brewer et al. [31]. Force Pilot's checks rest on the two best-supported findings: the release deficit, found in three studies though all from one group [24], [30], [144], and the lag on unpredictable waves [31], [154].

2.11 Dyslexia and syllable reading

Developmental dyslexia, a specific difficulty in learning to read, affects about 7 percent of primary school children [27], and across 178 studies adults with it showed very large gaps in reading and writing, larger for speed than for accuracy [155]. No reader with dyslexia played Syllables (Section 4.1), so this section gives only the basis of its design (Appendix G). Meta-analyses set the method. In a meta-analysis of randomised trials, phonics was the only treatment type with a confirmed effect for children with reading disabilities [25], and the National Reading Panel's meta-analysis found phonemic awareness taught with letters worked better than without [156]. A review of GraphoGame adds that a game alone is not enough: it worked only with close adult involvement [157]. Syllables pairs each written chunk with its sound, as phonics teaching does, and makes no claim to treat on its own. Phonological awareness develops from larger to smaller units [158]: about half of four-year-olds could split words into syllables and none into phonemes, against 90 and 70 percent by six [159]. That is why Syllables works at the syllable, with options that differ from the target inside it, and at several grain sizes because English spelling is inconsistent [160]. Speech rhythm [161], [162] and rapid naming [163], [164] lie outside what the game trains, and letter position dyslexia needs its own tests [165], so swapped-letter foils are not read as a sign of it.

Rhythm links reading to the fingers: paced tapping explained unique variance in reading and spelling at ten [166], and music training improved rhythm, phonological awareness and reading in a randomised trial [167]. The multisensory part of structured literacy has not been shown to add anything [168], so the vibration in Syllables is only a cue. Children are about 1.8 times slower than adults at 10 and reach adult speed by about 15 [169], [170], and choice reaction time slows again

through adulthood [171]; these set the age profiles. Children also understand synthetic speech less well than adults do [172], which shaped the voice.

2.12 Force sensing, signal processing and EEG event markers

The device reads force with SingleTact capacitive sensors in place of the force-sensing resistors of earlier Curtin devices [10], [11]. The 10 N part spans 512 counts, 51.2 counts per newton, after a tare [173], [174], and a calibrated sensor comes matched to a board that stores the maker's calibration [175]. How far its force in newtons can be trusted is less settled. The 2025 project found drift under 2.5 percent of full scale over 26 minutes [14], but a bench test of a calibrated 15 mm, 4.5 N sensor found accuracy errors near 3 percent of the load. Its repeatability and hysteresis, near 4 and 4.5 percent of full span, and its drift, up to 7.3 percent of span in 2 min, all exceed the maker's figures [176]. Neither tested a fingertip. The software relies instead on measures that do not use the conversion to newtons: the moment force crosses a person-specific threshold, and force in percent of a finger's own maximum. That is why each session is calibrated to the person's hand.

Fractionated reaction time splits a response where the muscle first fires, and only the part before that moved with the warning interval [177]; timing from the push splits it later, where the force starts to rise. When a press begins is itself a measurement choice: an onset picked from a rising signal shifts with how fast it rises and with the method [178], so the analysis finds each onset offline. A zero-phase forward-backward filter [179] low-passes the force, so it does not shift the onset, and a Savitzky-Golay filter smooths its rate of change [180].

Reaction time across blocks is usually modelled as a linear mixed model, in which a random intercept gives each participant their own speed and a random slope lets an effect differ between them [181], [182]. Without the slope, an effect that varies between people gets too narrow an interval (Section 4.3.1). Kenward-Roger or Satterthwaite degrees of freedom keep the error rate near nominal in small samples [183], [184], and block means are read as estimated marginal means [185]. An uncertainty is stated as the Guide to the Expression of Uncertainty in Measurement sets out [186]. Each source becomes a standard uncertainty, from repeated readings or from other knowledge such as a datasheet. A rectangular limit L, with every value

inside it equally likely, counts as L divided by $\sqrt{3}$. Independent sources combine as the root sum of squares, and a coverage factor $k = 2$ gives about 95 percent.

EEG event-related potentials are found by averaging many short sections of the recording, each lined up to an event, so every event has to be marked at the right moment. Guidelines ask studies to report when stimuli and responses occurred [23]. Timing varies between commodity computers, as Plant and Turner found with dedicated measuring hardware [187], so each laboratory measures its own. Bridges et al. timed the leading experiment packages with a photodiode, a sound recording and a robot pressing a button box. The best showed a visual lag of 4 to 7 ms on Windows and Linux and about a frame more on macOS, varying by under a millisecond. An ordinary USB keyboard can add 20 to 40 ms [67]. The device's own timing chain is checked the same way: a microphone for its sound delay and the raw force stream for its presses (Sections 4.2 and 4.3.2). A trigger pulse must last at least two amplifier samples, the usual rule for a digital trigger input [188]. The marker log follows the EEG-BIDS events file, giving each marker's onset and duration [189], with the byte stored as its value.

2.13 Validity and reliability

Validity is whether a measure reflects what it claims to; in a healthy cohort the evidence is construct evidence, reproducing the effects the literature predicts. Reliability is whether it gives the same answer when nothing has changed. For test-retest reliability, Koo and Li [22] recommend a two-way, absolute-agreement ICC, single-measure when single scores are used, computed as ICC(2,1) [190], [191]. They read it as poor below 0.5, moderate to 0.75, good to 0.9 and excellent above that, and grade a result on its 95 percent interval. With small samples this thesis grades the estimate (Appendix K). The standard error of measurement and the minimal detectable change turn this into the units of the measure [192], and Bland-Altman limits show whether disagreement is a shift or a spread [193]. With about ten people an ICC interval covers more than one band [194], and many checks on the same people need a correction such as Holm's [195]. An effect that replicates well can still measure one person poorly: across seven classic tasks, Hedge et al. found test-retest reliability of 0 to 0.82, because an effect everyone shows varies little between people [196]. Reproducing a known effect and following one person are therefore separate claims, and each mode here has to earn both. Precise reliability

estimates need about 50 participants and at least three trials [197], well beyond this study. Digital versions of laboratory tasks are checked against the original before they are trusted, as a tablet Corsi test [131], a three-minute vigilance test [198] and a mobile timing battery [66] were; the device's games need the same check.

2.14 Synthesis and novelty statement

The literature and the earlier Curtin projects point to one problem: a device practical enough for everyday use rarely collects research-quality data, and the devices that do stay in laboratories. Semester 1 built a platform that records the force and timing of every press; this thesis tests whether those numbers can be trusted. Table 7 links each game mode to the paradigm behind it and the effect a healthy hand should show, and each row became a check in the study. Where a reference value rests on a single study, such as the chord cost [86] or the Simon span [129], the check is set as a direction or a range rather than a match to a published number. The foreperiod effect, enslaving, the sequence rebound and the lead in tapping each rest on a review or on several studies [19], [64], [82], [84], [93], the touch errors on two [120], [121] and the release deficit on one group's studies [24], [30], [144]. The treatment evidence is weaker and in places conflicting, for bilateral training [75], [76] and force-tracking training [42], so the thesis makes no treatment claims.

The finger measures are not specific to stroke. The release deficit of Parkinson's disease is measured with light-force tracking like Force Pilot's [24], misplaced touches after nerve injury are mapped in a confusion matrix like Buzz Hunt's [121], and the four sensors suit GraphoGame's sound-to-print choice [199]. Extending the platform to these conditions needed software, not hardware, and Table 6 states what it claims for each.

Gamified tests are usually validated against their originals, and since a game can make one task draw on several abilities [200], each mode here is checked against its own paradigm. No other game device in the search combines three established checks: each measure checked against a reference outside the software, each game against an effect published for healthy hands, and each press timed from the start of its push on the raw force stream. Any force-based game device could apply them. Here they run on four force sensors with one per-press record and one calibration, at AUD 192 a calibrated sensor (Section 4.8.3). Table 8 compares the devices that

record finger force with what the method needs. None of the others reports a timing check against an outside reference. Only the 2025 Curtin version came near a known effect, without comparing it with the literature. AMBER is the closest: its full paper reports retest reliability, which this project's normative study must still supply, and correlations with standard tests, but no timing reference, known effect or markers.

Table 6: Conditions beyond stroke, the evidence behind each mode and what the platform claims.

Condition	Modes	Evidence	What the platform claims
Parkinson's disease	Force Pilot, Rhythm	Release deficit [24]; slow relaxation [201]; regular tremor [150]; gait cueing [202]	Measures only; healthy values on this device
Older adults	Force Pilot, Reaction, Syllables	Steadiness relates to function [153]; choice RT slows with age [171]	Measures only
Dyslexia	Syllables	Phonics works [25]; games need an adult [157]	Practice game, not a treatment

Table 7: Each game mode's paradigm and the effect healthy hands should show.

Mode	Paradigm	Expected in healthy hands	Clinical relevance	Key sources
Reaction	Choice reaction time with an exponential foreperiod	No anticipation; no practice effect on a second pass	Speed of initiation after stroke	[19], [118]
Adaptive	Challenge point	Pace rises while hit rate stays in band	Dosing practice at the right difficulty	[50], [51]
Muscle Memory	Serial reaction time task	Slower on random than on learned order	Implicit learning after stroke	[92], [105]
Chords	Enslaving and chord reaction time	Low leak at light force; harder chords missed more; about 126 ms per extra finger on keys	Loss of finger independence after stroke	[82], [84], [86]
Rhythm	Tapping to a beat	Taps anticipate the beat; spread in tens of ms	Rhythmic cueing, Parkinson's disease	[64], [65]
Force Pilot	Visuomotor force tracking	Irregular waves tracked worse; release equals press	Force control after stroke, in Parkinson's disease and with ageing	[20], [24]
Buzz Hunt	Tactile localisation	Near-ceiling accuracy; neighbour errors	Sensory loss after stroke	[120], [121], [123]
Echo	Corsi and Simon span	Span of about 5 to 9 on four lanes	Working memory after brain lesions	[128], [129]
Syllables	Phonics with sound and print	Ceiling for fluent readers	Dyslexia in children and adults	[25], [155]
Mirror	Bimanual in-phase coordination	Small gap between hands	Bilateral training after stroke	[73], [203]

Table 8: Devices that record finger force, against what the method needs, from each report or abstract (No: not reported).

Device	Each finger's force, every press	Timing checked against a reference	Known healthy effect	EEG markers
FINGER [8]	Index and middle, on blocked notes	No	No	No
Force tracking [41], [42]	Thumb with one finger, or the thumb alone	No	No	No
AMBER [46]	Fingers 2 to 5, in two of its three tasks	No	No	No
Individuation device [47]	Yes	No	No	No
Fingertip force device [48]	Yes, in three dimensions	No	No	No
Curtin device, 2023 to 2025 [10]-[17]	Uncalibrated, then four calibrated sensors	Not measured	Learning seen in 2025, not set against the literature	Code untested
This project	**Yes, at 200 Hz**	**Force stream, microphone and push onset; screen untimed**	**Sequence effect in four people**	**Markers in two recordings**

The 2025 project compared 50 random presses before and after 500 sequence presses in six adults, a difference that measures general practice as much as sequence learning. This year's score is the slowing when random order returns, which holds even against the faster of the two random blocks. Section 4.3 reports the first Curtin value of the effect itself. Four gaps remain. No device in the searches of Appendix J sets per-finger force games against known effects and outside references, and no Curtin version has had its measurements validated. Finger cueing is under-researched, and the other conditions' measures are not yet on one device. This thesis addresses the first two and supports the others.

Each of the three checks has its own literature: outside timing references in experiment software [67], [187], published effects as evidence of construct validity [21], [200] and onsets found on the force itself [177], [178].

Chapter 3. Experimental Procedure

The work followed the V-model of Figure 2, in which each stage of definition is checked by the stage opposite it. Semester 1 set the needs, with the progress report's constraint budget, failure-mode analysis and risk register [18]. The requirements and objectives (Table 9, Table 4) were built into the software, checked by automated tests and simulated players, then against references outside the software (Section 3.4) and last against effects the literature already knows (Chapter 4). In practice the stages overlapped, and every fault found later sent the code back down the V with a regression test, as IEC 62304 asks (Table 26).

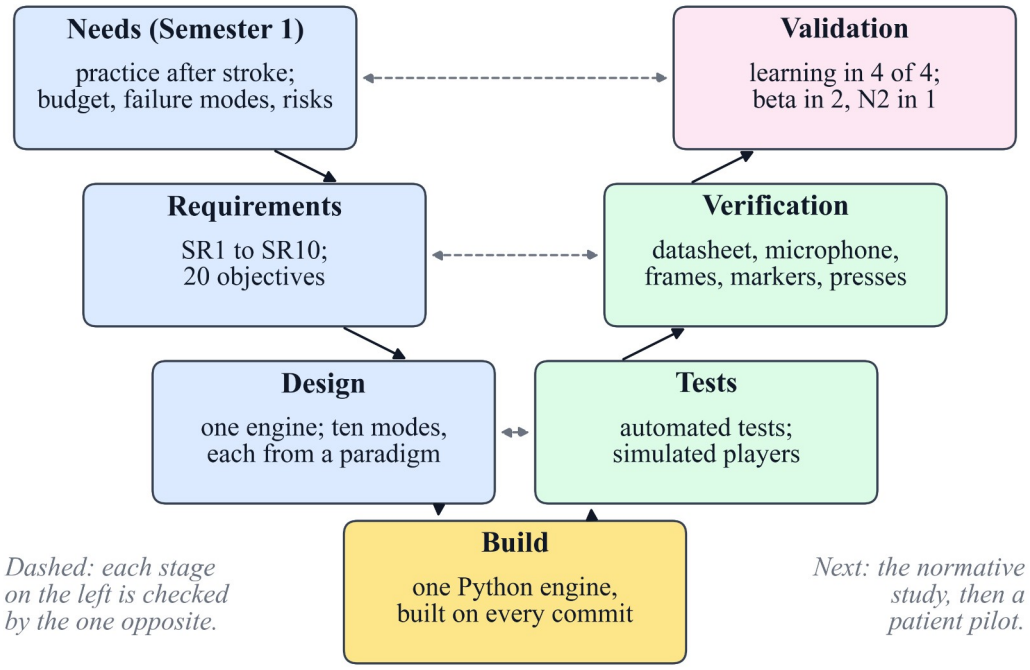


Figure 2: The project's development as a V-model. Each stage on the left is checked by the one opposite it.

3.1 System overview

Figure 3 shows the platform. A hand device holds four force sensors plus four vibration motors, one of each under each finger. An Arduino Nano reads the sensors, drives the motors and streams the readings over USB to a computer running the software. Every game mode runs on the author's engine, which reads each board or the keyboard through one interface, detects presses, cues by screen, sound and motor, scores each response and saves the result. Each game played is saved as its own session folder, and a scripted analysis notebook turns those folders into the study's

tables and figures. When the EEG laboratory configuration is loaded, the engine also sends one-byte event markers to a marker box connected to the EEG amplifier.

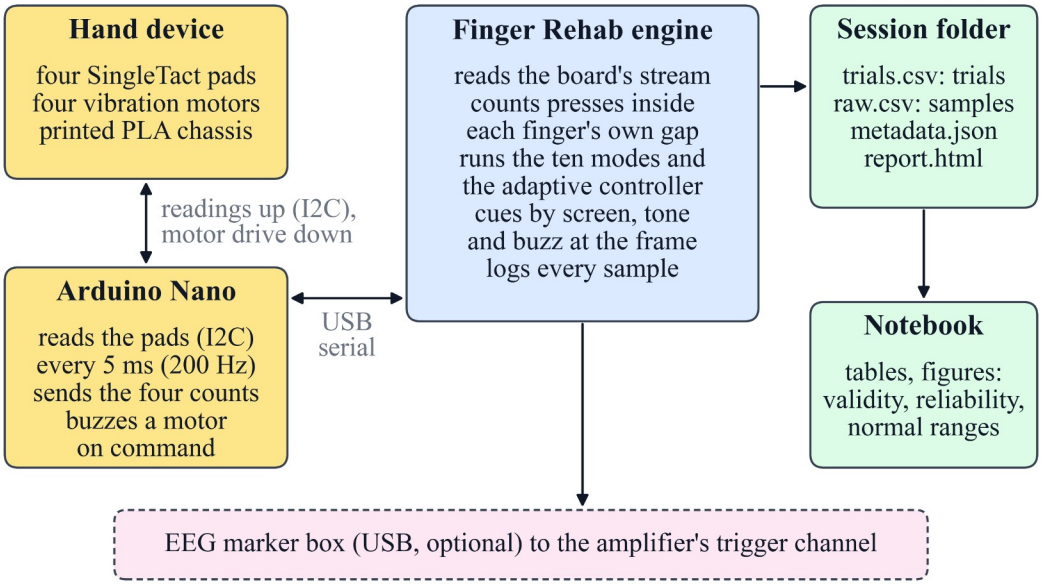


Figure 3: System architecture. One board per hand, one drawn; the second and the marker box are optional. The engine writes one session folder per game played.

Table 9 states the requirements the platform was built to and how each was verified, with the measured values of the progress report's constraint budget closed out in Table 16. The requirements are written as a user or a laboratory would test them.

Table 9: System requirements and their verification.

ID	Requirement	Verified by	Result
SR1	A press is detected from the calibrated force gap of that person's own hand, never from a fixed threshold	Calibration and detector tests; Figure 5	Met (Section 3.3.2)
SR2	Every trial is logged with its time, finger, outcome and force, and every sample of the force stream is kept	Logger tests; trials.csv and raw.csv in every session folder	Met (Section 3.3.1)
SR3	A hardware fault (a dropped board, a dead sensor) never counts against the participant	Dropout tests in every mode	Met in simulation: dropouts in tests, resting sensor on replay (4.3.2)
SR4	The study's eight modes play with their fixed settings in one session, on one board, inside a one-hour booking	Battery tests; 48 simulated sessions	Met in simulation, 44.4 to 44.9 min (Section 3.6); one real session, six of eight games in 14 min (Section 4.1)
SR5	The game starts on its own when a board is plugged in, within 2 s	Startup log; simulation	Met in part: first sample 1.75 s in simulation, 4.2 s as run (3.3.3); plug-in to start untimed

ID	Requirement	Verified by	Result
SR6	A session needs no technical help: log in with the session's length, calibrate, play	The participant sheet; the sessions of Section 3.6 run by the participants themselves	Met: Person 5's session needed no help (Section 4.1)
SR7	Windows and macOS installers and the laboratory folder come from one build of one commit	Continuous integration on every commit	Met (Table 26)
SR8	EEG markers leave on the display frame that shows the stimulus, one per event, coded by type	Marker tests; 1,886 of 1,886 markers in order	Met (Sections 4.2 and 4.4)
SR9	No recording is overwritten, and no name reaches a result	Session model tests; names replaced by codes on import	Met (Section 3.9.1)
SR10	The analysis recovers what a participant was given	Twelve simulated participants with known traits	Met in simulation: four traits r 0.90 or more (Figure 9); Echo 0.78

3.2 The hand device, as the software sees it

This section keeps only what the software depends on in the hand device (Figure 4), built in Semester 1 and described in the progress report [18]. The 2026 team built the boards, the printed circuit board and the firmware, and the finger rests, set by a linear actuator in Semester 1, are now set by hand on threaded rods. This designs out a failure mode the progress report rated high, a stuck linear actuator (Table 36), at the cost of powered resizing. Semester 2 used the device as it stood. Each fingertip rests on a SingleTact capacitive force sensor, 8 mm in diameter and rated to 10 N, read over the I2C bus. The factory-calibrated sensors read 19.5 mN per count within 2 percent of full scale [173]-[175]. Presses are detected from each person's own rest and light press (Section 3.3.2), and Force Pilot works in percent of each finger's maximum. Only forces reported in newtons therefore depend on that calibration (sensor figures in Appendix I).

Each hand's Arduino Nano reads its four sensors every 5 ms (199 Hz measured) and drives a vibration motor under each finger. A failed read is sent as zero, so a dead sensor logs as a dropout, not a resting finger.

3.3 Software platform

The software reads the board's four sensors at 200 Hz, counts each press against that person's calibration, runs the games and keeps every raw sample, so any score can be computed again.

3.3.1 Session model

The author wrote the software in Python, one code base for Windows and macOS [34]. It keeps the 2025 serial protocol and the design of its press detector, a moving resting level with thresholds above it [16]. It saves each game as a folder with its trial log, raw 200 Hz force stream, configuration and calibration, so any session can be re-analysed with its exact settings.

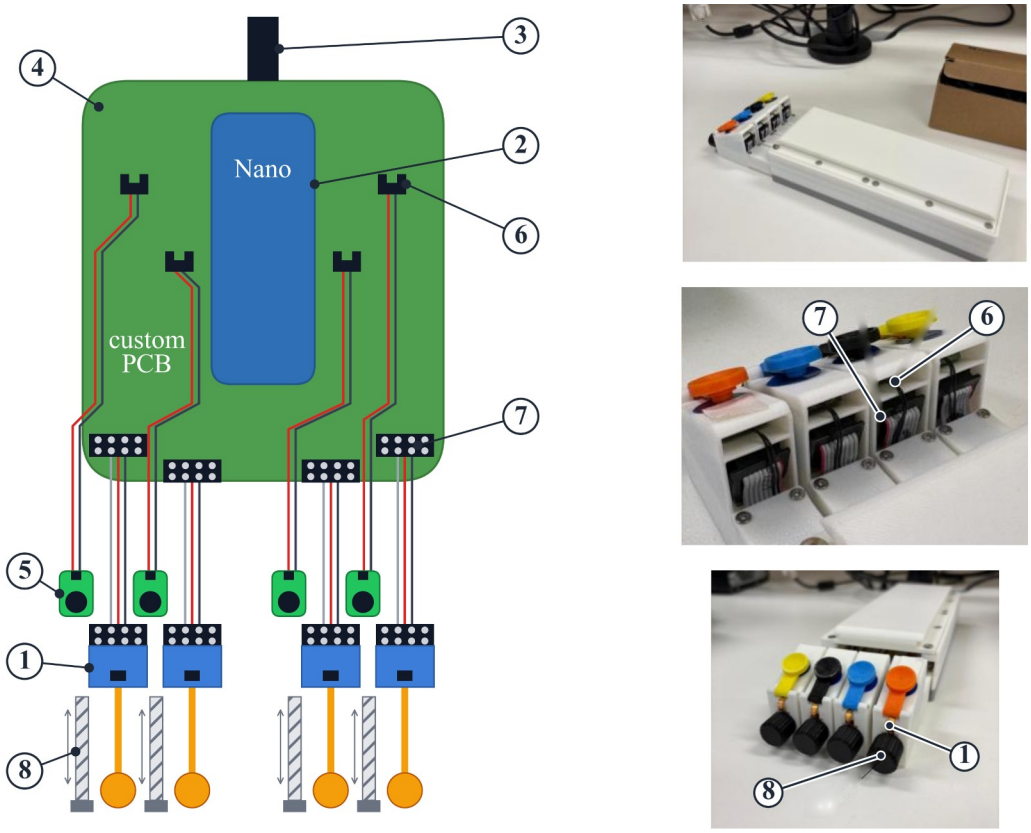


Figure 4: The hand device, drawn and photographed: 1 SingleTact sensor and board, 2 Arduino Nano, 3 USB cable, 4 custom PCB, 5 motor PCB, 6 motor connector, 7 sensor connector, 8 threaded finger-rest rod.

3.3.2 Press detection and per-person calibration

A press is detected from the force signal; there is no switch. A quick calibration at the start of every session, about 30 s per hand, records the sensors with nothing on them, the hand resting in position and a light press on each finger. On this device the little-finger sensor rests at about 30 counts and the index under 3, so one fixed threshold above the little finger's rest could be beyond a weak index's reach. Each finger's thresholds are therefore set inside its own gap g between resting and the light press, and a finger is asked to repeat if g is under 20 counts. Each raw sample x, in

counts at 200 Hz, is smoothed to s, and the baseline b follows s only while the finger is up, starting from the calibrated resting level:

$$s = 0.35 \times x + 0.65 \times s_{\text{prev}},\ b = 0.0005 \times s + 0.9995 \times b_{\text{prev}} \quad (1)$$

$$T_{\text{on}} = b + 0.30 \times g,\ T_{\text{off}} = b + 0.20 \times g \quad (2)$$

A press was counted when s rose through the press threshold and released when it fell back through the release threshold; Appendix I gives the refractory time, the threshold's floor and the caps on g and the release.

A sensor dropout is logged as its own outcome, never as a miss. Figure 5 replays one real press through the detector. Section 4.2 reports the count lag, how far the count trails the raw force. Figure 6 shows it grow on slower rises; every ramp to 3 s is counted.

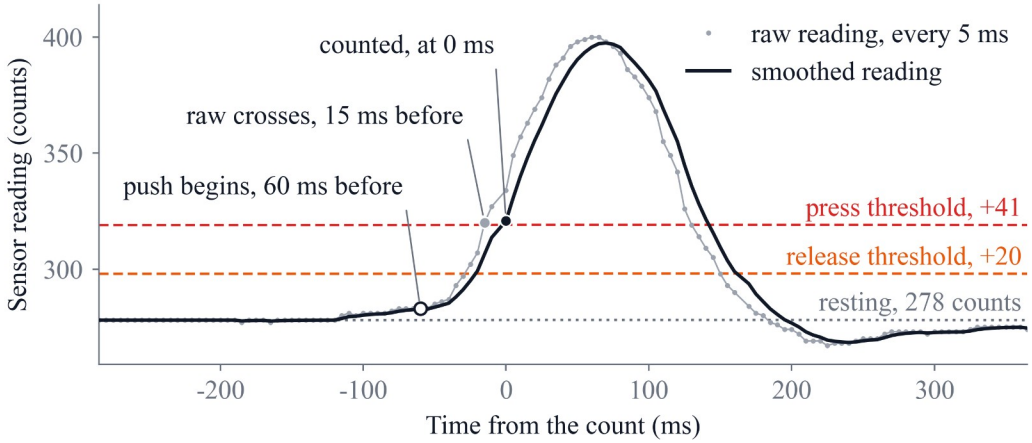


Figure 5: A slow real press through the detector: among the three slowest of eight presses in a 23 September 2026 block, replayed through its own code on the sensor's 5 ms grid with the levels it held; its onset is by the press check's rule. Raw counts (zero load about 255); the press threshold was at 40 percent of the gap, the study's at 30.

3.3.3 Zero-setup workflow

The software finds connected boards on its own and logs its startup, so a therapist connects the device and starts the software without changing a setting. On the laboratory computer the scan took the EEG trigger box, also an Arduino, for a hand board, so the box's port was set by hand (Table 15; Section 4.8.5). The author built the auto-launch the progress report planned: a watcher on Windows and macOS looks for a board every second and starts the game when one is plugged in. The first

force sample missed the 2 s target of OD1. Opening the port restarts the board. Its boot and 2.6 s self-test of four motors outlast the software's 3 s wait, so in 57 of 76 logged start-ups the first sample came 4.2 s after the port opened (Table 15). Streaming during the self-test, with no fixed wait, gives 1.75 s in simulation (Section 4.8.5). The watcher's time from plug-in to game start was not measured, and the laboratory sessions, started through PsychoPy, did not test it.

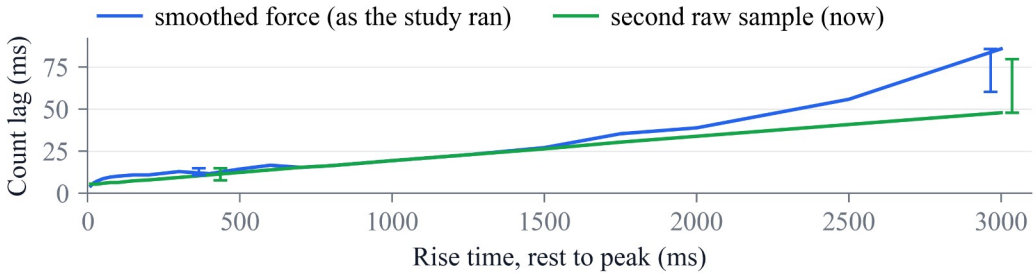


Figure 6: Count lag by rise time on simulated noise-free ramps in whole counts, from the ramp crossing the press threshold set at rest (30 percent of the gap) to the count: as the study ran, on the smoothed force (blue), and as the software now counts, on the second raw sample (green). Lines: a 70-count gap; bars at 400 ms and 3 s: gaps of 40 to 150.

3.3.4 The ten game modes

The author added six modes to the progress report's four. Each mode follows a published paradigm. Six of the eight in the study are built to reproduce a published effect in healthy hands, so their numbers can be checked against known values before use with patients (Table 10). Figure 7 shows five game screens and the results screen.

The adaptive controller adjusts the difficulty from how often the player presses the right finger and how quickly, aiming to hold the success rate between 65 and 80 percent. A hit lands within 90 percent of the beat interval. The pace starts at 30 BPM for a severely impaired hand, capped at 180 in the sessions. An optional climb raises it 20 percent a hit until the first miss, to 240 BPM (Section 4.2).

Rhythm's main problem was audio latency: the development laptop measured 87 ms against the 40 ms assumed in Semester 1, so measuring each machine's delays is now a set-up step (Table 16). For two hands, the bilateral runtime gives each hand its own board (Appendix I gives Mirror's pairing gate).

Table 10: The ten game modes and the construct each measures; Table 7 gives the paradigm and sources behind each.

Mode	What the player does	Construct
Reaction	Press the lit finger after a random wait	Choice reaction time
Adaptive	Press lit fingers at a pace that follows performance	Speed at a held success rate
Muscle Memory	Play takes of a piano riff with a hidden repeating sequence	Sequence learning
Chords	Press two to four fingers together, keep the others still	Finger individuation
Rhythm	Press on the beat of a song	Sensorimotor synchronisation
Syllables	Hear a word, press the finger under the spoken syllable	Sound-to-print matching
Mirror	Press the same finger on both hands together	Bimanual synchrony
Force Pilot	Keep a craft on a moving line with finger force	Force tracking
Buzz Hunt	Press the finger that vibrated	Tactile localisation and span
Echo	Watch a growing sequence, then play it back	Working memory span

Reaction replaced the Semester 1 Classic mode as the baseline block. It and the six new modes each follow one paradigm, and Appendix I lists the design choices their checks depend on, with sources.

3.3.5 Force Pilot for Parkinson's disease

In Semester 2 the author extended two modes to conditions other than stroke, with no new hardware. In Force Pilot one finger keeps a craft on the centre line of a scrolling corridor, with force shown in percent of that finger's comfortable maximum. A ladder of twelve waves runs from a slow sine to waves that cannot be predicted from one cycle. Its ramps and sine halves are symmetrical, so release error, the deficit the Parkinson's literature reports most often (Section 2.10), can be compared to press error. The notebook computes the published measures [31]-[33] from the raw 200 Hz stream, each read against a finger resting on the same sensor. Appendix F gives the waves, the measures and how they were checked.

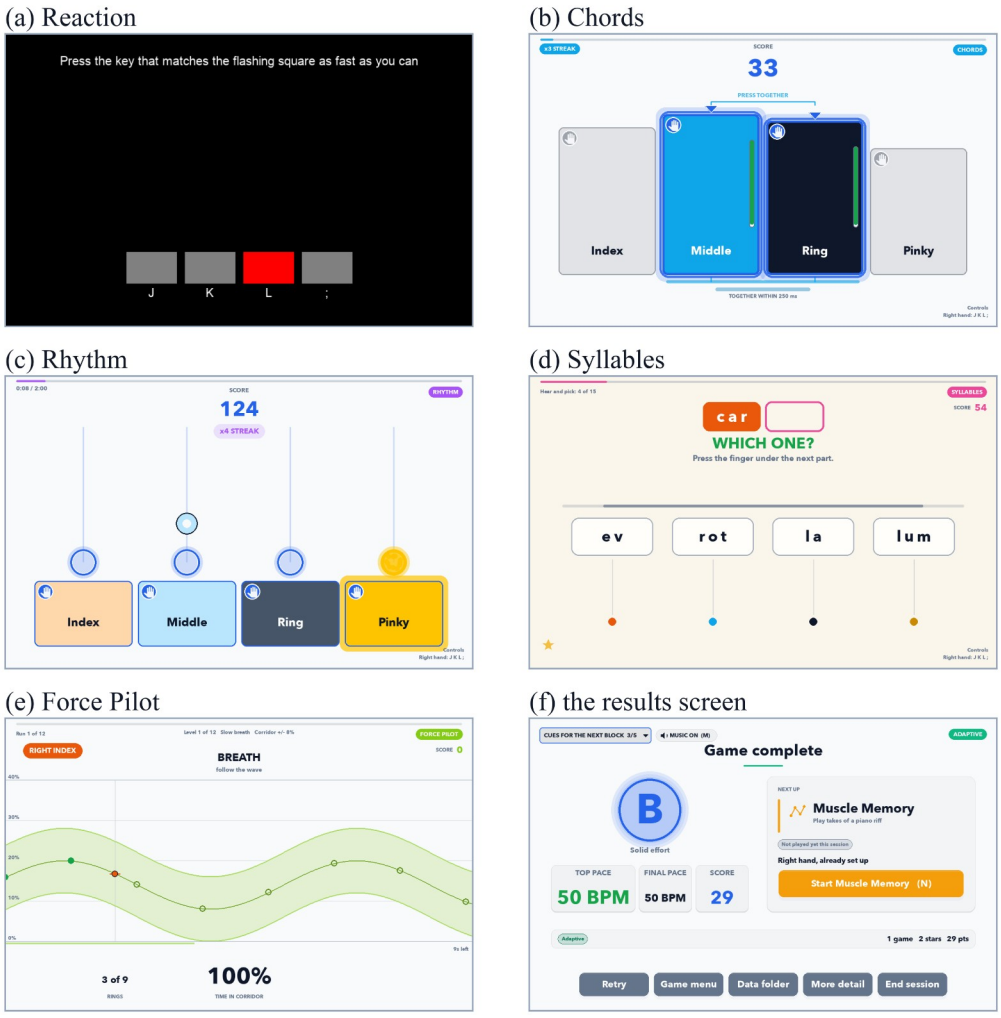


Figure 7: Five of the ten game screens and the results screen: (a) Reaction in the look of the EEG laboratory's own task, now its default; (b) Chords, whose finger cards the sequence task used; (c) Rhythm, notes falling towards the finger to press; (d) Syllables, a spoken word's next part; (e) Force Pilot, finger force keeping a craft in the corridor; (f) results after an Adaptive block.

3.3.6 Syllables for dyslexia

Syllables, first designed for children aged 6 to 9, speaks a word and shows four written chunks over the four fingers. The player presses the finger under the chunk just spoken; the wrong options come from the errors readers make. It now covers all ages with five profiles over 932 words, picked by the age entered at login (Appendix G). No reader with dyslexia was tested, and the mode is not part of the healthy session. The largest English trial of GraphoGame, whose sound-to-print matching it follows, found no effect on reading [204], so Syllables is reported as built and verified software, not as a treatment.

3.3.7 EEG event markers

Under the laboratory configuration the engine sends a one-byte marker for each event the EEG epochs are aligned to, held on the amplifier's trigger lines for at least 8 ms, twice the 3.9 ms two-sample minimum [188]. Stimulus markers leave on the display frame that shows the stimulus, and each marker is logged with its intended and actual time (codes in Appendix C). Figure 8 draws the timing chain from a cue to its logged press with the delays measured on the development laptop (Section 4.2).

3.4 Verification and testing

The author verified the platform at three levels. An automated suite of over 4,000 tests runs every mode through the real engine with simulated boards. It covers 87 percent of the game's lines, now about 69,000 lines of Python in 82 files against roughly 11,400 in 27 at the progress report. It catches a change that breaks working code, while the other two levels check the numbers themselves. A simulation harness plays whole sessions through the engine with a model hand and, in the cohort runs, players with traits of their own (Table 46). The harness gave the session timings. It also checks the analysis end to end. Twelve simulated participants, each given a known reaction time, tapping asynchrony, tracking lag and enslaving ratio, play the whole battery. The notebook has to recover what they were given. It does (Figure 9). On the 26 October code, $r = 0.904$ for reaction time, 0.987 for asynchrony, 0.997 for tracking lag, 0.965 for the enslaving ratio and 0.78 for Echo's mean span (Appendix K explains the small offsets). The real board ran nine modes, and the device's delays, noise and drift were measured against outside references (Table 16). A simulated session's 154,035 frames were checked for stale or leaked stimuli. Every fault found before the first participant was fixed with a test (Section 4.8.2), and the one found in use is in Section 4.3.2.

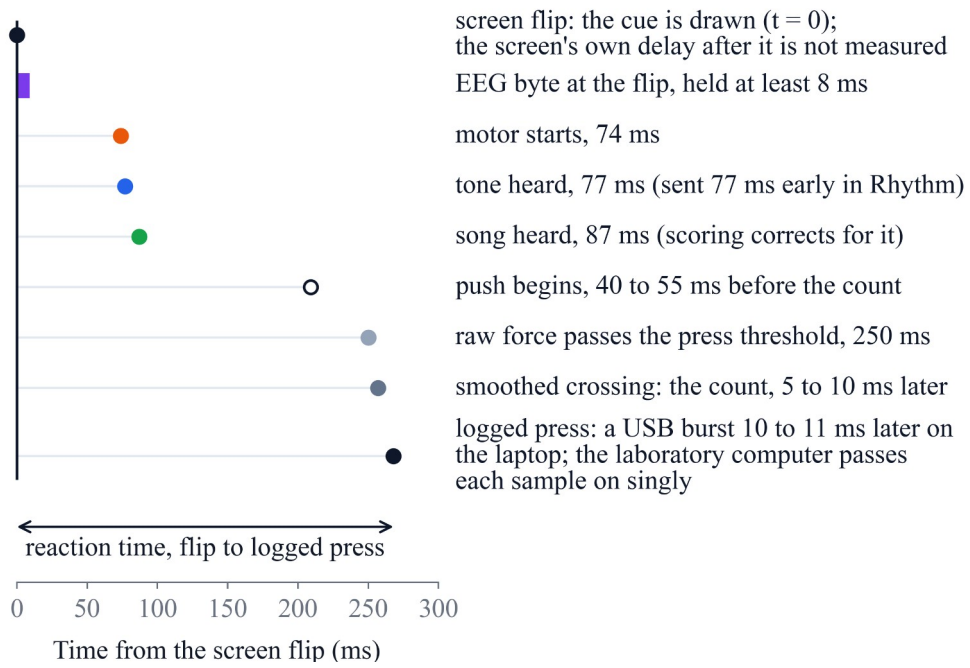


Figure 8: Delays from a cue to its logged press, for an example press whose raw force passes the press threshold 250 ms after the screen flip, from the study's presses and the development laptop; the laboratory computer passes each sample on singly.

The practices were set against the standards a clinical or research instrument would be held to (Table 26 in Appendix A). They include the software life cycle of IEC 62304 [205] and the risk management of ISO 14971 [206]. The project claims compliance with none of them, since that needs an audit it has not had. The author settled each main design decision, from the per-finger press threshold to markers sent on the display frame and the measured audio delay, by measurement (Table 41 in Appendix I).

3.5 Checking the platform in healthy adults

The platform was checked as a whole on the right side of the V-model (Figure 2). Each question of Section 1.4 was set against a reference outside the software (Table 11). Three checks used real people. The laboratory's sequence task gave a known answer, its published learning effect (Section 3.7). The force stream of those sessions timed each press against the start of its push, and one session was self-run with no one present (Section 3.6).

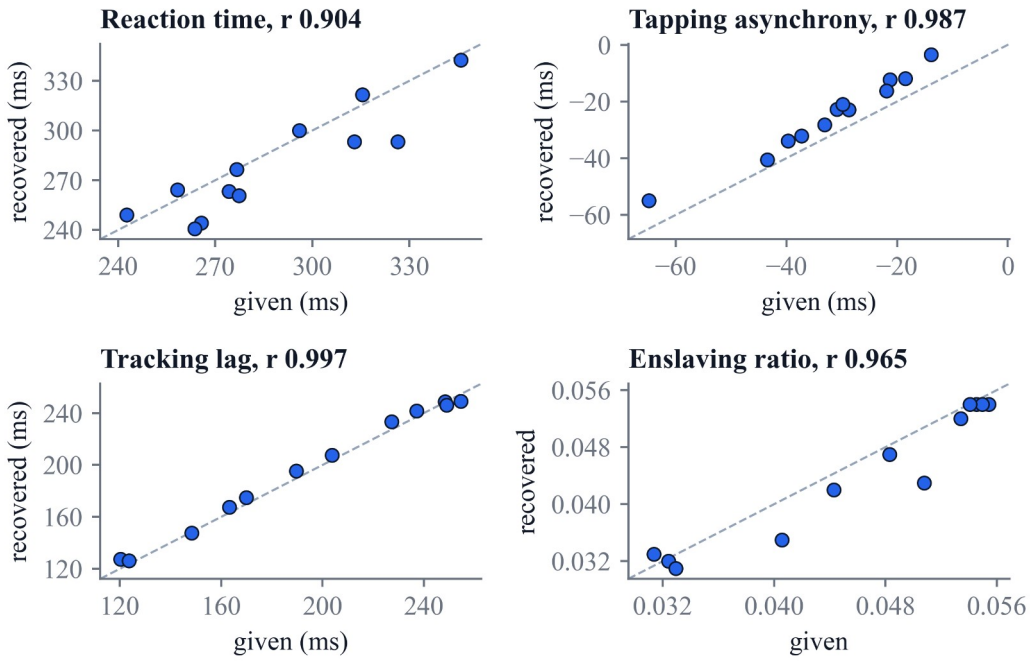


Figure 9: The notebook's recovery of twelve simulated players' traits. Each point is one player, recovered against given; the dashed line is equality.

Table 11: How each question is checked, and the reference outside the software that each check rests on.

Question	Check	Reference outside the software	Section
SQ1 Timing	Force and sound timing checked; every counted press set against the start of its push	Sensor datasheet; microphone loopback; the force stream; none for the screen yet	4.2, 4.3.2
SQ2 Function	A known learning effect reproduced on the device	The laboratory's sequence task and the published SRT effect [96]	4.3
SQ3 EEG	Every marker set against the game's log; known early and learning waves	The laboratory's amplifier; ERP CORE [207] and the SRT literature	4.4, Appendix C
SQ4 Readiness	The battery run without a researcher; the study kit rehearsed first	Simulated participants with known traits; the session's own logs	3.4, 4.1, 4.6.2
SQ5 Extension	The new measures run on synthetic traces with known answers; every chunk and word of each reading profile voiced	Published definitions of the measures; the profiles' sources	3.3.5, 3.3.6, 4.7

The platform delivers a study kit for the normative study that follows this project. It holds the battery, a pre-specified check for every mode and the analysis notebook that writes every result table. The author fixed the kit before any participant, in a design document public since 2 September 2026 [34], and every later change is dated in Appendix B. It was rehearsed end to end on simulated sessions. Syllables, built for

readers with dyslexia, would only show fluent adults at ceiling, and Mirror needs two boards, so neither is played.

Which results were planned matters for how far each can be trusted. The study kit's checks were fixed before any data were read. The sequence task's learning score, decided participant by participant on its bootstrap interval, follows the planned rule. The EEG's primary tests were settled on the first recording, before the second. The group t-test and mixed model, the further readings of Table 19, the press check with SQ1's lag check, and the P3 and beta checks were added once the data were in, each dated in Appendix B.

3.6 Participants and session procedure

Four people played the laboratory's sequence task: two students from the project team, recorded with EEG, and two of the laboratory's volunteers, all invited by the laboratory's head. Welber Marinovic, the laboratory's head, offered the self-run session to healthy adult volunteers from the laboratory's student group,. Ten were sought, and the sessions were left to them to test whether the software runs unaided. The offer was to play once on the laboratory's own computer, with the device left set up beside it. Participants could come at any time from 15 to 23 October 2026, without the author present. A one-page sheet by the computer gave the whole procedure, from setting the finger rests by their threaded rods to ending the session. The login screen records age, sex (optional), hand length and the writing hand, which stands for handedness, and asks for the session's length. Each file is written to a temporary copy and swapped in whole. Everyone plays the right hand. The number who took part and the lengths they chose are in Table 14.

Logging in with a session's length starts the whole battery, beginning with the quick calibration, in one of two set orders (Appendix K). The session waits for a dropped board and, if the application closes, resumes from the first unfinished block at the next login. With no researcher present, the software's logs record each session: the startup latency, the blocks completed, the minutes at the device and the rest taken. Over the 48 simulated 45-minute sessions, each simulated player's median time at the device was 44.4 to 44.9 minutes, the longest 45.6, inside the 50-minute hard stop (Table 47 in Appendix K). Figure 10 draws one in order A.

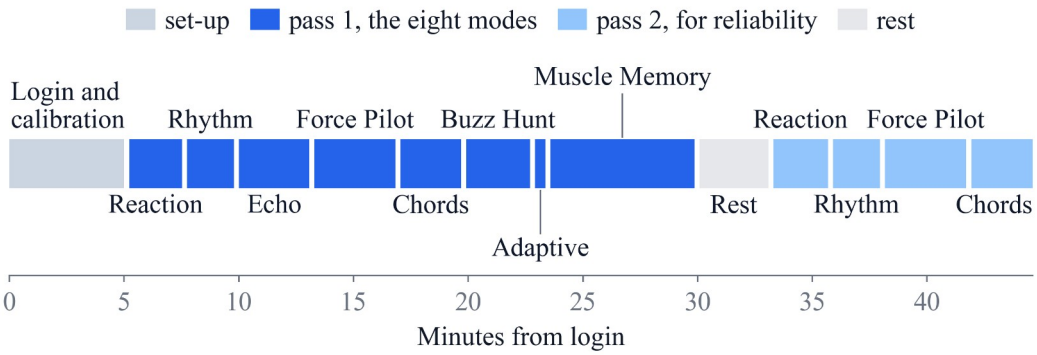


Figure 10: The 45-minute session in order A: login, seating and calibration, pass 1 through the eight modes, the rest and pass 2 of the four modes read for reliability; block lengths are medians of 48 simulated sessions (Table 47).

Every length plays the same full-length games, so a block from any length is the same measure (Appendix K gives each length's games and simulated timings). Only the 45 and 60-minute sessions play the second pass that R3 and the reliability results read, so the sheet asked for the longest session the participant's time allowed.

The data then follow a fixed path (Figure 11). A script replaces every name with a code and keeps nothing if a name remains. A second checks each session against its plan. The notebook computes the results from the coded files, and those files and results go to the supervisor. The whole path was rehearsed on ten simulated sessions before real data went through it (Appendix K gives each step).

3.7 EEG laboratory component

The Curtin EEG laboratory runs a serial reaction time task on a PsychoPy script of its own. Under the laboratory configuration, the platform's Reaction mode reproduces it trial for trial, on the finger cards rather than the script's squares (Figure 23 in Appendix C). A session has 48 random practice trials, eight learning blocks of the 10-item sequence 1-3-2-1-4-3-2-4-1-3 repeated ten times, 48 random post-test trials and a free recall. The planned rule keeps correct responses under 1000 ms, as the laboratory's studies do, but not each block's first trial. Appendix C gives the task's timing, the further readings and how recall is scored.

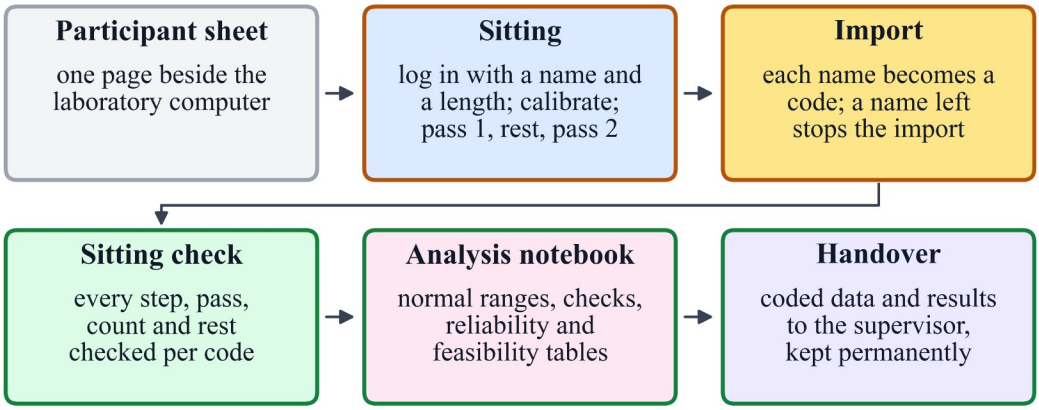


Figure 11: The study's data path. Names stay on the laboratory computer and the author's machine; from the import onwards every file, table and figure carries a code.

The recording is analysed in MNE-Python [208] and reported as the EEG guidelines ask [23], with every processing step and setting in Appendix C. Its windows and sites follow the SRT literature and ERP CORE [207] and were settled on the first recording. Two tests are primary, the N2 (random post-test against blocks 7 and 8) and the P3 (random practice against blocks 7 and 8), with Holm's correction across them. A result is called an effect only when its primary test passes; beta at C3 is secondary. The task's standard behavioural measure is its learning score [92], [93], scored by the planned rule. SQ2 is decided participant by participant, on each score and its bootstrap interval from 4,000 resamples of the trials. Across participants an exact sign-flip test checks the effect's direction; a one-sample t-test and a linear mixed model of the single trials give its size. Fitted by restricted maximum likelihood (REML) in the author's own code, the model gives each participant their own speed and learning effect [181]:

$$\mathrm{RT}_{ij} = m_k + u_j + v_j \times \mathrm{post}_i + e_{ij} \qquad (3)$$

where m is the mean of block k, the block holding trial i; u and v are participant j's own speed and post-test effect; post is 1 in the post-test and 0 elsewhere; and e is the trial's noise. Its post-test contrasts use three degrees of freedom, one fewer than the participants, as a paired t-test across them would. The speed-up per block is the linear trend of the means m over blocks 1 to 8. The two recordings are read together by averaging each person's own difference between conditions, with the trial labels

shuffled within each person 10,000 times, so that one person's level cannot stand in for an effect.

3.8 Measures and analysis

The unit of analysis is one value per participant, mode and measure from pass 1, with pass 2 used only for reliability. Where the analysis notebook re-scores a measure from the raw stream, the re-scored value is used rather than the in-game score. A press's force is its peak above the mean of the 250 ms before its cue, not before the press as in Rayyan Raiz's press zero [209]: a resting finger's level crept 0.27 N in Person 2's session. The peak is converted to newtons with the sensors' factory calibration (Equation 6). The enslaving ratio, a variant of the enslaving of Zatsiorsky et al. [82], is the mean peak of the fingers that should stay still over that of the pressed ones, each scaled by its gap. Force Pilot errors are in percent of each finger's own maximum. Equations 4 to 6 define the quantities used throughout:

$$\mathrm{ER} = F_{\mathrm{still}} / F_{\mathrm{target}} \tag{4}$$

$$\mathrm{asynchrony} = t_{\mathrm{press}} - t_{\mathrm{beat}} \tag{5}$$

$$F = 0.0195 \times (\mathrm{counts} - \mathrm{resting}) \tag{6}$$

where ER is the enslaving ratio of a single trial. The asynchrony is in milliseconds, negative when the press is early. F is in newtons, one count being 19.5 mN on the 10 N sensor, and resting is the mean count over the 250 ms before the cue.

3.8.1 How each mode is analysed

Table 43 in Appendix K gives each mode's measures, how each is calculated, its healthy value and its predicted reliability. Reaction times are summarised by the median rather than the mean, because their distribution is skewed [117], and Appendix K gives the scoring rules of each mode. The software's onset detector [178] finds the start of a push on the last rise before the counted press. It is the first sample where the low-passed force's velocity passes a threshold and the force keeps rising (settings in Appendix K).

3.8.2 Known-effect checks

Each mode has at least one pre-specified check (Table 44 in Appendix K). For six of the eight it is an effect the literature reports in healthy adults; Echo and Adaptive carry feasibility checks. Passing a check is construct evidence, not criterion validity

against a clinical test, and a large effect can pass despite a timing fault, which the press check (Section 4.3.2) tests directly.

3.8.3 What the study kit computes

The notebook also reads change within a block (W rows) and normal ranges. For five core measures (T1 to T5) it gives within-session reliability as ICC(2,1) with SEM and MDC95. Its directional checks use exact one-sided Wilcoxon signed-rank tests with Holm's correction, which cannot pass with fewer than eight participants; below that, each participant's values are described against the check's band. Appendix K gives the definitions, the predicted reliability classes, the power of the design and the rule that turns the checks and reliability into one verdict per mode.

3.9 Ethical, professional, cultural and environmental considerations

The study was designed as low-risk, self-run research; consent was given by logging in, and the sheet did not say what is recorded until a line was added after the sessions (Section 3.9.1). Table 12 gives each non-technical issue and its handling.

Table 12: Ethical, professional, cultural and environmental issues: who each affects and how the project handled it.

Issue	Who it affects	How it was handled
Discomfort and fatigue in a session of up to an hour	Participants	A 250 ms cue buzz, a stretch and a rest built in; the session is the participant's own, and Esc leaves it at any point (tested); one report of discomfort, while a sensor stuck (Section 4.3.2, Appendix M)
A session with no researcher present	Participants	A one-page sheet gives every step; the software needs no settings; a dropped board or a closed window resumes the session
Hidden sequence in Muscle Memory	Participants	Not disclosed beforehand because it would change the measure
Privacy	Participants	The login takes a name to resume a session and set its order; names become codes on import, before any analysis, and the import stops if one remains (tested; Figure 11); the data stay off the public repository
Overclaiming a clinical benefit	Patients, clinicians	No treatment claims; healthy values and measurement error reported as such
Finger rests set to the hand	Participants, patients	Threaded rods, set before the hand is placed; no powered adjustment
Attribution and licences	Past students, the EEG laboratory, the voice and music authors; the university	The 2025 serial protocol and press detector design, the laboratory's sequence task, Rayyan Raiz's bench analysis and the Kokoro voice credited where used, with the licence recorded with each asset of the open-source stack

Issue	Who it affects	How it was handled
Waste and repair	Environment, clinics	Reprintable PLA parts, AUD 30 a hand; older boards kept working; a part swapped rather than the device (Section 3.9.3)
Inclusion	Future users	Left-handers not excluded, though none took part and the one board is right-handed; plain-English screens and sheet
Extending beyond stroke	People with Parkinson's disease, children and readers with dyslexia in later studies	Force measures reported as research values and Syllables as practice, neither a diagnosis; no reader with dyslexia tested; music off and sessions planned around medication, tremor and fatigue; an adult in every session with children and an Australian voice first
Patient pilot hazards		
Testing an unvalidated device on patients	Stroke patients	Healthy validation first; each hazard below checked by the clinician before the pilot
Vibration on a hemiparetic hand's hypersensitive skin	Stroke patients	Buzz length adjustable, or the buzz off; strength fixed
A weak hand's fingers near the threaded rods	Stroke patients	Rods set before the hand is placed
Fatigue over a long session	Stroke patients	A rest, and the 15-minute session for a first visit
Sensor hygiene between patients	Stroke patients	An isopropyl wipe, since added to the sheet (Section 3.9.2)
A tremor counted as presses	Patients with a tremor	A press threshold above the resting tremor, checked in simulation (Section 4.8.5)

3.9.1 Participant welfare and privacy

The study was designed to meet the definition of low-risk research in the National Statement on Ethical Conduct in Human Research, where the only foreseeable risk is discomfort [210]: here, from vibration, light pressing and up to an hour of concentration. Because the invitation came from the head of the volunteers' own laboratory, a request could feel like an obligation. The self-run design eased that: each played in their own time with no one watching. The sheet gave the steps from setting the finger rests to ending the session with Esc, which works at any point, and the author's phone number for any problem. Taking part meant choosing to log in and play. The sheet did not say what is recorded or who receives it, so consent was implied rather than informed. The session sheet now states what is recorded and who receives it, so the normative study's consent is informed before login (Appendix L). Appendix L also drafts the patient pilot's information sheet and consent form, given through someone outside the teaching or care relationship. All four participants of the laboratory's task agreed beforehand that their data would be recorded, used and kept with this work.

Under the Australian Code for the Responsible Conduct of Research [211], every name becomes a code on import, before any analysis, so no name reaches a table, a

figure or the data handed to the supervisor. The login takes the name only to resume an interrupted session and to set its order. Persons 1 and 2, being on the project team, remain recognisable to colleagues from their roles and session dates; for them the codes protect only against readers outside the project. The named session folders and the EEG files, whose headers carry no name, stay with the author, off the public repository. The de-identified data go to the supervisor with this thesis and are kept permanently, beyond the Code guide's minimum, 12 months for an assessed student project or five years from publication [212].

Running patient trials before a healthy baseline would have asked vulnerable people for measurements that could not yet be interpreted. No one with Parkinson's disease or dyslexia and no child took part, and the software makes no diagnostic claim. A study with children would need a parent's consent, the child's agreement and an adult present.

3.9.2 Safety and professional practice

The device runs from a 5 V USB supply, at most 2.5 W, with no mains connection, and the finger rests are set by hand on threaded rods before the hand is placed. The sheet had no cleaning step; its first step has since gained an isopropyl wipe of the sensors, for the normative study. One finger rest's rod broke before the sessions, which went ahead as the rest still held its finger. The sheet asked participants to turn it gently; the rod needs replacing before use outside the laboratory. Software intended to diagnose, monitor or treat a condition is generally a medical device under Australian law, and the Therapeutic Goods Administration classifies it by its intended purpose and the harm it could cause [213]. The platform makes no diagnostic or treatment claim and was used only for research. Table 26 sets its practices beside those a regulated path asks for. Table 13 sets the project against the four principles of the Engineers Australia Code of Ethics [214]. Third-party material, including the music and the synthetic voice, is used within its licences.

3.9.3 Cultural and environmental considerations

Left-handed people were not excluded, though many neuroscience studies screen them out [215]. Even so, all six who played were right-handed, and the one board is a right-hand board. The normal values are only for right hands, and a left-hand board is needed before left-handers can be measured on the hand they favour. Across the

major heart, stroke and vascular diseases, First Nations Australians are hospitalised and die at 1.6 to 2.6 times the age-standardised rates of non-Indigenous Australians [2]. The patient studies that follow should therefore involve Aboriginal and Torres Strait Islander communities in their design from the start, through a reference group with an Aboriginal community controlled health service.

Table 13: The project against the Engineers Australia Code of Ethics.

Principle	What it asks	Where the project shows it
Demonstrate integrity	Act on a well-informed conscience; be honest and trustworthy; respect the dignity of all persons	Checks fixed before the data; failures and dropped checks reported; three progress-report claims corrected (Section 4.8.1); reused work credited; codes, not names, in results
Practise competently	Maintain knowledge and skills; represent competence objectively; act on adequate knowledge	Timing checked against independent references (Section 4.2); factory-calibrated force; healthy validation first; claims kept to the data
Exercise leadership	Uphold trust in engineering; support diversity; communicate honestly with stakeholders	Left-handers not excluded; the laboratory's task checked with its head; plain-language screens; a public kit for the next student
Promote sustainability	Engage responsibly with the community; support health, safety and wellbeing; balance present and future needs	A device a clinic can repair, a sensor at AUD 192 against AUD 830 to 860 a hand, drawing at most 2.5 Wh a session; older boards kept working; open code, so the next project builds on this one

Sensors and motor boards plug in through connectors (Figure 4), so a failed part is swapped rather than the device replaced. A sensor costs AUD 192 against AUD 830 to 860 for a whole hand, or less for a hand with uncalibrated sensors (Section 4.8.3). A session draws at most 2.5 Wh from the USB supply. The PLA chassis reprints, and keeping the 2025 serial protocol stops earlier boards becoming electronic waste. A retired board, sensor or motor goes to e-waste collection and the PLA to print recycling.

For stroke survivors, every cue arrives three ways by default, the lit card, a tone and a buzz under the finger to press, so a player with visual field loss still feels which finger to move. Each finger keeps one colour in every theme, and a high-contrast theme (21:1) is built in. A hit's green flash now carries a tick, so a colour-blind player can tell it from the grey of a miss. One gap remains for the pilot: the instructions can be spoken as well as shown, but only in English, so a player who reads or speaks little English needs them translated.

Chapter 4. Results and Discussion

Outside references corrected two constants (Section 4.2), and press timing was stable. The count came a median 40 to 55 ms into the push, the same within one 5 ms sample in the two blocks a learning score compares (Section 4.3.2). The EEG markers matched the log within one EEG sample (Section 4.4). With timing checked, all four participants learned the laboratory's sequence task on the device (Section 4.3.1). Pooled, both primary tests passed, though the N2 effect came from Person 1 alone and the P3 drop also fits time on task. In both EEG recordings, motor beta (a secondary measure) was lower for the learned order. The one self-run session saved all six blocks it played, and 4 of its 6 checks that can fail fell in band (Section 4.6).

4.1 Participants and feasibility

Six people used the device (Table 14), all invited by Welber Marinovic, all right-handed and aged 21 to 25. Persons 1 and 2, from the project team, played the laboratory's sequence task with EEG recorded (Section 4.4), and Persons 3 and 4, laboratory volunteers, played it without. Ten volunteers were sought for self-run sessions. Person 5 chose a 30-minute session and played six of its eight games in 14.4 minutes from the first block to the last, then chose to stop before the last two, Muscle Memory and Echo. The log shows the end-session dialog confirmed and no fault, and every block Person 5 started was finished and saved with its force stream, 13.8 minutes of play in all. Person 6 played one Force Pilot block. Muscle Memory, Echo, Mirror and Syllables were played by no one; their evidence is simulation and the faults of Appendix D.

Table 14: The people who played the device; the blocks stopped part way in the EEG sessions are left out (Appendix C).

Person	Date	Played	EEG
Person 1	8 October	The laboratory's sequence task, then Buzz Hunt	64 channels
Person 2	22 October	The laboratory's sequence task	64 channels
Person 3	15 October	The laboratory's sequence task	None
Person 4	19 October	The laboratory's sequence task	None
Person 5	20 October	30-minute session: Force Pilot, Chords, Buzz Hunt, Adaptive, Reaction and Rhythm	None
Person 6	15 October	One Force Pilot block	None

Persons 3 and 5 also answered the team's survey after their session (Appendix M). Both rated the sound and vibration cues and the game's engagement 5 of 5, and comfort and the spacing of the sensors 3 or 4. Person 5, who played alone, rated the on-screen instructions 2 of 5 and asked for sensors that slide sideways and a pause before the first trials. Person 3, whose middle sensor stuck (Section 4.3.2), gave 2 of 5 to both "no discomfort" and "no fatigue".

4.2 Engineering verification

Two constants were corrected against outside references: the force constant was 4.5 times too high and the audio delay 87 ms against 40 assumed. The sensors' noise and drift sit at 1.2 to 1.5 counts, under 0.3 percent of their range. As run on the study's computer, press to log met the progress report's target, settling met it in part and start-up missed it. The laptop's slowest presses took about 56 ms to the log. The sensor settles within 5 ms, but the smoothed count trailed the raw force a median 5 to 10 ms. Start-up took 4.2 s against 2 s, waiting on the board's restart and motor self-test. Table 15 sets the engineering objectives against the progress report's targets, and Table 16 the timing, delivery and build checks, each with how it was measured.

Table 15: Engineering verification of six objectives of Table 4: the progress report's target, what was measured and how.

Objective	Target	Measured	How
OA1 Adaptive band **Met in part**	Per-finger rate held in 0.65 to 0.80	Simulated, in band: 49 of 60 blocks as run, 39 with the climb at a real participant's trial SD of 94 ms; 48 and 58 at 25 ms. Person 5: 0.72, fingers 0.56 to 0.89	Closed-loop simulation (Figure 12); one session
OA2 Peak force per press **Met**	Every press, in newtons	In every trial row, 19.5 mN a count; Person 5's presses a median 0.45 to 0.57 N	Log inspection, unit test
OB1, OB2 Rhythm **Met in part**	Every press scored against the beat	Offset, SD and lag-1 per block; mean not read, the laboratory's sound delay untimed	Tests and log inspection
OC1 Board assignment **Met in simulation**	Automatic, with override	Scan 0.44 ms median, 1.22 ms at most; box set by hand; simulation finds it	Engine timing log; virtual ports
Board start-up (OD1) **Met in simulation**	Port open to first sample under 2 s	4.2 s in 57 of 76 logged start-ups, 3.0 s in four, a fixed reconnect fault in 15; 1.75 s in simulation	Port-open log; simulation

Table 16: Engineering verification of timing, delivery and the build: the checks, the progress report's constraint budget among them.

Check	Target	Measured	How
Rhythm cue timing **Met in simulation**	On the scored beat	Mean −0.15 ms, all cues within 4 ms (Semester 1: 9 to 11 ms late)	Frame test, simulated clock
Press to log, development laptop **Met in simulation**	Under 50 ms to the row	95 percent within 36 ms, 56 at most (smoothing 15 ms, 35 at most, then USB bursts of 10 to 11 ms, up to about 21; Table 17); in replay on the raw count (Table 21), about 26 ms	Replayed presses; board clock against the laptop's
Press to log, laboratory computer **Met**	Under 50 ms to the row	95 percent within 16 ms, all within 49 (35 ms of smoothing, 13.6 ms stamp at most)	Board clock against the computer's
Cue and press timing **Met**	Cue within a frame; press lag stated	Cue on the next frame, 0 to 16.7 ms; press a median 5 to 10 ms behind the raw force, 95 percent within 15 ms, more on slow rises (Figure 6)	Frame log; real presses replayed; simulated ramps
Visual timing **Met in simulation**	No stale or leaked stimulus	154,035 frames of a simulated session checked; none found	Frame-by-frame check
Laptop output latency **Met**	Measured, then subtracted	Song 87 ms over five takes (SD 3.9 ms; 40 assumed), short sounds 77 ms, motors 74 ms	Loopback, less macOS's stated, unchecked input delay
EEG marker delivery **Met**	Every marker, in order	1,886 of 1,886 in order through the laboratory route; 82 of 82 through the simulator; 1,859 of 1,859 and 1,794 of 1,794 in the two recordings, within 1.1 ms of the fitted clock	Windows test; simulator; laboratory
Press timing on real presses **Met**	Same lag in the blocks a score compares	Counted 40 to 55 ms after the push began, block 8 and the post-test within one sample, all blocks within 15 ms; at most 0.2 percent were a touch counted apart from its push	Force stream of four sequence-task runs (Section 4.3.2)
Analysis against a second analyst **Met**	Agrees	Block mean RTs within 5 ms on 9 of 10 blocks (8 October); the tenth kept different trials	A team member's own analysis of the session
Flashing on a new computer **Met**	From the installed application alone	Every build passes the tools check on clean Windows and macOS machines; a real board flashed and verified in 13 s	Continuous integration; real board

Check	Target	Measured	How
Test suite **Met**	All passing	Every test passes at submission; one runs on Windows only	Automated run
Sensor settling (constraint budget) **Met in part**	Under 5 ms after the rising edge	Sensor under 1 ms, read every 5 ms; the count a median 5 to 10 ms later as run, one sample in replay	Datasheet [174]; replay
Force resolution (constraint budget) **Met**	A 50 g change told apart	19.5 mN (2 g) per count, so 50 g is 25 counts against 1.2 to 1.5 counts of noise	Factory calibration; sensors at rest
Sample rate (constraint budget) **Met**	200 Hz per board	199 Hz	Board clock
Game loop (constraint budget) **Met**	120 frames per second	Loop capped at 120 ticks per second; a cue lands on the next display frame, 0 to 16.7 ms at the display's 60 Hz	Code; frame log
Actuator stall shut-off (constraint budget) **Not applicable**	Stop within 200 ms	No actuator: the finger rests moved to threaded rods	Appendix E

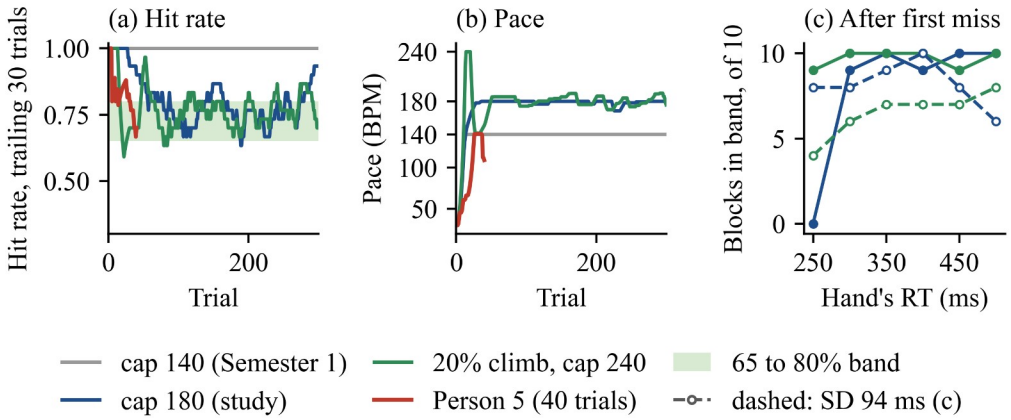


Figure 12: The adaptive controller on simulated players, trial SD 25 ms unless stated: (a) trailing hit rate and (b) pace, one run of a 280 ms hand at the 140 and 180 BPM ceilings and with the climb, and Person 5's real block; (c) how many of ten simulated blocks per hand, 250 to 500 ms, stayed in the band over the 32 trials after the first miss, as run (blue) and with the climb (green), solid at SD 25 ms and dashed at 94: 48 and 49 of 60 as run, 58 and 39 with the climb.

In simulation the controller used in the sessions brought hands of 300 to 500 ms into the band within 30 to 45 trials and held 0.75 to 0.77 over their last 100 trials. A 250 ms hand, near Person 5's Reaction speed (Table 20), stayed above the band at the 180 BPM cap (0.98). At a real participant's trial SD of 94 ms the controller kept 49 of 60 blocks in band over the 32 trials after the first miss. With the optional climb it kept

39 of 60 (Section 3.3.4; Figure 12), so the pilot should leave the climb off until it is tuned. Person 5's one block rose from 30 to 141 BPM over its first 30 trials and averaged 0.72 over its 40 trials, inside the band.

Table 17 gathers the timing and force parts into an uncertainty budget for one reaction time and one peak force. Read at the count, a reaction time includes 41 to 56 ms of the push (the lag plus a 1 ms stamp) and the screen's and sensor's delays. These cancel when two blocks share a lag, as a learning score's blocks do. The lag's trial-to-trial spread, 7 to 11 ms (SD), accounts for 0.6 to 1.4 percent of the variance of a participant's single trials (SD 94 ms in the mixed model). The maker's flat error limits on the 10 N full scale, combined as independent parts by root sum of squares (Section 2.12), give a force in newtons a standard uncertainty of about 0.3 N, or 0.6 N at about 95 percent ($k = 2$). Loading each sensor with known weights would replace this estimate (Section 6.3). Two presses of similar size on one sensor differ by little more than its repeatability, 0.1 to 0.2 N (Section 4.7, item 5), so the games read force against each person's own presses (Section 3.2).

Table 17: Uncertainty budget for one reaction time and one peak force, as standard uncertainties (k = 1) unless stated.

Part	Size	How it is known
Flash against the display	At the flip in the sequence task; up to one 16.7 ms frame early in Reaction	Code; frame log
The screen's and the sensor's own delays	Sensor under 1 ms; screen not yet timed	Datasheet [174]; a clip tool (Section 4.8.5)
Push to counted press	Participant medians 40 to 55 ms, interquartile range 10 to 15 ms; longer for a slow rise	Force stream, four participants (Section 4.3.2)
The onset rule itself	Lags move 5 to 10 ms and push-start scores at most 4 ms with the velocity threshold at 10 or 30 percent of peak velocity, not its 18; on simulated smooth pushes with the sensors' noise, the start found 7 ms early to 15 ms late for rises of 100 to 200 ms, later for slower rises	The press check rerun under each rule; simulated pushes
Sample to computer over USB	0.8 to 0.9 ms, 99 percent within 2.4 ms and 13.6 ms at most, on the laboratory's computer; bursts of 10 to 11 ms, up to 21, on the development laptop	Board clock against the computer's
Reaction time in all, sequence task	Systematic 41 to 56 ms (press lag plus 1 ms stamp) plus the screen and sensor; varying 7 to 11 ms (SD); in Reaction 4.8 ms (SD) more for the frame	The lag's interquartile range over 1.35, an estimate of its SD, and the stamp's SD, as independent parts
Calibration, linearity and hysteresis	Within 2, under 2 and under 4 percent: 0.12, 0.12 and 0.23 N	Maker's calibration and datasheet [174], [175]
Repeatability	Under 1 percent of full scale (1 SD), 0.1 N	Datasheet [174]

Part	Size	How it is known
Noise, drift and resolution at rest	1.2 to 1.5 counts, at most 1.5 counts in 60 s, 19.5 mN a count: under 0.03 N	Sensors measured at rest; datasheet
Drift under a held load	2 percent of full scale in a minute at half load: 0.12 N, for a hold, not a brief press	Datasheet [174]
Peak force in all	About 0.3 N absolute (0.30 N, the parts above but drift), 0.6 N at $k = 2$; about 0.1 to 0.2 N between similar presses on one sensor (Section 4.7, item 5)	Root sum of squares, the limits as rectangular bounds on the 10 N full scale

4.3 Learning on the laboratory's sequence task

All four participants learned the laboratory's task on the device, with their presses counted at the same lag, within a 5 ms sample, in the blocks each score compares; Section 4.6 reads the device's own games.

4.3.1 The laboratory's sequence task in four people

Four people played the laboratory's sequence task, 896 trials each, scored by the planned rule (Section 3.7). All four got faster from block 1 to block 8, by 25, 23, 10 and 41 percent at the count, and slower again when the order turned random (Figure 13, Table 18). Each learning score, the post-test median minus block 8's, was positive, though Person 3's bootstrap interval reached down to 1 ms. The mean was 91 ms (95 percent CI 42 to 140; $t(3) = 5.96$, $p = 0.009$, dz = 3.0). With four people an exact one-sided sign-flip test cannot go below $p = 0.0625$, so the group result rests on the t-test and the model. Table 19 reads the score five more ways, chosen after the data like the group tests (Appendix B). The four that score each participant are all positive, and only Person 3's interval without reversals reaches −24 ms.

Fitted to all 2,984 kept trials, the mixed model of Section 3.7 put the post-test 99 ms above block 8 (95 percent CI 38 to 160, $p = 0.014$; Figure 13b, black square). Participants differed in that effect by 34 ms (SD). Over blocks 1 to 8 the model's linear trend fell 9.1 ms a block (95 percent CI 7.5 to 10.6). The two random blocks did not differ: post-test minus practice was −4 ms in the model (95 percent CI −67 to 59) and +5 ms between the medians. A random intercept alone gives 100 ms with too narrow an interval, 82 to 118 ms, as it treats the four participants as sharing one effect.

Table 18: The sequence task: median RT per block and learning score with bootstrap interval, in ms, early presses (within 100 ms either side of the target's flash) on the correct finger, and recall: items in place from the best starting point (share of correct triplets).

Person	Random practice	Block 1	Block 8	Random post-test	Score (95 percent CI)	Early, correct (percent)	Recall, best start (triplets)
Person 1	298	264	199	280	+81 (50 to 114)	101 of 111 (91)	10 of 10 (1.00)
Person 2	316	346	268	369	+101 (47 to 154)	13 of 19 (68)	8 of 10 (0.75)
Person 3	311	259	234	289	+55 (1 to 87)	240 of 254 (94)	6 of 10 (0.75)
Person 4	297	294	174	301	+127 (93 to 157)	31 of 33 (94)	10 of 10 (1.00)
Mean, pooled	305	291	219	310	+91 (42 to 140)	385 of 417 (92)	

Table 19: The learning score, post-test minus block 8 in ms, read six ways, all but the planned rule chosen after the data. Each p is from a one-sample t-test across the four participants, the last from the mixed model; the push reading uses the trials whose push start was found.

Reading	Person 1	Person 2	Person 3	Person 4	Mean	p
The planned rule	81	101	55	127	91	0.009
Against the faster random block	81	48	55	123	77	0.020
Without reversal trials	54	65	25 (−24 to 82)	118	66	0.044
Without presses under 100 ms	64	78	43	127	78	0.022
At the start of the push	80	97	47	137	90	0.017
Mixed model of the single trials					99	0.014

After 800 trained trials the mean is close to the 91 and 111 ms that Destrebecqz et al. report after 1,152 [96]. The random-order medians, 280 to 369 ms, are faster than a published four-choice norm of 390 ms at ages 18 to 25 [216], even with each press counted 40 to 55 ms into the push (Section 4.3.2). A fixed wait may explain the speed: each flash follows the last press after 500 ms, which a player can time [19]. Random blocks here hold twice as many reversals (36 against 18 percent of trials), A-B-A runs that are slower to answer [98], so the score was also read without them. The learning was at least partly explicit. Scored from the best starting point, Person 1 and Person 4 typed back the whole order, and all four recalled more correct triplets than guessing would ($p = 0.029$ or lower) [96].

The laboratory's script counts any press after the flash as correct, even one 2 ms after it, which can only come from knowing what was next. These correct presses under 100 ms were 62, 9, 137 and 27 of the four participants' correct presses, mostly in blocks 4 to 8, and Person 3's 137 explain the dip in Figure 13a. They pull block 8's median down, so the score was also read without presses under 100 ms.

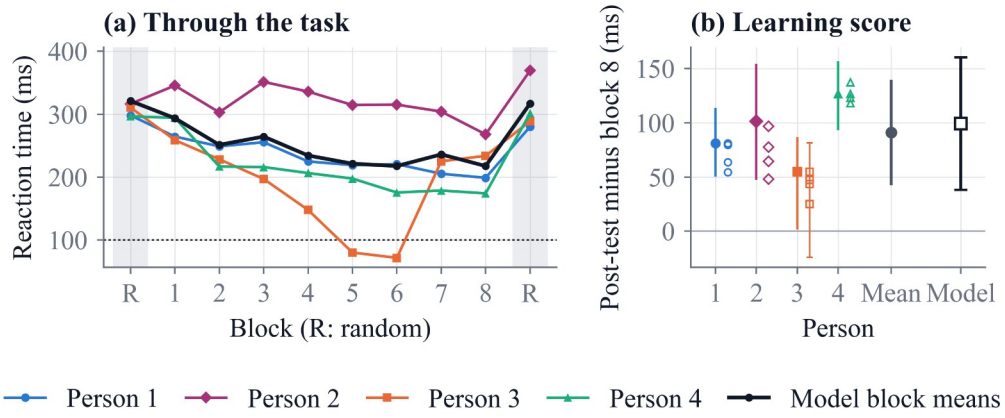


Figure 13: The laboratory's sequence task in four people. (a) Each participant's median reaction time per block (colours) and the mixed model's block means of single trials (black), above the medians where slow trials pull a block's mean up, random blocks shaded; Person 3's dip below the dotted 100 ms is presses begun before the flash could be seen. (b) Each learning score with its 95 percent interval, beside it the score's other four readings of Table 19 (small, hollow, with the table's one interval), the mean of the four with its t-test interval (grey) and the mixed model's score, post-test minus block 8, 99 ms (95 percent CI 38 to 160; black square).

Early presses (Table 18) also show what the participants knew; this reading and the error rates were added after the data (Appendix B). In the sequence blocks they hit the right finger 68 to 94 percent of the time, against one in three for a guess avoiding the finger just pressed (p = 0.002 or lower). Three of the four also beat 60 percent, the hit rate from knowing only which finger usually comes next (each $p < 0.001$). In the random blocks only 2 of 375 answers came that early, and neither was right. Errors rose in the post-test to 15 to 21 percent, against 5 to 21 in block 8 and 2 to 9 in random practice. They were higher than both in every participant (Person 3, the closest, 21.3 against 20.7 percent in block 8), so the slowing was not bought with accuracy.

4.3.2 Is each press timed consistently?

Within a score, yes. In all four participants each press was counted a median 40 to 55 ms after the push began. The lag shortened with practice in three participants but stayed within a 5 ms sample in the two blocks each learning score compares, so the scores are not an artefact of when the press is counted.

The laboratory asked for this press check on 22 October 2026. It looks for two faults. If the time from the start of the push to the count changed as a participant sped up, the learning curve would bend. A light touch before the real push would time the press from the touch. The software's onset detector (Section 3.8.1) found the start of the push on the board's force stream for 98 percent of the 3,463 counted presses. The reaction time was read from both points (Figure 14). The same sensors time both points, so the check shows how the count moves against the push, not the delay from the screen (Section 6.3).

Trial by trial, the press was counted a median 45, 50, 40 and 55 ms after the push began. Each block's median lag stayed within 10 to 15 ms of the others (Figure 14e). In Persons 1, 3 and 4 the lag fell 10 to 15 ms from the practice block. Their pushes grew 1.5 to 2.2 times steeper by the post-test, so they crossed the threshold sooner, while the smoothing delay stayed one or two samples. Between block 8 and the post-test the lag changed by 0, +5, 0 and 0 ms, within one sample, as the added check asks. Timed from the push, the block 1 to 8 gain is 8, 30 and 13 percent smaller in Persons 1, 3 and 4 and 3 percent larger in Person 2, whose lag rose (gains in Figure 14f).

Read at the push instead, the scores moved by 10 ms at most: 80, 97, 47 and 137 ms, against 83, 101, 53 and 127 ms at the count on the same trials. The two reaction times correlated at $r = 0.984$ to 0.997. Both sets differ slightly from Table 18: they use only the 3,402 presses with a found onset. Other onset thresholds move the scores by 4 ms at most (Table 17; rerun in the repository, Table 25). On simulated pushes the rule finds a slow push's start late, which shortens the lags of the slower early blocks, so the lag's fall is if anything understated.

Light touches were rare. No counted press was let go within 60 ms. Of the 15 times the same finger pressed again within 0.3 s, only 2 began with a light touch, one each

in Person 1 and Person 3. So at most 0.2 percent of each participant's counted presses were a light touch counted apart from its push.

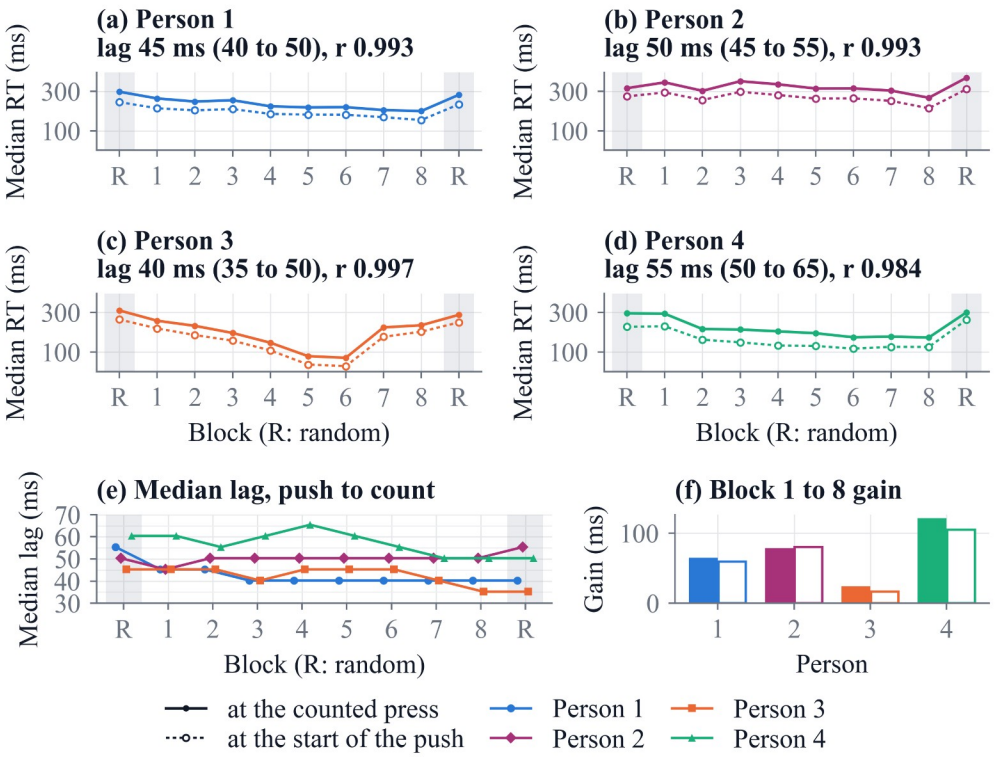


Figure 14: Each press against its push in four participants. (a) to (d) Each participant's median reaction time per block, at the counted press (solid) and at the start of the push (dotted); each title gives the median lag with its interquartile range and the trial-by-trial correlation. (e) Each participant's median lag per block, spanning 10 to 15 ms per participant and within one sample from block 8 to the post-test. (f) Each participant's block 1 to 8 gain at the count (filled) and at the push (hollow).

Person 3 wrote afterwards that the middle sensor sometimes stopped toward the end (Appendix M), and the log shows why (Figure 15). Before block 7 that finger's presses were held a median 0.21 s. From block 7 the sensor rested 6 to 17 counts or more (about 0.1 to 0.3 N) above its pre-press level, past its release threshold at 5 counts above it. The finger therefore stayed down for 4 to 80 s at a time, 9 times (7 drawn in Figure 15), so a fresh press on it could not be counted. Of the middle cues, 11 went unanswered and 24 were credited to another finger, 17 of them to the index.

The group result does not rest on that finger. Without the middle finger's trials, Person 3's score was 45 ms (−19 to 77) rather than 55 ms, and the mean of the four

89 ms ($t(3) = 5.09$, $p = 0.015$). As the sensor stuck, Person 3's correct presses under 100 ms fell from 33 of 54 kept trials in block 6 to 10 of 73 and 4 of 69 in blocks 7 and 8, perhaps lowering that score (Figure 13a).

The detector now releases a finger held 2 s (the hold rule) or resting 1.5 s between its release threshold and half its press peak (the rest rule). It then resets its resting level. Synthetic traces test both. Replayed over Person 3's force stream with the settings that ran, the old rules reproduce every one of the 988 presses the detector logged in that session, practice and recall included. The new rules answer all 11 unanswered middle cues on the middle finger and move back 15 of the 24 credited to another finger. The other three participants' press counts change by at most six of about a thousand, and Person 2's two unanswered middle cues become answered. Re-read by the planned rule, the learning scores are 81, 104, 55 (5 to 88) and 127 ms, mean 92 ($p = 0.010$), against 81, 101, 55 and 127 (Table 18). The result is unchanged. The 2 s hold outlasts any tap or chord the modes ask for, and Force Pilot's longer holds are scored on the raw stream.

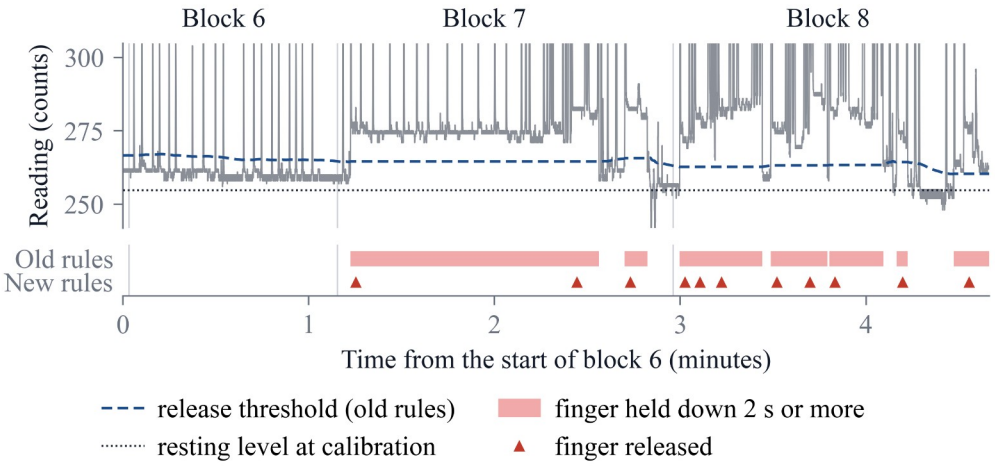


Figure 15: Person 3's middle sensor through blocks 6 to 8 of the sequence task: its reading (grey) and, on the strip below, what the detector decided. From block 7 the sensor rested above the release threshold the old rules held (dashed), so under them the finger stayed down (bars where 2 s or more) and fresh presses on it were not counted; the rest and hold rules now release it (triangles). The axis is cut so the resting level shows; the press peaks run off the top.

4.4 EEG laboratory sessions

The two EEG recordings answer SQ3: every marker matched the log, and the flash's P1 kept its latency. Persons 1 and 2 were recorded in the Curtin EEG laboratory (BioSemi ActiveTwo, 64 scalp channels at 512 Hz) while playing the sequence task.

The markers were checked before any effect was read, as the laboratory advised, and passed in both recordings. Every marker matched the game's log (1,859 of 1,859 and 1,794 of 1,794). The scatter about the fitted clock, 0.56 ms, is what rounding to a 512 Hz sample produces by itself, and it did not grow from the first half of either recording to the second (Figure 24). Every response byte named the log's finger (894 of 894 and 894 of 894). The flash's P1 wave peaked 121 and 117 ms after its marker in random trials and held that latency through each session. It vanished from averages taken at random times, as it should if the markers are right.

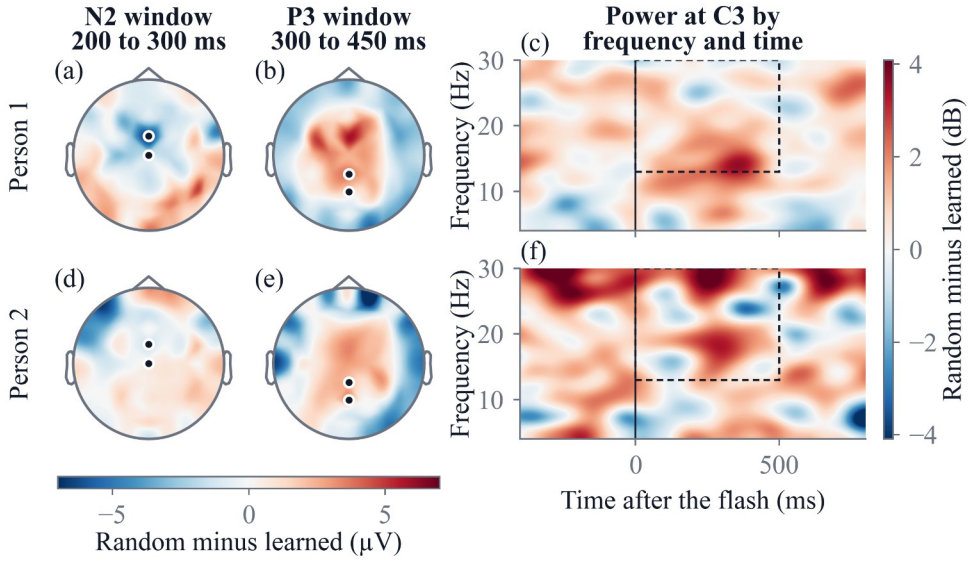


Figure 16: Random flashes against the learned order, Person 1 above and Person 2 below, on one scale: red where random flashes gave more, blue where they gave less. The heads are seen from above, nose up, with each channel's mean voltage difference in the N2 window (a, d; random post-test minus blocks 7 and 8, the late blocks) and the P3 window (b, e; random practice minus the late blocks); the dots mark the sites each wave is read at, where a larger N2 shows blue and a larger P3 red. Panels c and f give the power at C3, over the left motor cortex, by frequency and time (random post-test minus the late blocks); the dashed box is the beta window of Figure 17.

On those markers the averages showed the expected responses, mapped over the scalp in Figure 16 (Appendix C.1 gives every measure, figure and check). Pooled across the two people, both primary tests passed, the N2 at −2.3 µV (Holm p = 0.038) and the P3 at +2.5 µV (Holm p = 0.006), each with a caution. Beta over the left motor cortex, a secondary measure, was higher on random flashes than on the learned order in both people, by 1.1 and 1.3 dB (p < 0.001 together), as Lum et al. found [111]. It was already higher before the flash, by 0.6 and 1.6 dB, so it marks the random block more than each flash (Appendix C.1). Random post-test flashes were answered 75 to 81 ms later, which alone could shift beta. Yet beta stayed higher against learned trials matched in reaction time (pooled p = 0.020; Person 2 alone p = 0.126) and before the press (pooled p < 0.001; Figure 17).

Person 2's recording, held out from settling the windows, is the one test on new data. Its beta difference held (p = 0.033) and its P3 drop matched Person 1's (2.5 against 2.4 µV), but its N2 did not differ (p = 0.740). The pooled N2 therefore rests on Person 1 alone (−4.2 µV, p = 0.020), the recording its windows were settled on. And the P3 shrank from random practice to the late blocks but did not recover in the random post-test (0.0 µV, p = 0.973), so time on task explains the drop as well as learning.

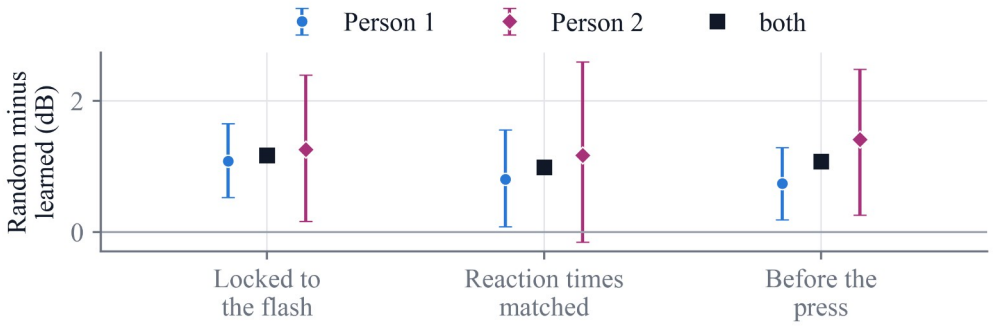


Figure 17: Beta at C3, random minus learned, per person with its bootstrap interval and both together (squares): at the flash, with reaction times matched, and before the press.

4.5 Retest consistency of the sequence task

One session gives the measure's error but too few participants to grade its consistency. Timed from the start of the push, each participant's random practice and post-test medians, 800 sequence trials apart, differed by −11, +37, −14 and +36 ms.

That gives a retest ICC(2,1) of 0.50 (95 percent CI −0.67 to 0.96), an SEM of about 20 ms and an MDC95 of about 55 ms (Equations 7 and 8, Appendix K). Read at the count, which carries the lag's change with practice, the ICC falls to 0.37 (−1, its floor, to 0.95; Appendix K). With four participants both intervals span every band. No one played the second pass, so agreement between passes on the five reliability measures (T1 to T5) waits on the normative study (Section 6.3); Table 45 shows those rows working on simulated players. A participant's learning score carries a confidence interval of 30 to 54 ms either side, and such scores retest at only about r = 0.28 across sessions [101].

4.6 The device's own games in one session

4.6.1 Values on healthy hands

Each mode played returned plausible values beside its reference (Table 20), except Rhythm's asynchrony, which waits on the computer's sound delay. Normal ranges await the normative study (Section 3.8.3).

Time in the corridor says little on its own, since a finger held still keeps 54 to 100 percent of a run there (Section 4.7, item 6). The tracking error says more (Table 20, Figure 18): 2.0 and 3.0 percent of maximum, 56 and 33 percent below the 4.5 percent scored by holding the best constant force. The gain on the sine waves was above the value reported for young adults on a visual tracking task [24], [30]. So was the ratio of release to press error on the symmetric waves, though the published one uses RMS rather than mean absolute error. Over all waves (F3), release and press error differed by under 0.1 percent of maximum (Figure 19). With one or two people, neither difference can be told from an effect of the task's design.

Table 20: Values on healthy hands from one session and two single blocks, against the value each is read against: Person 5's value first, then Person 6's for Force Pilot or Person 1's for Buzz Hunt (65 trials in the EEG session).

Mode	Measure	Value	Healthy value
Reaction	Median reaction time, ms	260	Device-specific [116]
Chords	Added time per extra finger, ms	36	About 126 [86]
Chords	Median enslaving ratio	0.042	8 to 10 percent of total force at a quarter of maximum, a different ratio [84]
Force Pilot	Time in corridor	0.95, 0.88	Design: above 0.8 on most runs; a finger held still 0.54 to 1.00

Mode	Measure	Value	Healthy value
Force Pilot	Tracking error, percent of maximum	2.0, 3.0	Device-specific; holding the best constant force scores 4.5
Force Pilot	Release over press error, symmetric waves	1.05, 0.96	0.83 to 0.90 in young adults, of RMS error [24], [30]
Force Pilot	Gain on the sine waves	0.99, 0.93	About 0.83 in young adults [24]
Buzz Hunt	Localisation accuracy, percent	100, 94	Design: above 90 percent
Buzz Hunt	Median localisation RT, ms	427, 490 (every trial)	Device-specific
Adaptive	Top pace, BPM	141	Design: none; development peaks 90 to 144, cap 180
Rhythm	Mean asynchrony, ms	−91 as logged; not read	About 0 to −20 ms to music [66]; needs the computer's sound delay

Chords' added time per extra finger, 36 ms, was well under the 126 ms Verwey reported for chords typed on keys [86], where Section 2.5 expected an unpractised session to cost more. On the sensors each finger already rests on its sensor and is counted at a light threshold, so a chord needs no key travel; one session cannot tell the sensors from the hand. The same may hold for Reaction's median of 260 ms against a four-choice norm of 390 ms [216], and there each cue also buzzed the finger to press, which in one study flattened the cost of choice [115]. Rhythm's timing was not read. Its mean asynchrony, −91 ms as logged, rests on the sound delay of the computer it played on, measured on the development laptop (Section 4.2) but not on the laboratory's. Person 5's one Rhythm block therefore gives its spread, an asynchrony SD of 50 ms inside Rh2's 10 to 100 ms band, and a lag-1 correlation of 0.22, positive as partial error correction predicts [64], [69]. Table 39 (Appendix H) gives the Force Pilot measures that the Parkinson's disease literature reports on the same kind of task.

4.6.2 Each game against its check

Person 5's values fell inside the expected band on 13 of the 16 checks one session can read, including 4 of the 6 that could fail, and Persons 6 and 1 inside all 7 of theirs (Figure 19). The checks one session cannot read need a second pass, a Muscle Memory or Echo block, the computer's sound delay, or eight people.



Figure 18: Force against target on Storm, the fixed irregular wave, in percent of each participant's maximum press, with the corridor shaded at 4 percent either side, the ladder's narrowest; each title gives that participant's mean absolute error on Storm, and over the whole ladder it is 2.0 and 3.0 percent for Persons 5 and 6 (Table 20), highest on Chop for Person 5 and on Open ocean for Person 6 (Appendix F).

Of Person 5's three values outside their bands, one is the irregular-wave check (F1), whose known weakness was recorded before the data: 8 s of preview shows an irregular wave coming (Appendix B). Person 6, as F1 predicts, followed irregular waves less closely by correlation. The catch-trial check's 10 percent band (B3) can be met on three catch trials only with no false alarm, and Person 5 made one. The ring-finger check (C3), reported rather than tested, found Person 5's largest leak on the little finger. Within a block Person 5 slowed 28 ms on Reaction, the reference (W1), and 18 ms on Buzz Hunt, 10 ms less (W5).

4.7 Discussion

All five questions of Table 3 are answered yes, each with its limits.

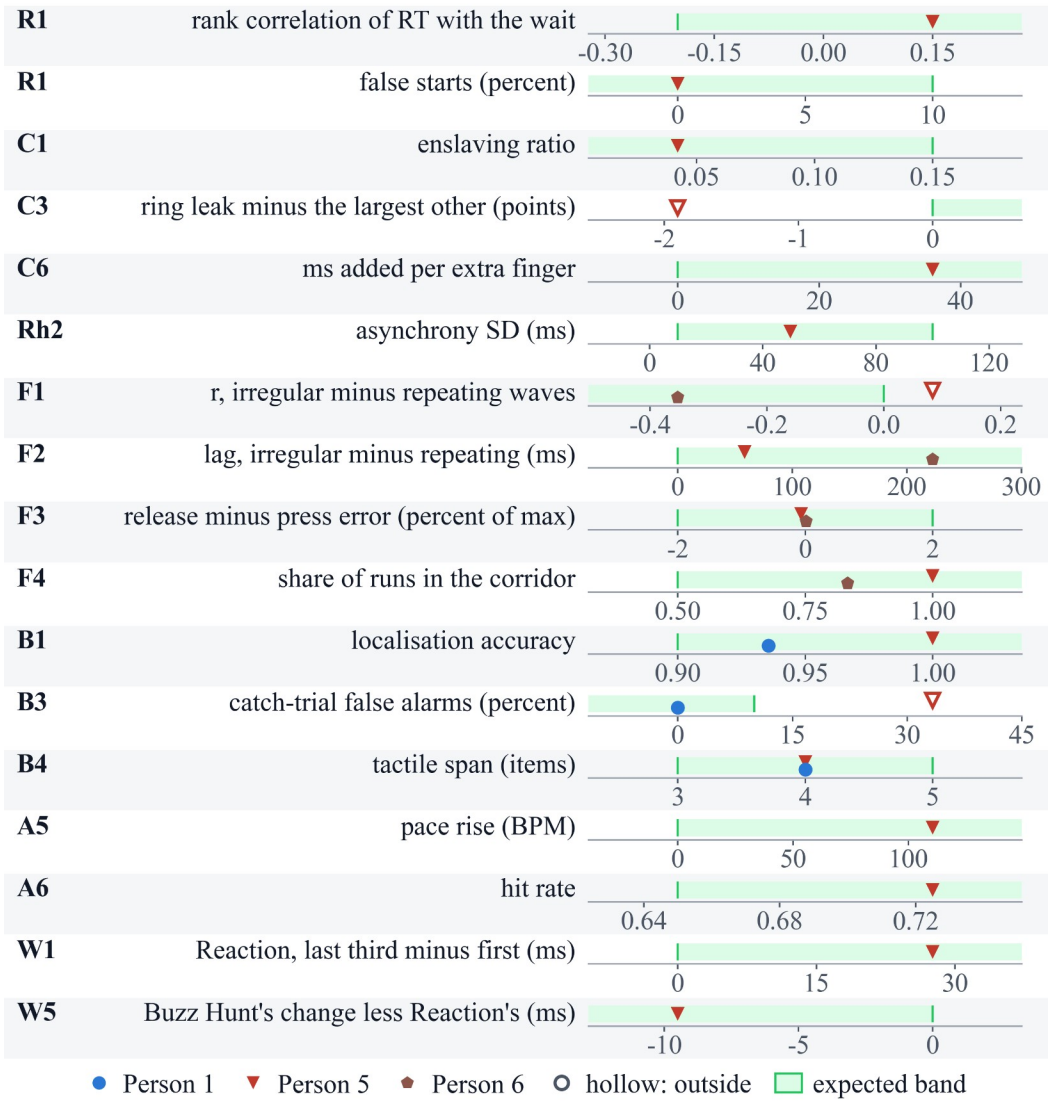


Figure 19: Every check one session can read, each person's value against its band, R1 on two rows: the band shaded, a point per person in that person's colour, hollow where the value fell outside; C3 is in percentage points. None is tested yet, as each check needs eight people; C2 and W4 are read across people, so one session has no value for them. The six that could fail are R1, C6, F1, F2, B1 and B3 (Appendix K).

SQ1 Timing. Yes, for presses and markers; one detector fault. References outside the software found two errors and both were corrected: the force constant was 4.5 times too high, and the audio delay was 87 ms, not the 40 ms assumed. On the laboratory computer, where the study ran, every press was logged within about 49 ms of its threshold crossing, inside the 50 ms bound, and 95 percent within 16 ms. On the development laptop 95 percent fell within 36 ms, but the slowest took about 56 ms (Table 16).

On the study's own presses the count came a median 40 to 55 ms after the push began, with block 8 and the post-test within one 5 ms sample (a check added after the data, Appendix B). At most 0.2 percent of counted presses were a light touch counted apart from the push that followed it. Because the lag grows as a push rises more slowly, its shrinking with practice made up 8 to 30 percent of three participants' speed-up at the count. Every simulated push to 3 s is counted at the device's settings (Section 4.8.4). The fault showed in use. A finger resting a little above its release threshold stayed down, so its next presses went uncounted until it lifted. The detector now ends such a hold, checked by replay (Section 4.3.2). The sound path behind Rhythm on the laboratory's computer is untimed; the loopback reads planted delays to 1.5 ms in tests.

SQ2 Function. Yes. All four showed the laboratory's sequence effect, each score above zero without presses under 100 ms. Without the reversal or stuck-sensor trials every score stayed positive, but Person 3's intervals reached −24 and −19 ms (readings chosen after the data). The mean, 91 ms after 800 trained trials, matches the 91 and 111 ms published after 1,152 [96]. Neither practice, fatigue nor the timing of the count made it: every score stayed above zero against the faster random block and moved 10 ms or less when read at the push. The learning also showed in which finger the participants chose and in errors that rose on random order. Recall did not set the score: 81 and 127 ms for the two who typed back the whole order and 101 and 55 ms for the two who recalled part. But all four knew some of the order, so the match with the published values is one of size, not of the kind of learning.

SQ3 EEG. Yes, for the markers and the trial labels. In two recordings two weeks apart every marker matched the log within one sample. Averages cut at the game's markers reproduced the sensory waves and a beta difference between random and learned flashes, which held with reaction time matched and before the press. A delay common to every marker is still untimed (the clip tool reads planted delays to 0.4 ms, Section 4.8.5), and the ERN needs more errors than one session gives.

SQ4 Readiness. Yes, in one session run with no one in the room. The volunteer logged in, calibrated and played six games in 14.4 minutes, and every block was saved with its force stream. The notebook, rehearsed on simulated sessions, read the values against the checks (Sections 3.6 and 4.6.2). OV5 is met in part, since six of eight games were played and the longer lengths only simulated. The volunteer rated

the on-screen instructions 2 of 5 (Appendix M); every mode can also speak them. Two readiness objectives are met only in simulation (Section 4.8.5). The sessions' first sample took 4.2 s against the 2 s objective (OD1). The scan also took the trigger box for a hand board, so its port was set by hand.

SQ5 Extension. Yes, in software. Force Pilot's Parkinson's measures read a planted 5 percent release error to within 0.5 percent of maximum on synthetic traces and were read on two healthy blocks (Appendix H). Syllables runs five age profiles over 932 words, every chunk and word voiced, and in simulation it read back each reader's planted accuracy (Section 4.8.5). Neither has yet been played by the people it is for.

What this means for patients. In healthy hands the device recorded the gain that practising a repeated sequence brings, in every person who played the task, read at the count and at the push. Patients will press more slowly and with less force, so their reaction times should be read from the start of the push, which the software found for 98 percent of counted presses (Section 4.3.2). In simulation every weak or slow push to 3 s was counted (Section 4.8.5). Alongside a patient pilot, the normative study needs its eight or more people and its second pass, which give the normal ranges and the MDC95 that a patient's change will be read against.

Limitations. The results carry eight:

1. A press was timed as the smoothed force crossed the press threshold, a median 40 to 55 ms after the push began on the study's presses (Section 4.3.2). The count's own lag behind the raw force grows on slower rises, to 11 to 15 ms at a 400 ms rise and 60 to 86 ms at 3 s in simulation (Figure 6). The study's presses sit far inside that: their count was at most 35 ms behind the raw force. In simulation a slow push is still counted, but late. A reaction time taken at the count therefore compares with other force-sensor studies, not keyboard norms, and a patient whose force rises slowly will look slower unless timed from the onset (Section 2.12).
2. The checks used healthy young adults, so they show the platform works on healthy hands, not its sensitivity to impairment. Two came from the project team, and knowing the device may have made the sequence easier to notice. The reliability analysis has run on simulated sessions only, since no one played

the second pass. Four participants can show only an effect this large (dz 3.0), and the two EEG recordings speak for those two people.

3. Adaptive describes the controller in healthy hands, not a player's speed threshold, and Syllables was played by simulated readers, not by any reader with dyslexia.

4. The Force Pilot values are healthy values on this device, not a Parkinson's test. The waves ask for at most 23.6 percent of maximum per second, below the relaxation rate even early Parkinson's disease reaches when asked to go fast [145]. Feedback is never removed, and short holds and the power from 8 to 12 Hz sit at the sensor's own floor.

5. Force in newtons rests on the maker's calibration of each sensor, which holds full scale within 2 percent for a load pressed squarely and evenly over the sensor [175], [217]. A fingertip only approximates that load, so newton values are safest compared within this device. With the bench test's non-drift errors (up to 4.5 percent of span, Section 2.12) as limits on the 10 N span, the budget's 0.3 N becomes about 0.4 N and its 0.1 N between similar presses about 0.2 N. Person 5's single-finger presses peaked at a median 0.45 to 0.57 N by mode, so the absolute 0.3 N is more than half a light press. Because of this, the measures rest on each person's own thresholds and percent of each finger's maximum, which do not use the conversion (Section 2.12).

6. Some measures are coarse or differ from their published form. Chord span lasts a few of the board's 5 ms samples. The enslaving ratio divides gap-scaled peaks in brief light presses, where the published value is a share of total force at a quarter of maximum [84], so the two agree in direction only. A still finger keeps 54 to 100 percent of a run in the corridor, and most of the enslaving ratio is built before the press. This means T3 and T4 describe a steady hand more than tracking or independence (Appendix B).

7. The sequence task's participants recalled much of the order, so their learning was at least partly explicit, and the task as run cannot separate implicit from explicit learning [96].

8. The laboratory's order is not second-order conditional [95]: knowing only which finger usually follows the last one would put 60 percent of early presses

on the right finger. Three participants exceeded that (each p < 0.001); Person 2's 13 of 19 was above it but not significantly ($p = 0.308$).

4.8 Critical review

Each issue below comes with its fix and any measured result: three progress-report claims corrected, 41 faults fixed with tests before any participant, one more in use, two in the final analysis and two in simulation, and the outcome of all fifteen planned tasks.

4.8.1 Semester 1 review, carried forward

The Semester 1 software worked as designed in development testing, and most of what it taught came from using it. The adaptive controller's starting pace of 60 BPM proved too fast even for a healthy hand, so the start and minimum fell to 30 and 10 BPM. Its penalty was revised once pressing several fingers at once was found to raise the score. The rhythm mode scored correctly timed presses as late until a 40 ms allowance for audio latency was subtracted. The progress report called that value measured by a calibration at startup. In fact the Semester 1 code used a fixed default, the typical delay of its audio library. The first real measurement, made with a microphone in Semester 2, found 87 ms on the development laptop (Section 4.2).

The report also credited the target band of 65 to 80 percent to Hatem et al. [218]; that review supports adapting difficulty but does not set the band, which is a design choice a little below Wilson's optimum (Section 2.2). The report's constraint budget also marked the 5 ms settling target as met; measured in Semester 2, the count lagged the raw force a median 5 to 10 ms (Section 4.2), one sample in replay. Fixed finger sequences in the bilateral mode became predictable and were replaced by adaptive selection. Automatic device detection worked on Windows and macOS, observed on the author's computers rather than counted, once a set of startup and file-handling faults was fixed and tested. The lesson was that many problems only appeared once the system was used. The author's own hand also cannot stand in for other people's, which is why Semester 2 turned to a study with other hands.

4.8.2 Semester 2 faults found before data collection

Before any participant, the author found and fixed 41 faults that would have changed the data or a session, each with a test that fails on the old code. Sixteen of them came

from running and measuring the software, or from the sensor's datasheet: simulated sessions, a frame-by-frame check of a full session's screen, nine modes played on the real board, the laptop's delays measured with a microphone, and the first real Force Pilot and Syllables runs. Ten follow under four headings. Table 33 lists the other six with their values before and after the fix, and Figure 20 draws seven.

Timing. The rhythm cue was scheduled one game-loop tick late, about 8 ms at 120 ticks per second, 9 to 11 ms on a simulated clock, so players would have learned to press that late. Now drawn on its frame, it has a simulated mean offset of −0.15 ms, every cue within 4 ms. The laptop's audio delay had been assumed to be 40 ms; measured with a microphone it was 87 ms, and measuring and subtracting each machine's delay is now a set-up step.

Force. The force constant assumed a 45 N part, so every force in newtons read 4.5 times too high; the factory calibration of the 10 N sensors, 19.5 mN per count, replaced it. Force Pilot's re-scoring set its zero as the press began, so it differed from the game's own score by up to 5.6 percent of maximum. With the zero at rest, as the game sets it, the two agree to a median 0.03 percent. The literature's 12 Hz filter let sensor noise into Force Pilot's rates, up to 3.7 times the rate asked for, so a 5 Hz filter, which passes a 2 s pulse whole, replaced it.

Calibration and data. Calibration was lost at the start of every block, so every block after the first ran on default thresholds. The profile now persists, and all six blocks of Person 5's session ran on the one calibration at its start. Sensor dropouts were scored as misses, so a loose plug would have read as a patient getting worse; a dropout is now its own outcome in every mode (SR3).

The games. The adaptive controller's ceiling kept a fast simulated player out of the target band for 300 trials; raised, it had the player back in the band by trial 53 (Figure 12a, b). In simulation Buzz Hunt's worst-case block ran 19.7 minutes with 6 of 28 trials near threshold; capped at 15 minutes it holds 25 of 28, and the study's shorter block has 23 trials (Table 27). A late flash in Adaptive leaked into the next trial; none appeared in a simulated session's 154,035 frames after the fix.

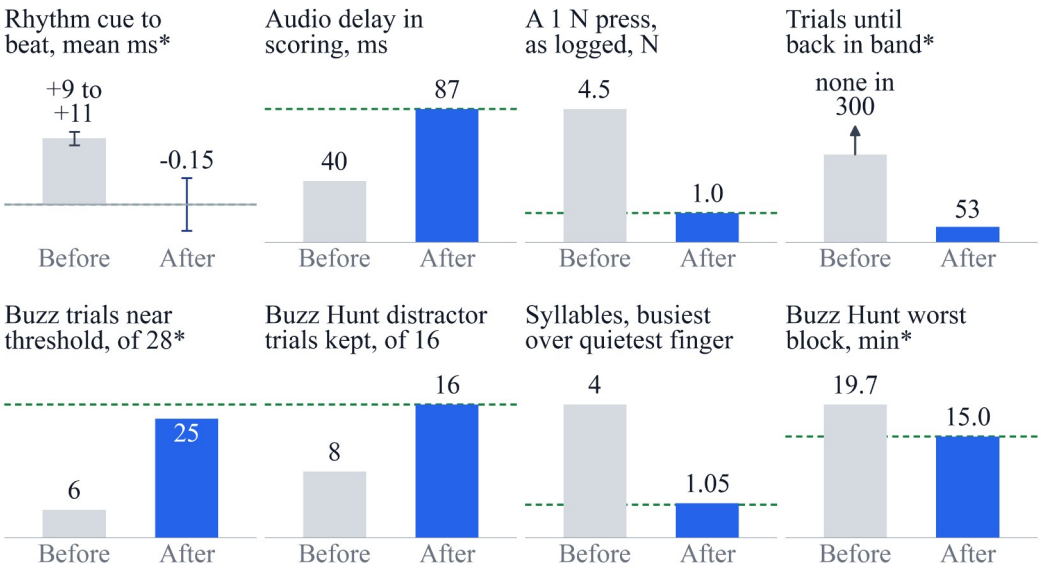


Figure 20: Seven of the faults of Section 4.8.2 and Table 33, in eight panels (two for Buzz Hunt's block), before (grey) and after (blue) the fix. Dashed: aims, where set; whiskers: the cue's range; starred: on a simulated clock or player.

Before the sessions the author ran a design review of each mode and its analysis, and counted a suspected fault only once a test reproduced it. Sixteen of those findings would have changed a study number. Among them, a board drop was scored as a lapse of attention, the one rest the retest needs was logged as zero and a session ran on the previous participant's calibration. Appendix D lists them with seven faults found when the last changes were checked. The development-block analysis found one more: force read as the press registered, a median 0.76 N against a true peak of 1.75 N. Reading each mode against its literature found another: two silent misses in one Echo game ended the whole block; they now end that game only.

4.8.3 Project management against the plan

Of the fifteen Semester 2 tasks in the progress report, five were done, four replaced by another check, two done in part and two not run. Joshua Bojanjac took over the two print tasks with the hardware (Appendix E).

Meetings with the supervisor ran fortnightly, and with Welber Marinovic as often, almost weekly in his laboratory in mid-semester. Figure 21 sets each workstream against the planned weeks and the 24 September re-plan to one board, two passes. The patient trials set for weeks 9 and 10 became the healthy sessions of weeks 13 and 14, and the hardware tasks of weeks 1 to 6 gave way to the platform work that filled them. The two not run, the RAS comparison and the convergence study,

needed the patient trial. Of the two done in part, the survey was answered by two healthy participants instead, and the trial paperwork became the patient pilot's draft information sheet and consent form (Appendix L; Table 35). Appendix E also closes out the report's seven risks and six failure modes. The EEG risk turned around once the laboratory joined the project, and the Windows risk occurred in part: the scan took the laboratory's EEG trigger box for a hand board until that port was set by hand. The stuck-actuator failure was designed out with the threaded finger rests.

The plan's weak point was time. Growing the platform from four modes to ten, eight with a pre-specified check, used the time the second board needed, so the register's one-person capacity risk occurred. It was handled as the register said, by deferring the lowest-value item. When the patient trials did not fit the timeline, Rhythm, built for the RAS comparison, became a timing measurement. The last change of plan was the collection itself, left to volunteers with a one-page sheet to test whether the software runs unaided. Few came forward unprompted, so the result across people rests on the laboratory's task. Such a collection needs bookings or the author present for part of each day, and the growth to ten modes took the weeks booking would have needed, a lesson Section 6.2 takes up.

The progress report planned about AUD 40 of hardware for Semester 2 after about AUD 160 in Semester 1, and the software stayed free and open source. The author spent none of either budget; the team members who built the hardware bought its parts. From the parts list [34], a hand costs about AUD 830 to 860: four calibrated sensors at AUD 192, an Arduino Nano at AUD 15 to 48, a USB cable at AUD 17 and PLA at AUD 30. The electronics team's circuit board and motors were not costed; uncalibrated sensors with boards, AUD 115 each, would cut the sensors' AUD 768 to AUD 460.

4.8.4 Reflection

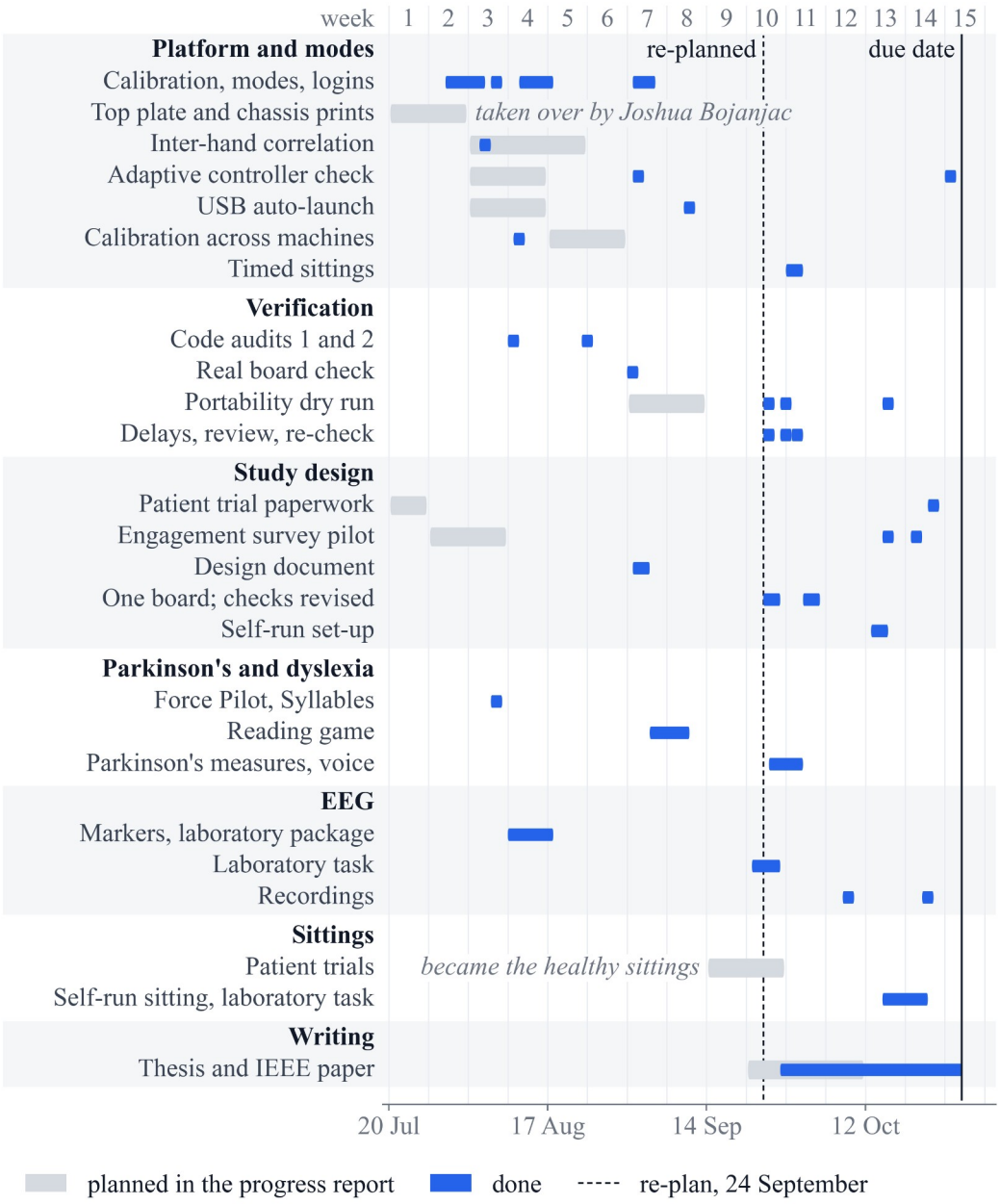


Figure 21: Semester 2 against the progress report's plan: fourteen of its fifteen tasks as ten grey bars in their planned weeks (the fifteenth is the fortnightly supervisor meetings), the work done in blue, from the commits and the project log, a task met by another check on the dates that check ran, the re-plan of 24 September 2026 dashed and the due date before its extension to 6 November; Table 35 closes all fifteen.

The most useful habit of the year was treating every number the software produces as a measurement needing a known answer: the sensor's datasheet, a microphone or a simulated hand with planted traits. Unit tests had passed constants coded as intended but wrong; the force constant 4.5 times too high is the clearest case. None of the 41

faults that would have changed a study number or a session's course was caught by a test that existed. Each kind of check found faults the others missed. Running and measuring the software found 14 (simulated sessions, the frame-by-frame check, the real board). The design review found 16, the final check 7, outside references 2, the block analysis 1 and the literature 1. The 42nd fault, the resting middle finger, showed in use. Two tests would have caught the two largest: that the force constant equals the datasheet's figure, now in the suite, and a microphone loopback on each new computer, now a set-up step. Simulation carries the same risk: the climb, tuned on steady simulated hands, lost to the sessions' controller in simulation at a 94 ms spread (Section 4.2).

Running the analysis on simulated sessions before the data found software faults the data would have carried. With the checks fixed in advance, every later change was dated with its reason (Appendix B). The EEG analysis taught the same lesson: its first analysis reported an error negativity and a beta rebound that its own tests did not support, so it now states an effect only when its primary test passes. The checks after the data, outside the 41, found two more analysis faults that no test had caught. First, the analysis had been scoring every correct press rather than the trials the planned rule keeps (Section 3.7); applied before any result was reported, that rule moved the four learning scores by 4 to 20 ms. Second, a random intercept alone gave the learning score too narrow an interval for an effect that varies between people, so the model gained a random slope (Section 4.3.1).

Designing for a session with no researcher present moved every instruction onto the screen and let the session resume after any failure, though the one self-run participant rated the instructions 2 of 5.

The author chose to move into software and beyond stroke; fixing the battery of games before writing the checks contained that scope risk. The move deferred the second board, the bench sensor comparison and the two objectives that need that hardware (Section 6.3), and gave ten modes on one verified detector, logger and notebook. On the sensors a published effect kept its direction but changed size. Chords cost 36 ms per extra finger against 126 ms on keys (Section 4.6.1), so the normative study sets the chord band from its own data.

The press check also found something that applies to any force-sensor device, as the onset literature expects (Section 2.12). A threshold count trails the push's start by the

time force takes to reach the threshold. That time shortens as presses quicken, and on slow rises the resting level's creep lengthens it (Figure 6). A learning curve read at the count therefore includes that shortening. From block 1 to block 8 the four gained 64, 78, 23 and 120 ms at the count and 59, 80, 16 and 104 ms at the push on the same trials (Figure 14f). In Person 3 the lag's fall made up nearly a third of the gain.

An onset rule run during play would remove the lag but start presses on a tremor or a light touch, which the threshold ignores, so the pilot keeps the threshold and reads the onset offline. The re-press test misses a light touch that runs into its push without letting go; only a search of the stored force for a second rise finds it.

Every session ran the resting level at a 10 s time constant, which counts every simulated ramp to 3 s (Figure 6); the code's unused 0.25 s default counted no simulated rise slower than 0.8 s. Like the force constant, it was a default that differed from the value the device used, and a test now holds the code's defaults to the configuration.

The press check itself came from the laboratory. The author's 7 October pilot had measured the lag, but only the block-by-block check the laboratory asked for showed it changing with practice. That finding matters for timing patients.

4.8.5 Checked in simulation

Since the device went back to the team on 23 October 2026, eleven parts, besides the rest and hold rules of Section 4.3.2, were checked in simulation or on a replay of the study's force streams (Table 21). All earlier results in this chapter came from the sessions' software. Of the eleven parts, only the start-up, the count and the adaptive controller could alter them. Each check runs against something known: the logged boots, the study's own streams, the sensors' measured noise, planted delays, sensor errors or a planted reading accuracy (Table 21). None models a patient's hand: a real tremor is not a pure 4 to 6 Hz sine, nor a slow press a ramp. Nor does the adaptive model match a healthy hand. At a 94 ms spread on the sessions' controller, its 250 to 300 ms hands hit 0.81 to 0.86 over 40 trials, against Person 5's 0.72. They never press a wrong finger, the cause of 9 of Person 5's 11 misses. These checks found two more faults. A code default differed from the device's resting level (Section 4.8.4). Mirror's hand comparison took the second hand's peak at its threshold crossing, so it misread a planted 0.50 asymmetry (Table 21).

Table 21: Parts checked in simulation or on a replay, each by an analysis script or an app test; each fix has a test that fails on the old code.

Part	Checked on	Result
Start-up: firmware streams during its self-test; no fixed wait (OD1)	A board simulated from the logged boots	First sample at 1.75 s, or 0.41 s on the 3.0 s boards
The count: the second raw sample in a row over the press threshold	The four sequence-task streams, replayed	Scored one sample after the raw force; 4,006 of 4,009 presses kept, practice and recall included; log within 19 ms, about 26 on the laptop; scores moved 4 ms at most
The code's default resting level, held to the device's by a test	Linear ramps of 0.1 to 3 s	Every ramp counted; at 3 s 60 to 86 ms behind as the study ran, 48 to 80 on the raw count
Trigger and release set clear of a resting finger's tremor	Ten minutes of a 4 to 6 Hz sine as rest tremor; weak, slow (to 3 s) and shaking presses	No false press to 0.3 N, the largest tried; 510 of 510 presses counted once
Adaptive: a climb of 20 percent a hit to the first miss; a 240 BPM ceiling (OA1)	Hands of 250 to 500 ms, trial SD 25 or 94 ms, ten seeds each	In band after the first miss: 58 of 60 blocks at SD 25 ms (48 as run), 39 of 60 at 94 (49 as run)
The trigger box found as the one USB port with no force stream (OC1)	Virtual serial ports	Box found; no byte sent to any port
Mirror: each hand's peak taken from its release (OC2, OC3)	Eight players on two simulated boards	Lead in 8 of 8, within 3.5 ms; planted 0.50 read within 0.01 (before: 0.90 or 0.08)
Clip and loopback tools for the screen's and the sound's delays (SQ1, SQ3)	240 fps clips and loopback recordings with planted delays	Screen within 0.4 ms, each edge timed to a fraction of its 4.2 ms frame; sound within 1.5 ms
Instructions said aloud; an 8 s wait before the first game	The engine, run headless	Said in every mode; first wait 8 s
An uncalibrated sensor (OV4)	The study's presses through twelve simulated sensors with gain and curve errors to 10 percent	99 percent counted on both, within 1.3 ms on average
Syllables with simulated readers (OE2)	A fluent and a struggling reader on all five profiles	Planted accuracy read back: 0.96 to 1.00 (0.97 planted) and 0.69 to 0.81 (0.75)

Chapter 5. Conclusions

5.1 Answer to the project question

The results show that the device is ready for a patient pilot, with a normative study carried out alongside it. The main requirements were met for press timing, EEG event markers, data recording and reproducing the sequence-learning effect in healthy adults. The sequence task was the only known-effect check tested across several participants during this study. Table 22 summarises the results for each engineering question.

Table 22: Answers to the five engineering questions.

Question	Answer	Main evidence	Section
SQ1 Timing	Yes for presses and markers; laboratory screen and audio delays still need measuring	Two constants corrected, study presses logged within 49 ms, median push-to-count delay 40 to 55 ms and scored blocks within one sample. A stuck sensor issue was corrected and tested by replay	4.2, 4.3.2
SQ2 Function	Yes, although Person 3's result was less certain	Learning effect in all four participants, mean 91 ms (95% CI 42 to 140). Early correct presses ranged from 68 to 94%, compared with 33% by chance. Person 3's confidence intervals crossed zero under some additional analyses	4.3
SQ3 EEG	Yes for event markers; screen delay not yet independently measured	Every marker within one sample across two recordings. Beta power was lower for the learned order (secondary measure, $p < 0.001$)	4.4
SQ4 Readiness	Yes, based on one independent session	Participant completed a session without assistance and all six blocks were saved. Startup took 4.2 s during testing and 1.75 s in simulation	4.1, 4.6.2
SQ5 Extension	Yes, in software	Parkinson's-related force measurements checked using synthetic recordings and calculated for two healthy blocks. All five Syllables age profiles included spoken feedback	3.3.5, 3.3.6, 4.7

Checking the software against independent references identified and corrected two important errors. The force conversion constant was 4.5 times too high, while the audio delay was measured at 87 ms rather than the assumed 40 ms. Every study press was logged within 50 ms of crossing its threshold, and all EEG event markers matched their corresponding game logs within one sample across the two recordings. The press-timing check also compared each registered press with the start of the finger's force rise. This was completed for 3,402 real presses.

The device was then tested against a known sequence-learning effect. All four healthy participants showed slower responses when the repeated sequence changed back to random order, with an average difference of 91 ms. This was similar to the

effect reported in previous research after longer training. Both EEG recordings also showed the expected difference in motor beta power between learned and random trials. The independent session showed that someone could use the device without a researcher present, with all six blocks played successfully saved.

The additional game modes also worked through the same software engine. Force Pilot recovered a deliberately added 5 percent release error to within 0.5 percent of maximum force on synthetic signals, and its measurements were calculated from two healthy blocks. Syllables was developed with five age profiles and spoken feedback for all words and syllables. However, neither mode has yet been tested with people who have the condition it was designed for.

The main work remaining is to establish normal measurement ranges, reliability between repeated passes and the expected results for each game mode. This will require the normative study alongside the planned stroke patient pilot. Absolute force measurements in newtons also still rely on the manufacturer's sensor calibration, as discussed in Table 17.

5.2 Objectives

Table 23 gives the final status of all project objectives. An objective is marked as met when a participant's results or a dated project record show that it was completed. Objectives are marked as partly met when only some of the requirements were achieved. Those tested using simulation or replayed study recordings are marked as met in simulation (Section 4.8.5). Two objectives, OA3 and OD3, were deferred because they depend on further hardware work.

Of the 18 objectives, five were met, three were partly met, eight were met in simulation and two were deferred. The eight objectives met in simulation have been implemented in software and passed their tests. However, they still need physical hardware or further participant testing. This includes testing the second hand device (OC1 to OC3), comparing uncalibrated sensors (OV4), checking the startup time on the actual board (OD1) and testing repeated passes or participants with Parkinson's disease or dyslexia (OV3, OE1 and OE2).

Section 6.3 outlines the work needed to complete these checks. Looking back at the four main areas from the progress report, the adaptive controller (A) achieved a success rate of 0.72 during Person 5's block and kept 49 of 60 simulated blocks

within range when using reaction-time variation similar to the study. Rhythm (B) recorded one participant's timing variation, with a standard deviation of 50 ms. The bilateral system (C) was tested using two simulated hand devices.

The deployment system (D) recorded startup times for all 76 logged starts.

Table 23: Final status of every project objective.

ID	Objective	Status	Evidence
OA1	Adaptive controller keeps each finger between 65 and 80% success	Partly met	Person 5's overall result was within range, but individual fingers varied from 0.56 to 0.89. At 94 ms trial SD, 49 of 60 simulated blocks were in range as run and 39 with the climb
OA2	Record peak force for every press	Met	Every trial contains peak force calculated using the factory calibration of 19.5 mN per count
OA3	Chassis fits hands from 5th to 95th percentile	Deferred	Six hands were fitted using threaded rods, but the full size range was not measured
OB1 to OB3	Rhythm engine, beat statistics and timing windows	Partly met	Timing SD and lag-1 correlation measured for one participant; mean beat lead tested in simulation; four timing windows
OC1	Automatic device assignment	Met in simulation	Trigger box identified, with 0.44 ms median scan time using simulated ports
OC2	Mirror therapy mode	Met in simulation	Two simulated boards; movement lead measured in all eight simulated participants
OC3	Bilateral asymmetry and correlation	Met in simulation	Asymmetry recovered within 0.01 of the planted value
OD1	Setup under 2 s	Met in simulation	1.75 s in simulation, compared with 4.2 s during actual testing
OD2	Record startup delay	Met	76 startup times recorded (Table 15)
OD3	Wrist and forearm positioning	Deferred	Included in the Semester 1 design, not measured during this study
OV1	Normative study kit	Met	Tested on ten simulated sessions and completed once by a real participant, saving all six blocks
OV2	Predefined checks for every mode	Met	21 checks fixed before data collection, including 11 that can fail
OV3	Reliability analysis	Met in simulation	Tested using twelve simulated participants (Section 4.5; Table 45)
OV4	Agreement between calibrated and uncalibrated sensors	Met in simulation	Timing comparison completed in Section 4.8.5
OV5	Assessment without assistance	Partly met	One independent session completed six of eight games
OV6	EEG event-marker timing	Met	Every marker matched the log across both recordings
OE1	Parkinson's-related force measurements	Met in simulation	Planted 5% release error recovered within 0.5% of maximum; two healthy blocks analysed
OE2	Syllables for ages 6 to over 60	Met in simulation	All 932 words voiced and planted reading accuracy recovered

5.3 Significance and lessons

The three-part testing method developed in this project (Section 1.5 and Figure 22) helped identify measurement problems that the automated software tests had missed. The two largest errors had both passed their original unit tests, even though the force conversion and audio timing were incorrect (Section 4.8.4). The problem with Person 3's middle-finger sensor also showed why counted presses need to be checked against the recorded force signal (Section 4.3.2). The software now includes checks covering these different areas.

Each counted press can be compared with the beginning of its force rise. During the sequence task, the start of the push was identified for 3,402 of the 3,463 counted presses, with the count occurring a median of 40 to 55 ms later. EEG event markers are sent when the stimulus appears on the display frame. Across two recordings, all 1,859 and 1,794 markers matched the log within one sample.

The analysis was also tested using simulated participants with known characteristics. Four of the assigned measurements were recovered with correlations of at least r = 0.90. The press detector was further tested to identify how slowly someone could apply force and still have a press recorded. At the device's settings, all simulated force ramps lasting up to 3 s were detected (Figure 6 and Table 24). Using the original planned scoring method, the device also reproduced the expected sequence-learning effect in all four healthy participants.

The literature review and the 19 papers examined in Appendix J did not identify another finger rehabilitation device that combined these three types of checking. However, the approach is not limited to the Curtin device. Other force-based systems could use their sensor datasheets, independent timing measurements and raw force recordings to carry out similar checks. AMBER, the individuation device and the fingertip force device compared in Table 8 all record individual finger force, so the method could be applied directly to them.

For example, AMBER could compare its force recordings against an independent clock, check its tasks against known finger enslaving effects and measure each press from the beginning of force application. FINGER and other force-tracking devices measure only one or two fingers, so the same checks could be used for the fingers

they record. Table 24 shows what the current measurements could provide during one patient session and the size of the errors that would need to be considered.

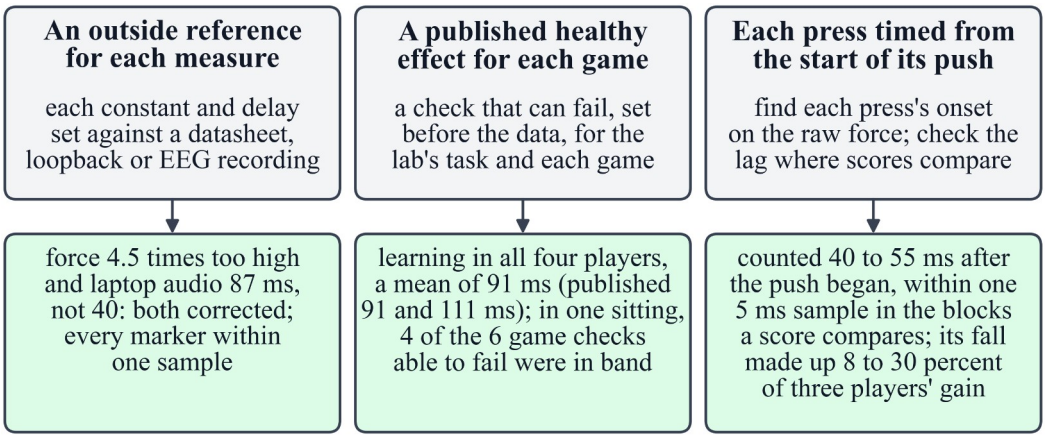


Figure 22: The three-part measurement method from Section 1.5, showing the checks that could be used on another device and the results found during this project.

Table 24: Measurements available from one patient session and their expected errors.

Measurement	Error of one reading	What one session could measure
Learning score	Confidence interval approximately 30 to 54 ms either side of the estimate (Section 4.5)	A learning score larger than this uncertainty; healthy group effect was 91 ms
Individual reaction time	7 to 11 ms SD, representing 0.6 to 1.4% of participant trial variance (Table 17)	Reaction time for every press, with relatively little variation added by the device
Slow force rise	Updated detector counted simulated presses 8 to 15 ms after raw threshold crossing for a 400 ms rise, and 48 to 80 ms at 3 s (Figure 6)	Each press timed from the beginning of its force rise
Peak force	Approximately 0.8 to 1.1 N between sensors and 0.3 to 0.6 N on one sensor at 95% confidence for the difference between two readings (Table 17, limitation 5)	Force changes larger than the expected measurement uncertainty

The normative study should use booked sessions lasting at least 45 minutes. Shorter sessions do not include the second pass needed to measure reliability, while the 15-minute version also leaves out four of the eight study games (Appendix K). Overall, rehabilitation devices that use force thresholds to calculate learning scores should either measure reaction times from when the finger first starts applying force or report the delay for each block alongside the learning results.

Chapter 6. Future Work

6.1 Problems that were not anticipated

Several unexpected issues came from using the device in the EEG laboratory. The automatic board scan initially identified the laboratory's EEG trigger box as a hand device because both used an Arduino. This meant the trigger box's port had to be selected manually. The automatic launch feature also was not used during the laboratory sessions because they were started through PsychoPy instead. Another issue was that the sessions ran on the laboratory computer, where the audio delay had not been measured.

The one Rhythm block therefore provided information about variation between presses, but not their absolute timing relative to the beat. Before using the software on another computer, the audio delay should be measured using microphone loopback. The middle-finger sensor problem was another unexpected result. During one participant's sequence task, the finger's resting force remained slightly above its release threshold. The software then treated the finger as still pressed, causing further presses to be missed until it lifted (Section 4.3.2).

Recruitment for independent sessions was also more difficult than expected. Of the ten volunteers sought, only one completed a session without being prompted, while another completed a single game block. However, the completed session followed approximately the same timing as the simulations. Its six games took 13.8 minutes of actual play, similar to the first-pass estimates in Table 47. This supports the session-length estimates used in Section 3.6, although more participants are still needed.

The patient trials originally planned in the progress report could not be completed within the available time. These were replaced with the healthy participant checks carried out during Semester 2.

6.2 What would be done differently

If the project were started again, the author would plan the measurement checks from the first week rather than leaving most of them until after the software platform had been developed. Important constants and delays would also be checked against independent references before building the first game mode. Testing the first four

modes with healthy adults during Semester 1 would have provided reference measurements that could then be used when developing the remaining modes.

Recruitment would also be organised differently. Booking participants in advance would likely have been more effective than leaving the study available for volunteers to complete in their own time. With the author available in the laboratory for part of each day, two booked afternoons could have allowed eight participants to complete a 45-minute session each, including the second pass needed for reliability. Instead, expanding the platform from four modes to ten took much of weeks 2 to 8 (Figure 21), reducing the time available for testing.

For the next project, no further game modes should be added until the current eight study modes have established normal ranges. Any future mode should also have its expected result, acceptable range and independent measurement reference defined before development begins.

6.3 Priorities for the next project

The next project team plans to add a camera-based hand tracker. This would allow participants to play some of the games by moving their fingers in the air without wearing or resting their hand on the device. The software already receives input through one shared interface, used by the current board, an optional second board and the keyboard. A camera tracker could therefore be added as another input source without needing to rewrite the existing game modes, logging system or analysis checks.

Finger movement, tapping and tremor in Parkinson's disease could potentially be measured using the camera, although actual force measurements would still require the sensors. The camera's timing would also need to be checked against an independent reference before it was used with patients. The following work is listed in order of priority, with several areas large enough to form separate thesis projects. Before the patient pilot begins, three additional checks should also be completed: filming the laboratory screen at 240 frames per second, measuring its audio delay using microphone loopback and testing each force sensor against known weights.

1. **Normative study.** Run the planned battery with at least eight healthy participants using booked sessions rather than relying on volunteers to attend independently. Participants should complete the second pass and return for a

second session one to two weeks later. This would establish normal ranges and calculate ICC(2,1), SEM and MDC95 for the different measurements. Eight participants are the minimum under the planned statistical method because Holm's first correction across nine checks requires p below 0.0056, while the exact test with eight people can reach 1 in 256 (Appendix K).

2. **Hardware and press detector checks for the patient pilot.** Add finger sensors that can slide sideways to suit different hand widths, as suggested in the participant survey (Appendix M). The updated rest and hold rules and resting-tremor threshold should also be tested on an actual patient's hand. The changes listed in Table 21 need to be checked on the physical device (Section 4.8.5), and the adaptive controller used during the study should remain in place until the optional difficulty climb has been adjusted.
3. **Stroke patient pilot.** Use MDC95 to help decide whether a patient's performance has changed by more than the expected measurement error. An initial conservative value is 55 ms for the median reaction time of a block measured from force onset (Section 4.5). The pilot should compare patient measurements with healthy reference ranges and track whether performance changes beyond measurement error across repeated sessions. An Aboriginal community controlled health service should be involved from the beginning of the study design (Section 3.9.3).
4. **EEG study.** Measure the screen flash, audio tone and vibration against the event markers using the clip-based timing tool and microphone loopback (Section 4.8.5). Future testing should also compare fixed and variable intervals between trials. Previous research reported sequence-learning reliability of approximately r = 0.3 between sessions and 0.6 for split-half testing [101]. An underlying effect size of d = 0.8 would therefore appear closer to 0.62 and require approximately 42 participants per group.
5. **Force Pilot and Parkinson's disease.** Compare people with Parkinson's disease with age-matched healthy adults, using the more affected hand and testing both on and off medication. A second-day session would also allow reliability to be measured. Based on a release effect size of d = 1.1, approximately 15 participants per group would be needed. This could form a separate thesis project with a movement disorders clinic. Force Pilot is still an

early version, and the fast-release pulses, force tapping and memory-guided holds described in Appendix F could be added later.

6. **Syllables and dyslexia.** Test the game with children and readers with dyslexia using individual case studies, alongside a standardised reading assessment and control group [25]. An adult should be involved during each session, as suggested by previous research [157]. Initial testing should use probe-only sessions introduced at different times for different readers. Future development could include rhyme tasks, individual speech-sound activities based on phonics teaching [25] and word lists that adapt to each reader, similar to GraphoGame [199]. Each new activity should have its own check before being included.
7. **Home-based use.** Test whether participants can use the device regularly at home while a clinician reviews their performance records remotely. Adherence will be particularly important, since only one of the ten independent sessions sought during this project was completed without prompting.
8. **Sensors and the second hand device.** Test each calibrated sensor using known weights and compare the results with its expected resolution of 19.5 mN per count. Finish the left-hand device, check chassis adjustment across different hand sizes and measure the intended wrist and forearm angles (OA3 and OD3). Mirror and the bilateral hand-comparison measurements should then be tested using real hardware rather than simulation. An uncalibrated sensor costing AUD 115 with its board should also be compared directly with a calibrated sensor costing AUD 192, particularly for press timing, where the two agreed in simulation.

The software, analysis notebook, import scripts and participant information sheet are available in the project repository [34]. The de-identified study data will be provided to the supervisor along with this thesis.

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Appendix A: The software's development practice

The source is at https://github.com/basil2fxs/Hand-Rehab-Thesis [34] (Table 25). Each board sends every sample as one line, "FSR: a,b,c,d", the four sensors' readings, at 115200 baud, and buzzes motor n when sent "STIM:n". The firmware is uploaded with avrdude, a flashing tool bundled with the software, which tries the Nano's newer bootloader at 115200 baud and then the older one at 57600.

Table 25: Top-level structure of the project repository.

Folder or file	Contents
app/finger_rehab/	Engine, game modes, screens, hardware and logging
app/config/	Every setting, and the laboratory's changes
app/assets/	Firmware, music, word lists, voice and tones
app/arduino/	Board firmware and the sensor address tool
app/builds/, app/scripts/	Installers; bench, timing and simulation scripts, among them the model hand (measure_battery.py) and the simulated cohort (simulate_cohort.py); the import step that replaces names with codes
app/tests/	The automated test suite
app/docs/	Laboratory set-up, the study documents, research per mode
analysis/	The notebook, the sequence-task and press checks behind Chapter 4, the force-stream replay of the detector fix, the sensor and adaptive figures, the onset-rule rerun (onset_sensitivity.py) and its check on simulated pushes (onset_accuracy.py), the detector's lag against rise time (detector_lag_vs_rise.py) and the resting-level hold's replay (hold_replay.py); eeg/, behind Section 4.4
EEG_Lab/	The laboratory computer's game, settings and launcher

Table 26 sets the project's practices against published standards and guidelines; each row says where the practice can be checked.

Table 26: The project's practices against published standards and guidelines.

Practice	Standard or guideline	How the project applied it
Requirements traced to verification; configuration management; every problem fixed with a regression test	Medical device software life cycle, IEC 62304 [205]	Each requirement verified (Table 9); one public repository; installers built from one commit; a failing test before every fix
Hazards identified, controlled and reviewed	Medical device risk management, ISO 14971 [206]	A board drop scored as its own outcome; finger rests set by hand before the hand is placed; no treatment claims (Table 12)
Code under version control and tested; analysis written as code	Best practices for scientific computing [219]	An automated test suite; a notebook that regenerates every table; the analysis checked on simulated participants with known traits; the mixed model checked against a paired t-test, which the Satterthwaite and Kenward-Roger methods approach when an effect varies between participants (Section 2.12), and its random-intercept form against the lme4 intervals of the bench analysis [209]

Practice	Standard or guideline	How the project applied it
Hypotheses, statistics and exclusions fixed before data	Preregistration [220]	Every check written before any participant; each later change logged with its reason (Appendix B)
Measurement properties named in a standard taxonomy	COSMIN [21]	Construct validity by known effects, reliability by ICC, and measurement error by SEM and MDC95
Reliability reported in full	GRRAS [221]	ICC form stated, with intervals, SEM, MDC95 and Bland-Altman limits
EEG methods and stimulus timing reported	Keil et al. [23]; COBIDAS MEEG [222]	Marker timing measured against the amplifier; every processing step and parameter listed; the code public

Appendix B: Study configuration and design changes

The study battery is the software preset healthy_one_hand_v3: twelve blocks, eight in pass 1 and four in pass 2, with a 45-minute budget and a 50-minute hard stop. The 15, 30 and 60-minute sessions are the presets trial_15_v3, trial_30_v3 and trial_60_v2 (Section 3.6). While the battery runs, the preset replaces the shipped configuration's values with the settings in Table 27. Every setting it changes is a trial count, a rest or a difficulty level held fixed; none changes a press threshold or a scoring rule, and an automated test fails if the list changes.

Table 27: Study settings per mode.

Mode	Setting in the study
Reaction	Choice task, 20 scored trials, response window fixed at 2.0 s
Rhythm	One fixed track of about two minutes, medium difficulty, vibration on the beat
Echo	Four lanes, Simon rule, two games, read by their mean for the normal value and by the better game for E1, the same sequences for everyone, capped at 10 items
Force Pilot	One climb of the fixed twelve-wave ladder
Chords	Two sub-blocks of 20 trials (4 single fingers, 10 pairs, 5 triples and 1 four-finger chord each)
Buzz Hunt	16 localisation, 3 catch and 4 span trials, one 2.0 s window
Muscle Memory	Short form: a 12-press warm-up, a 32-trial random take, then seven takes of three sequence cycles (36 presses), the fourth and sixth of them untrained probes
Adaptive	40 trials
Syllables, Mirror	Not played

The changes in Table 28 were made before any session's data were read, except the last ten, dated 22 to 27 October. Most came on 24 September 2026, when the design moved to one board and two passes, and on 28 September, when Syllables left the healthy battery. A review of each mode's literature on 1 October settled most of the rest, and later changes are dated in the table.

Table 28: Changes to the pre-specified checks (codes as in Appendix K), reference values and analysis, with the reason for each; all before any session's data were read except the ten of 22 to 27 October.

Check	Change	Reason
R2, C4, C5, Rh3, M1, M2, W2 (hand comparisons and Mirror)	Dropped	They need both hands or two boards
R3, no practice effect	Restored as pass 2 against pass 1	The second pass makes it computable
E1, Echo span	Band set to 5 to 9 on four lanes; a feasibility check	The Simon game has four buttons; a healthy cohort cannot fail it from below

Check	Change	Reason
E3, errors beyond chance	Dropped	On four lanes every lane is a sequence lane
T1 to T5	Added	The second pass
C6, chord cost	Added	Single fingers dealt among the chords give the baseline
P2, learning on a second block	Tested only in the 60-minute session	Muscle Memory plays twice only in the 60-minute session
E2, Hebb repetition in Echo	Dropped	The Simon rule has no hidden repeated sequence
W6, Force Pilot trend	Dropped	On one climb, level and run order are the same, so the residual trend is zero
S6, S7, Syllables	Dropped	Syllables left the healthy battery; adult readers sit at ceiling
Rhythm, second go	Added, exploratory	Played in the minutes Syllables gave back; no class predicted
R3 and F3, the equivalence checks	Read on the 90 percent interval, two one-sided tests at 5 percent [223]	The standard form; also requiring the 95 percent interval made each a 2.5 percent test on each side
R3, basis	The practice caveat added [224]	Choice reaction time improves with repeated testing, so a fail is read as practice
R1, the wait	One-sided: fails only when RT falls as the wait grows	With catch trials RT rises a little with the wait [225]
Rh1, the lead	Share of presses later than 150 ms, under 10 percent	Tapping to music shows less lead than tones [66]
F1, F2	Irregular waves against repeating ones	The ladder narrows the corridor more than it speeds the target, and 8 s of preview puts lags near zero
F3	A feasibility check	Its 2 percent margin is met even by a true release excess of 1 percent of maximum
F4	A feasibility check	A finger held still keeps 54 to 100 percent of a run in the corridor
C2, W4, C3	C2 and W4 decided on hit rate; C3 and the enslaving halves reported	At light presses the enslaving ratio is mostly built before the press
C6	Mean onset time, not the first press	Verwey's chord times are mean key times [86]; the first press barely moves
B2, B5	B2 exploratory; B5 dropped, the window fixed at 2.0 s	Two or three wrong-finger errors expected; the window ladder only counted trials
A6	A feasibility check	The near-certain hits of the slow start hold the hit rate above 0.65
OV4, sensor agreement	Moved to future work; met in simulation (Section 4.8.5)	The uncalibrated set was not in a device in time
Reaction instruction	Shown on the GET READY card	The sessions ran with no researcher to say it (15 October)
Reading rule	Validity and consistency combined into one verdict per mode (Appendix K)	Answers the research question mode by mode (15 October)
Reading rule, low bars	A mode whose passed checks are only R1, Rh1, B1 or B3 is ready on feasibility only	Appendix K already names them as bars a working mode clears (15 October)

Check	Change	Reason
Verdict words	Fixed as Ready, Ready on two measures, Valid only, Feasibility only, Not tested and Not ready; a column says whether one measure is both valid and consistent, from exploratory ICCs of the measures behind C6 and P1	The rule lets a mode's checks and reliability rest on different measures, and the verdict now says when they do (16 October)
Adults over 50, and any second session	Kept apart from the study's values	Age changes several of the measures, and a second session is a retest rather than a new person (16 October)
Mean asynchrony reference	About 0 to −20 ms to music, where the design had 0 to −40	Dalla Bella et al.'s music values reread from their figure; tones lead by 30 to 40 ms (18 October)
Release over press reference, and F3's basis	0.83 to 0.90 in young adults, where the design had about 0.9; F3 rests on the young-adult studies	Davidson et al.'s ratios recomputed; the older and Parkinson's studies do not test equal release and press error (18 October)
Sequence task scoring	The rule of Section 3.7 applied: correct responses under 1000 ms, each block's first trial dropped	The analysis had scored every correct press; found after the data, before any result was reported (22 October)
Press timing check	Each counted press set against the start of its push on the force stream	Asked for by the laboratory once the data were in (22 October)
SQ1's lag check	Same lag in a score's two blocks	Lag seen to fall from practice (27 October, after the data)
Sequence task, fast presses	The learning score also read without presses under 100 ms (78 ms, $p = 0.022$) and under 150 ms (69 ms, $p = 0.034$)	The script counts a press 2 ms after the flash as correct (22 October, after the data)
P3 against the random post-test	Added as a check outside Holm	The primary P3 contrast runs from random practice to blocks 7 and 8, so time on the task could shrink it (22 October, after the data)
Sequence task across participants	A one-sample t-test on the four participants' scores, and a linear mixed model of the trials: block as a fixed effect, with a random intercept and a random post-test effect per participant	The laboratory's task is not one of the device's checks; chosen once the data were in (22 October)
Sequence task, further readings	The score against the faster random block; the two random blocks set side by side; early presses scored by the finger chosen; error rates	Each separates learning from practice, fatigue or a trade of speed for accuracy (22 October, after the data)
Beta with reaction time taken out	Learned trials matched to each random trial's reaction time, and beta in the half second before the press	Random flashes were answered later, which moves the movement's beta drop (22 October, after the data)
Learning score, all readings	Set side by side in Table 19, with a one-sample t-test across the four for the faster random block and the start of the push	The other readings already had one (23 October, after the data)
Recall against chance	Random recalls drawn without the same key twice in a row	No recall of the sequence repeats a key, and the early-press chance already left the repeat out; the first draws allowed it (23 October, after the data)

Check numbers follow the design document, so a gap in them is a check dropped before any participant. The naming of a failed R3 was clarified on 15 October, before any session's data were read, without changing whether it passes (Appendix K).

Appendix C: The EEG laboratory sessions

Figure 23 shows the first recording, of 8 October 2026, under way; the second, of 22 October, used the same laboratory, task and settings. All four participants of the sequence task saw it on the finger cards the other games use, as each block's settings record, rather than on the script's four squares of Figure 7(a), which the software now draws by default. The outer cards sit about three times further out than the squares, so both recordings were checked for eye movements (below). Table 29 is the marker map, version 1.7. Each code is held on the trigger lines for 8 to 25 ms, until the first frame at least 8 ms after it was sent, and then returned to zero. A response byte leaves 2 to 24 ms after its press, when the trial closes; the log keeps both times, and the clock is fitted on the time each byte left. Figure 24 plots every marker of both recordings against the clock fitted through it.

Table 29: EEG marker codes sent by the software.

Code	Meaning
20	GET READY at the start of a block
21	Reaction: the wait before the finger lights begins
22	Rhythm: vibration sent ahead of the beat
23, 24	Force Pilot: run start; section edge
25	Reaction: when a catch trial would have lit a finger
30 to 37	Stimulus: 30 screen only (in an SRT block, the flash and its tone), plus 1 with a tone, plus 2 with a vibration, plus 4 when the finger is not shown
38	Buzz Hunt: the vibration is the stimulus
40, 41	Muscle Memory: sequence item; random item
50 to 53	Syllables: option set; option set after a miss; prompt vibration; the four words of a speed check
100 plus finger	Correct press
110 plus finger	Wrong finger (the finger pressed)
120 plus finger	False start (pressed before the stimulus)
130	No press before the deadline
131	Press with no trial active
140, 141, 142	Feedback shown: hit, miss, near
200 plus mode	Block start (219 if abandoned)
220 plus mode	Block end
240 to 245	Session start, end, pause, resume, rest start, rest end

Fingers are numbered 0 to 3 from index to little on a one-hand block, and 4 to 7 for the left hand when both hands play. Mode numbers: Reaction 0, Adaptive 2, Rhythm 3, Mirror 4, Muscle Memory 5, Chords 6, Syllables 7, Force Pilot 8, Buzz Hunt 10, Echo 12 and the laboratory SRT 13.

The task's timing follows the laboratory's script: each target is flashed for 100 ms with its own tone and has a 2.6 s response window. As in Nissen and Bullemer's first experiment [92], the next target follows a set 500 ms wait after a response. The

script's timing groups [99] and musical experience question [29] are kept. Alongside the planned rule, the analysis also scores the learning effect without reversal trials, which are slower and more common in random blocks [98]. Recall is scored by position, by correct triplets [96] and by the longest correct run, each against 10,000 random recalls drawn from the four keys, never the same key twice in a row, as in the sequence. Each EEG file is paired with its game block by the order of the marker bytes, and the exported event file gives each press marker its offset to the force onset (Section 2.12).



Figure 23: The recording of 8 October 2026: the laboratory's 64-channel cap and amplifier cables, a Reaction trial in the sequence task with the index card lit, and the right hand on the device's finger rests.

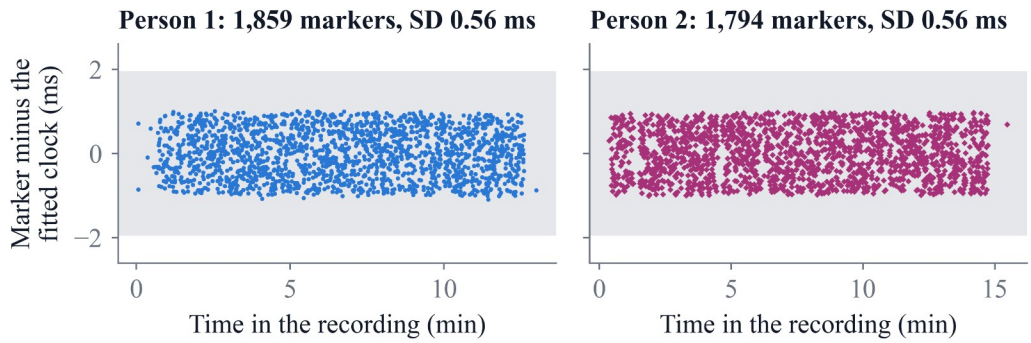


Figure 24: Each marker against the clock fitted between the log's send times and the EEG; the band is one 512 Hz sample either way.

Table 30 gives the processing behind Section 4.4 in the order it runs, so the analysis can be repeated from either recording.

Table 30: EEG processing settings.

Step	Setting
Reference	Common average of the 64 channels, set first, since BioSemi records without a fixed reference and the bad-channel checks need one
Filters	0.1 to 40 Hz, zero-phase FIR; notches at 50 and 100 Hz
Bad channels	Flat, noisy or unlike their neighbours (r under 0.4); rebuilt by spherical-spline interpolation
Eye artefacts	ICA (Picard, 99 percent of the variance, fitted at 1 to 40 Hz); components that correlate with an eye channel at an absolute z over 3 removed
Epochs	Sections of the recording around each event, dropped above 150 µV peak to peak
Baselines	200 ms before a flash or buzz; 400 to 200 ms before the force onset of a press, as for the ERN in ERP CORE [207]
Conditions	Random practice; sequence blocks 1 and 2 (early) and 7 and 8 (late, the learned order); random post-test; correct trials only
Windows and sites	N2 200 to 300 ms at FCz and Cz; P3 300 to 450 ms at Pz and CPz; after the force onset, ERN 0 to 100 ms at FCz and Cz and Pe 200 to 400 ms at CPz and Pz
Band power	Morlet wavelets from 4 to 30 Hz with half as many cycles as hertz; theta 4 to 7 Hz at FCz and Cz, beta 13 to 30 Hz at C3, 0 to 0.5 s after the flash, in dB

In the first recording three channels were rebuilt in the Reaction file and six in Buzz Hunt, all right central or temporal, and ICA removed two blinks and a horizontal eye movement from each file. The flash-locked comparisons kept 43, 176, 139 and 38 correct trials in random practice, early and late sequence and random post-test, and 32 localisation buzzes. The Buzz Hunt block's 65 trials were 32 localisation buzzes, three catch trials, 20 one-buzz-or-two trials and 10 span trials, and 60 of its 62 buzzes survived rejection, every localisation buzz among them. Pooled over the blocks, the press-locked epochs held 687 correct and 83 wrong presses, of which 78 and 9 were in random blocks.

The second recording ran 15.6 minutes. A Force Pilot block in each session and a Buzz Hunt block in the second were stopped part way and are left out, as the analysis's rule for abandoned blocks says. Five channels were rebuilt (F5, P9, AF4, FT8 and P4). ICA found 26 components, none of which met the eye criterion; Person 2 blinked 4.3 times a minute. Removing the two most eye-like components, 2 and 7, left the N2 difference at −0.4 µV ($p = 0.692$) and the P3 at +2.4 µV ($p = 0.027$), close to the uncorrected values in Table 31, so the results are given without removal. Rejection kept 720 of 896 flash epochs, against 851 of 896 in the first recording, and the flash-locked comparisons kept 30, 161, 122 and 21 correct trials in the same four conditions. The press-locked epochs held 584 correct and 70 wrong presses. In the sequence trials the flash's P1 was smaller and broader than in the random ones, peaking at 158 ms against 117 ms, and the same in both halves of the session. Both kinds of flash share one marker path, so the difference lies in the brain's response, not the markers.

C.1 The EEG results of the two recordings

Table 31 gives each measure of the two recordings of Section 4.4, per person and pooled, with terms as in the Nomenclature; with two people, the results describe these two alone.

Table 31: EEG results of the two laboratory recordings, with p in brackets. Both: the two people's differences averaged, with trial labels shuffled within each person (Section 3.7); Holm's correction runs across the N2 and P3; (check): added after the data (Appendix B).

Measure	Person 1	Person 2	Both (p)
Markers matching the game's log	1,859 of 1,859	1,794 of 1,794	
Marker scatter about the fitted clock, SD (largest), ms	0.56 (1.09)	0.56 (1.01)	
Reaction time, recording against log: r, offset in ms	0.99997, −1.0	0.99997, −0.9	
Flash P1 at O1 and O2, random trials, ms	121	117	
Learning score, ms (Section 4.3)	+81	+101	
N2, random post-test against late sequence, µV [109]	−4.2 (0.020)	−0.4 (0.740)	−2.3 (0.038; Holm 0.038)
P3, random practice against late sequence, µV [110]	+2.4 (0.023)	+2.5 (0.048)	+2.5 (0.003; Holm 0.006)
P3, random post-test against late sequence, µV (check)	+0.5 (0.644)	−0.5 (0.758)	0.0 (0.973)
Beta at electrode C3, random post-test against late sequence, dB [111]	+1.1 (<0.001)	+1.3 (0.033)	+1.2 (<0.001)

Measure	Person 1	Person 2	Both (p)
Beta at C3, late sequence trials matched in RT (check), dB	+0.8 (0.038)	+1.2 (0.126)	+1.0 (0.020)
Beta at C3 in the 0.5 s before the press (check), dB	+0.7 (0.010)	+1.4 (0.012)	+1.1 (<0.001)
Theta at FCz and Cz, dB [112]	+0.1 (0.686)	+0.4 (0.512)	not pooled
ERN, wrong against correct press, µV [226]	−1.9 (0.067)	−0.8 (0.260)	not pooled
Pe, wrong against correct press, µV	+0.2 (0.830)	+3.2 (<0.001)	not pooled

Both recordings showed a beta difference (Figure 17 in Section 4.4). Beta over the left motor cortex was higher on random flashes than on the learned order, by 1.1 and 1.3 dB ($p < 0.001$ and $p = 0.033$ in this secondary test; $p < 0.001$ together), as Lum et al. found shortly before the response [111]. This test rests on fewer epochs than the waves: 37 and 129 random post-test and late-sequence epochs in Person 1, 12 and 86 in Person 2. Beta was already higher in the 400 ms before the flash, by 0.6 and 1.6 dB ($p = 0.048$ and $p = 0.012$), so the difference reads as the state of a random block more than a response to each flash. The C3 panels of Figure 16 are differences of absolute power, with no baseline, so they show this difference before the flash as well as after it. Beta near 30 Hz can also carry muscle activity, which a 40 Hz low-pass filter does not remove, so tension in a random block could add to the difference. The random post-test flashes were answered 75 to 81 ms later than those of the late sequence blocks, which could move the movement's own beta drop out of a window fixed to the flash. This gap pools blocks 7 and 8 over the trials the EEG kept, so it sits below the block 8 scores of Table 18. Two checks added after the data took reaction time out (Appendix B). Beta stayed higher against learned trials matched in reaction time ($p = 0.020$ for both people together) and in the half second before the press ($p < 0.001$). Table 31 gives each person; Person 2's matched reading alone did not pass.

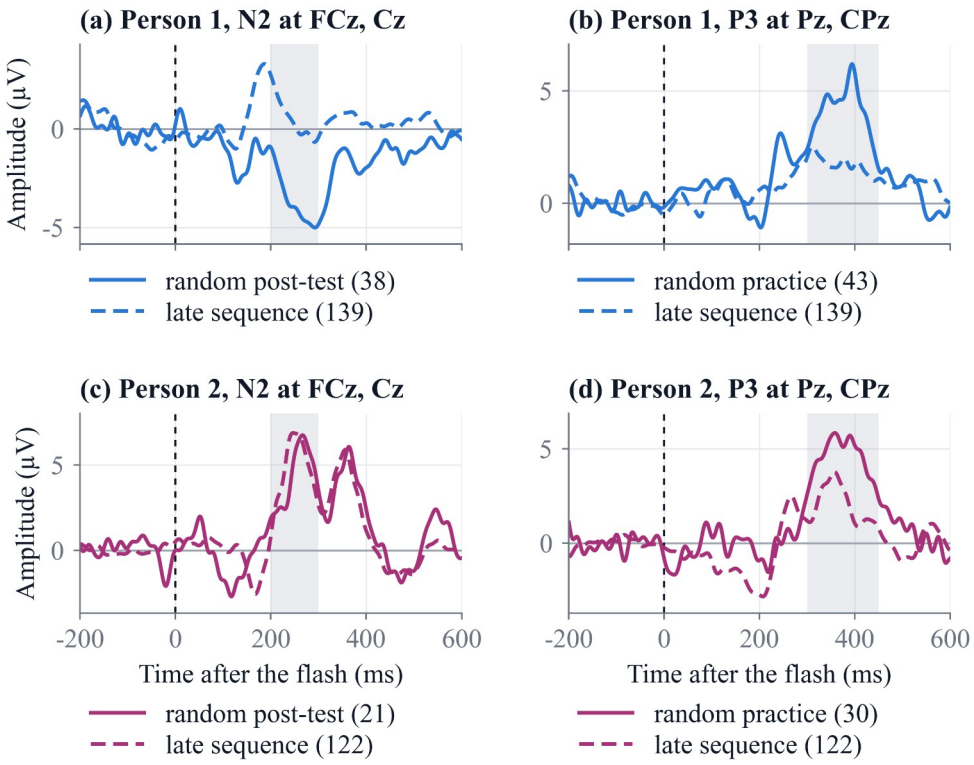


Figure 25: Each person's response to the flash in that person's colour, random flashes (solid) against the late sequence blocks (dashed), with the number of trials kept in each legend. (a, c) The N2 at FCz and Cz, random post-test against the learned order. (b, d) The P3 at Pz and CPz, random practice against the learned order. Dotted line: the flash; grey: each test's window.

Person 2's recording, held out while the windows were settled on Person 1's, is the one test of the pipeline on new data (Section 4.4): its beta difference held and its P3 dropped by 2.5 µV against Person 1's 2.4. The P3 did not recover in the random post-test (0.0 µV against the late blocks, pooled, p = 0.973), so time on the task explains the drop as well as learning does; this check was added after the second recording (Appendix B). In the scalp maps of Figure 16, Person 1's N2 difference sits over the fronto-central midline and Person 2's does not. In both, the C3 spectrum carries more beta power on random flashes, most from 200 to 400 ms after the flash and also before it. The N2 grew for random flashes after learning in Person 1, by 4.2 µV (p = 0.020), and held when a regression method removed the overlap between neighbouring trials' responses [227]. In Person 2 it did not change (p = 0.740), so the two-person N2 result rests on Person 1's recording (Figure 25). Eimer et al. found the

N2 effect in people who could report the sequence [109]; Person 1 typed back the whole order and Person 2 eight of its ten items. Person 2's post-test also kept 21 flash trials against Person 1's 38, so its N2 contrast had about half the trials. Theta did not differ in either recording; it was read after the flash, whereas Lum et al. locked it to the response [112]. The ERN went the expected way in both but passed in neither, and the Pe after a wrong press was clear in Person 2 only.

C.2 Further results of the first recording

As the laboratory suggested, a separate press-locked set at FCz kept random and sequence trials apart. Cleaned apart from the flash-locked set, it held slightly different trials, among them 82 of the 87 sequence-trial errors, which reached −3.9 µV against a CRN of −2.8 µV (p = 0.40). The ERN of Table 31 differs because it pools both kinds of trial at FCz and Cz, timed from the start of the push. With the flash response of fast guesses regressed out, the error waveform reached −3.0 µV 78 ms after the force onset. Table 32 gives the Buzz Hunt measures.

Table 32: Buzz Hunt measures of the first laboratory session (Person 1). Opposite and same side: the hemisphere relative to the buzzed hand; percent: change in band power.

Measure	Result	p
Touch N1, opposite against same side	−3.6 against −1.5 µV	0.14
Mu after the buzz [228]	−35 percent at electrode C3	0.024
Beta rebound after the press	+6 percent at electrode C3	0.58
P3, buzz against flash [229]	+6.4 against +1.6 µV	< 0.001
P3 half-area latency, buzz against flash [230]	533 against 385 ms	< 0.001
RT, buzz against flash, as the EEG analysis kept them	491 against 291 ms	< 0.001

In Buzz Hunt the P3 was clear and mu over the hand's motor area fell after the buzz; with 32 buzzes, the touch N1 and the beta rebound did not pass. Figure 26 sets the two cues side by side, using only Reaction's random flashes, 80 of which survived this comparison's own epoch rejection. The buzz drew a larger and later P3 and slower answers, as its longer interval between targets and slower evaluation predict. Its first cortical response, over the opposite hemisphere, peaked at 275 ms, about 75 ms of which is the vibration motor's start-up delay. The games differ in pace and task as well as in the sense they cue, and the screen's delay was not timed, so this one recording compares the two games, not the senses themselves.

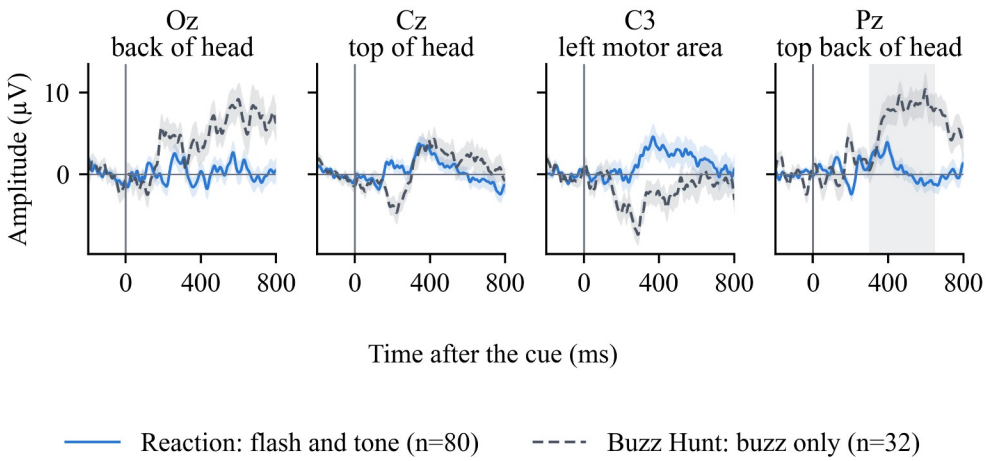


Figure 26: Person 1's response to each game's cue at four sites, Reaction's flash in blue and Buzz Hunt's buzz in grey (dashed): mean (line) and standard error (shading). Grey: this comparison's P3 window, 300 to 650 ms, wide enough for both games' P3; Reaction's own tests use 300 to 450 ms (Table 30).

Appendix D: Faults found before data collection

Besides the faults of Section 4.8.2, running and measuring the software found the six in Table 33, each with its value before and after the fix. Table 34 gives the sixteen faults from the design review of each mode that would have changed a study number, and the last paragraph the seven found when the last changes were checked. Each was fixed with a test that fails on the code as it was. The review's full list is in the repository [34].

Table 33: The other faults found by running and measuring the software, before data collection: what each would have done and the result of the fix.

Finding	What it would have done	Before, after
Force Pilot lag read 0 ms on ramps	Check F2 could not fail	0 ms on every ramp; plausible lags
Buzz Hunt distractor trials voided	Half the data discarded	8 of 16 discarded; none
Syllables finger use unbalanced	One finger did most of the pressing	Spread 4 to 1; 1.05 to 1
Syllables foils that sounded like the answer	A correct hearing scored as an error	About 6.7 percent of vowel foils on unstressed syllables; none
Syllables helper buzz gave the answer away	A player could wait for the buzz and stay on the easiest level	32 of 44 bench sets answered after the buzz; no buzz for adults
Speech file named after a Windows device	The Windows build failed	Build failed; builds pass

Table 34: The sixteen faults from the design review of each mode, all of which would have changed a study number, by mode.

Mode	Finding	What it would have done
Reaction	A board drop mid-trial was scored as a miss and a lapse	A loose plug read as a lapse of attention
Reaction	Fingers were dealt from a bag of four	A quarter of the cues could be worked out from the three before
Force Pilot	A run restarted after a pause took its zero from the force frozen at the pause	A perfectly flown run scored 10 percent time in corridor
Force Pilot	Runs the game had voided for lost signal were re-scored beside their replays	The same step counted twice, once from a dead board
Chords	Forces in newtons were divided by references in counts	Pressing too hard could never be flagged
Chords	The row label followed the first finger's speed, not the outcome class	Most healthy clean chords read as late; the clean hit rate was a speed rate
Rhythm	The cue tone was played at the beat and heard 77 ms after it	A second pacing signal 77 ms after the music and the buzz
Rhythm	Notes missed while the board was away counted as misses	A 2.5 s drop read as three misses and a broken streak
Muscle Memory	A trial lost to a board drop counted as a miss and as fatigue	A 12 s drop forced a rest; a second drop ended the block
Buzz Hunt	Localisation RT ran from the frame before the buzz command	Every RT 17 ms long at 60 Hz, against the EEG byte

Mode	Finding	What it would have done
Buzz Hunt	Catch trials were drawn per trial inside the sixteen	A fifth of blocks had no false-alarm measure
Echo	A silent turn while the board was away spent the spare life	A four second drop ended a game at span 0
Syllables	A right answer given after the prompt buzz lowered the foil level	Eight of twelve simulated blocks drifted off the level the design pins
Mirror	The analysis counted wrong-finger trials as clean pairs	The synchrony gap disagreed with the game's own summary
Battery	A board picked as Left at login was renamed but the hand was not switched	The calibration could not finish and the session ran on the previous participant's thresholds
Battery	A relaunch inside the rest started the next block at once	The one rest the retest depends on was logged as zero

The last check, on 29 September 2026, found seven more. A Nano with the older bootloader was read as a busy port, and a failed read-back with the older flashing tool as the wrong chip, so the messages would have blamed the wrong cause. A pause on a loud trial, one played louder than the rest, dropped the louder sound while the log still called the trial loud. A Syllables fallback could offer a vowel foil that sounds the same as the right answer. The length picked on the game menu carried over to the next login, and a finished session left its start button ready, so a second session would have shared the first one's folder. A working Windows board was read as driverless, which would have failed the new-computer check.

Appendix E: Semester 2 plan against what happened

The progress report committed to fifteen Semester 2 tasks; Table 35 records what happened to each. Figure 21 in Section 4.8 shows the semester's workstreams in time.

Table 35: The progress report's Semester 2 tasks and their outcome. Threads of the progress report, named in Section 5.2: A the adaptive engine, B the rhythm engine, C the bilateral runtime, D clinical deployment; Cross: tasks across them.

Thread	Planned task	Outcome
A	Print and fit-check the magnetic top plate	Taken over by Joshua Bojanjac with the hardware
A	Replay the adaptive controller against the 2025 cohort's logs	Replaced by closed-loop simulation through the real engine (Section 4.2)
A	Write up the replay and trial results in the thesis	Adaptive results in Section 4.2
B	Pilot the engagement survey	Not run with patients; answered by two healthy participants (Appendix M)
B	Fixed-cadence against RAS comparison	Not run: no patient trial; Rhythm became a timing measurement
B	Write up the RAS results in the thesis	Replaced by the Rhythm checks
C	Print and fit-check both mirrored halves	Taken over by Joshua Bojanjac with the hardware; right half in use
C	Replace the inter-hand correlation placeholder	Done
C	Bilateral convergence study	Not run: no patient trial and one board
D	Auto-launch when the USB is plugged in	Done for Windows and macOS
D	Calibration that persists across machines	Replaced by the per-person calibration at every session
D	Portability dry run before trials	Replaced by 48 simulated sessions and the rehearsal of Section 3.6
Cross	Ethics paperwork for the patient trials	In part: the patient pilot's information sheet and consent form drafted (Appendix L); the healthy sessions ran as a low-risk activity with the safeguards of Section 3.9.1
Cross	Fortnightly supervisor check-ins	Held through the semester
Cross	Draft the final thesis	This thesis

Table 36 closes out the progress report's risk register and failure-mode analysis.

Table 36: The progress report's risks and failure modes at submission.

Risk or failure mode	At the progress report	At submission
Supervisor not yet shown the finished software	Low; the report and the repository as the written record	Closed by that record and fortnightly supervisor meetings
Magnetic top plate slips past the report	Deferred to Semester 2	Closed: the prints were taken over by Joshua Bojanjac with the hardware (Section 4.8.3)

Risk or failure mode	At the progress report	At submission
No EEG hardware in 2026	Happened; marker code degraded gracefully	Turned around: the Curtin EEG laboratory's amplifier and trigger box, with recordings on 8 and 22 October 2026 (Section 4.4)
Curtin Windows permissions block the app	Untested on Curtin machines	Occurred in part: the scan took the EEG trigger box for a hand board, and the planned fix, a port set by hand, let the recordings and sessions run
Too many threads for one person	Happened on one thread	Occurred again: the second board, the lowest-value item, was deferred as the register planned
Ethics approval for the patient trials runs late	Pre-writing in Week 1	Closed with the patient trials, which the healthy checks replaced (Section 3.9.1)
Second mirrored chassis half does not fit	Both halves moved to Semester 2	Not reached: Mirror needs the second board, which was deferred
Failure mode: stuck linear actuator	High until Semester 2	Designed out: the finger rests are set by hand on threaded rods
Failure mode: sensor disconnected mid-session	Low, mitigation in place	Seen on the bench; a dropout is now its own outcome and never a miss (requirement SR3)
Failure mode: serial port stops	Low, mitigation in place	Seen on the bench; the block now waits for the board, and a closed window resumes the session at login (Section 3.6)
Failure mode: two boards out of step	Medium	Not tested: Mirror needs two boards
Failure mode: display driver fails	Low, mitigation in place	Unchanged: the error is logged and the game exits cleanly
Failure mode: a session file write fails	Low, mitigation in place	Unchanged: each file is written to a temporary copy and swapped in whole

Appendix F: The Force Pilot wave ladder

Table 37 gives the twelve waves of Section 3.3.5 and what each is read for. From the raw stream the notebook computes the error scaled to the target, the time within 5 percent of it, and release against press error on the symmetric waves. It also takes the error in the hold after each release, the rates of rise and relaxation, and the gain and lag from a sine fit. Last come steadiness in the holds, tremor-band power from 2 to 8 and 8 to 12 Hz, the fading across the Heartbeat pulses, and the force that leaks into the three fingers that should stay still.

The player sees about 8 s of the corridor ahead. The task runs between 8 and 31 percent of the finger's maximum, the light-force range of the Parkinson's studies of Section 2.10 in proportion, though not in newtons. The corridor's half-width narrows from 8 to 4 percent of maximum up the ladder. In published force tasks, a 20-minute retest showed a practice effect [231], and area-type errors reproduced poorly between sessions [232]. Automated tests feed the measures synthetic force traces with known answers, and the first real ladder runs, 22 from two people, were re-scored before any participant played. They exposed two faults specific to this sensor, a zero taken as the press began and a lag read as 0 ms on every ramp (Section 4.8.2 and Table 33), and a missing rule. Hands-off runs, read as the worst tracking on the ladder, would have passed F1 on their own, so the analysis now drops a run whose finger never pressed. A finger resting on the sensor wanders by 0.26 to 1.09 percent of maximum, the floor a measure must clear. On the development runs the tracking error, the rates and the lags cleared that floor; the steadiness of short holds and the power from 8 to 12 Hz did not, so on this device those two describe the sensor rather than the hand. Later Parkinson's work should turn the music off, since rhythm cues movement in Parkinson's disease [202].

A Parkinson's study should add the tasks the current ladder cannot reach, to measure the disease's slowness, rate and memory-guided deficits. These are fast-release pulses, a force-based tapping task, and holds that give 5 s of visual feedback and then hide it for 8 s, since force only starts to fall 1.5 to 2.5 s after vision goes [233].

Table 37: The Force Pilot wave ladder and what each wave is read for.

Level	Wave	Shape, in percent of the finger's maximum	Read for
1	Slow breath	Sine at 0.15 Hz, 8 to 20	Gain and lag on a predictable wave
2	Tide	Ramps at 5 per second with a 3 s hold at 28	Release against press; steadiness in the hold
3	Swell	Sine at 0.2 Hz, 8 to 26	Gain against the published young-adult value
4	Stairs	Holds at 14, 20, 26, 20, 14 and 8	Steadiness and settling after each step
5	Hills	Ramps at 5 per second, up and down twice	Release against press; pauses within a ramp
6	Beach waves	Sine at 0.33 Hz, 8 to 22	Gain and lag
7	Heartbeat	Four 2 s pulses from 10 to 25, peak slope 23.6 per second	Rise and relaxation rates; fading across pulses
8	Dunes	Up at 4 and down at 12 per second, twice	Fast release, reported apart
9 to 12	Chop, Open ocean, Storm, Uncharted	Two, three, eight and eight sine components	Irregular against repeating waves; tremor band

Appendix G: Syllables profiles

Table 38 gives the five age profiles of Section 3.3.6. The structure of a session came from a second review of the reading literature. Examples that differ from the target in one feature taught young children faster when other features were held constant at first [234]. Answering until correct taught no more than being told the answer [235], and one syllable programme that built up from syllables to whole words had a positive trial [236]. A session therefore runs in parts: hear and pick, build the word from its syllables and, for readers aged 10 to 59, a speed check in which a whole word is flashed, masked and picked from four. Options come from one confusion family at a time (vowels, onsets, codas, letter order, affixes and, for children, reversible letters), each with its own level that rises after three right first presses and falls after an error. Two wrong presses show the answer, and each finished word is shown before it is heard, so the reader decodes it first [237]. The voice says each syllable as it sounds in that word, then blends them, as in graphosyllabic analysis, which helped struggling adolescent readers decode and spell long words [238]. The mechanics follow GraphoGame's matching of sound to print [199], a fading time-delay prompt [239], [240], spaced return of words [241], [242], distractors from real errors [243], made-up words for decoding [244] and a fixed-step staircase [245]. From 16 to 59 a staircase near 85.8 percent correct sets the pace [245]; a free British voice [246] speaks every chunk and word, with a kit to record an Australian one.

Table 38: Syllables profiles, chosen from the reader's age in years. Time per set: the time to answer each set of options; tiles: the options on screen.

Profile	Words	Time per set	Helper buzz	Word shown first
6 to 9	Child bank, two to four syllables	6.0 down to 4.5 s by level	Yes, fading	Up to level 3, sound 175 ms later
10 to 12	Harder child words and teen words, 20 percent made up	4.8 down to 3.0 s	Late in the set only	Up to level 2
13 to 15	As for 10 to 12	4.2 down to 2.4 s	Late in the set only	Up to level 2
16 to 59	Rarer derived words, three to five syllables, 40 percent made up	Staircase, 1.0 to 4.5 s, from 3.6 s	None	Never; the word is only heard
60 and over	As for 16 to 59, up to four syllables, 30 percent made up	4.2 down to 2.4 s	None	Never; tiles 1.3 times larger

Appendix H: Force Pilot measures for Parkinson's disease research

Table 39 gives the Force Pilot measures of Section 4.6.1 that Parkinson's disease research uses, with the values of Person 5's and Person 6's Force Pilot blocks, single blocks kept for a later study to compare against. Two measures of Table 39 stand out. Person 5's lags from the sine fit, −143 ms, and from F2's cross-correlation, −28 ms, put Person 5 ahead of the target, which 8 s of preview allows. Person 6 let go of the Heartbeat pulses at 2.18 times the target's rate, the opposite of the slow relaxation of Parkinson's disease. SQ5's planted-error test runs in the software's test suite. A simulated player follows Tide (Table 37), a ramp up to a hold and back down, exactly, except that on the way down it stays 5 percent of maximum above the target. The game's scoring reads a release error of 5 percent of maximum, within 0.5 percent, and a press error under 0.5 percent.

Table 39: Force Pilot measures used in Parkinson's disease research, from the two healthy blocks on this device.

Measure	Person 5	Person 6	Published value
Release over press error (MAE ratio)	1.05	0.96	0.83 to 0.90 in young adults, as ratios of relative root mean square error (RRMSE) [24], [30]
Gain on the sine waves	0.99	0.93	About 0.83 in young adults [24]
Lag from the sine fit, ms	−143	−4	Near zero in young adults, as a phase shift in percent [24]
Rise on Heartbeat over the target's	0.57	0.87	No slower in the pinch grip in early Parkinson's disease [145]
Relaxation on Heartbeat over the target's	0.61	2.18	Slower in the pinch grip in early Parkinson's disease [145]
Decrement across the Heartbeat beats, per beat	−0.007	−0.021	Present in Parkinson's disease, absent in health
SD of a hold, percent of maximum	0.90	0.70	Higher in Parkinson's disease; near the sensor's floor here

Appendix I: Design choices of the platform and the game modes

Table 40 gives, for Reaction and the six modes added in Semester 2, the design choices their checks depend on and the reason for each.

Table 40: Design choices of the game modes and why each was made.

Mode	Design choice	Why
Reaction	An exponential wait [19]; fingers dealt from a bag of twenty, never the same twice in a row	The cue can be neither timed nor predicted, which R1 tests
Muscle Memory	A second-order conditional sequence [95] hidden in a piano riff; learning read from probe takes of unfamiliar material	Learning shows as the cost of the probes, not as faster fingers overall
Chords	The fingers that should stay still scored with those that press; difficulty from finger adjacency and the ring finger's dependence [82], [83]	Feedback on independence reduces enslaving [91]
Force Pilot	The 200 Hz force in percent of each finger's maximum; one climb of a fixed twelve-wave ladder (Appendix F)	Release, rates and lag come from the raw stream, and the conversion constant cancels
Buzz Hunt	A fixed 150 ms pulse [122]; a 2.0 s response window in the session (Table 27)	Pulses of 50 to 200 ms rate best on phone-type motors
Echo	A growing sequence on the finger lanes, the Simon rule with one spare life	A span from a single attempt is noisy [132]
Syllables	A helper buzz on the correct finger that fades as the player improves [239]; a free synthetic British voice [246]	A time-delay prompt faded with skill; a voice for every chunk at no cost

Table 41 gives the platform's main design decisions, the alternative each replaced and the result that settled it.

Table 41: Design decisions, the alternative replaced and the deciding result.

Decision	Chosen, over the alternative	What settled it
Press threshold	Each finger's calibrated gap (Equation 2), over one fixed threshold	The little-finger sensor rests at about 30 counts and the index under 3
Marker timing	The display frame that shows the cue, over the moment the game asks for it	The screen lags a request by 0 to 16.7 ms; on the frame every marker matched the log within one sample
The laboratory's task	A trial-for-trial copy in Reaction mode, over its keyboard script	Every press keeps its force stream (Section 4.3.2); block means within 5 ms of a team member's analysis on 9 of 10 blocks
Force in newtons	Factory calibration, 0.0195 N per count, over a 45 N part's 0.0879	Against the datasheet the old constant read every force 4.5 times too high
Audio delay	A set-up measurement on each machine, over the 40 ms assumed	Song 87 ms, short sounds 77 ms and motors 74 ms on the development laptop
Detector settings	Weights 0.35 and 0.0005; press at 30 percent of the gap, over 40; release at 20 percent, at most 0.6 of the press and 10 counts below it	Time constants of about 12 ms and 10 s; release below press stops chatter; at 40 percent, a firm calibration press meant pressing hard all session

The detail behind Sections 3.2 and 3.3 sits here. A 10 N SingleTact sensor spreads its rating over 512 counts, 51.2 counts per newton or 19.5 mN per count after a tare, with a linearity error under 2 percent and hysteresis under 4 percent [173], [174]. The factory calibration holds full scale within 2 percent, against about 10 percent for a standard sensor, and changes nothing about drift or timing [174], [175]. Each hand's Nano buzzes a finger for 150 ms on command. The detector counts a press only if it comes at least 100 ms after that finger's last event. The press threshold has a floor, the largest of 12 counts, eight times the sensor's noise, and its resting load plus three times its noise. The gap g is capped at three times the smallest light press its sensor allows, at least 66 counts, so that a hard calibration press cannot make the game hard. The software logs force as the rise above the baseline times 19.5 mN per count. After several misses in a row the adaptive controller lowers the pace, never below 10 BPM, and favours the player's strongest finger for a while. Rhythm sends the cue tone 77 ms early and the buzz 74 ms early, their measured delays, so on the laptop every cue lands on the beat. Mirror counts two presses as a pair when they land within 350 ms of each other, far wider than the tens of milliseconds a healthy chord spans (Section 2.3); this gate rejects 0.4 percent of simulated pairs. An eleventh mode, Lighthouse, was retired before any participant as impractical to play.

Appendix J: The device literature search

Each device kept in Table 8 was read against what the method needs, from its report or abstract. The search behind Sections 2.1 and 2.14 was run on 15 October 2026 in PubMed, on titles and abstracts, as two searches:

1. finger AND rehabilitation AND force AND (game OR games OR gaming OR gamified), which returned 9 records;
2. finger AND rehabilitation AND force AND (device OR robot* OR game*) AND (validity OR reliability OR psychometric*), which returned 5.

The 14 PubMed records hold 13 papers, since one is the preprint of another, and the same terms in Europe PMC, on title or abstract, added nothing new. On 16 October 2026 the two searches were run in IEEE Xplore over titles, abstracts and index terms. Of their 100 records, the three nearest are added to Table 42; apart from Knill et al., already found, none recorded each finger's force with a stated validity or reliability. Of these three, the Virtual Eggs Test is the closest in design, with healthy reference values and a retest in 79 adults, but on grip force [247]. On 4 November 2026 OpenAlex returned 84 records for the two searches on titles and abstracts, 13 already found; the three nearest of the rest are added, one a doctoral thesis that adds training games to Knill et al.'s individuation device. Each abstract was read for what the device records and whether the validity or reliability of its measurements in people was reported. A device counts as closer when it measures the force each finger presses with and reports either, as four of the 19 papers do, on three devices (Figure 27). Eleven papers were read in full and none checks its timing against an outside reference. Of the nine read on 4 November 2026, FINGER's tests measured the robot's bandwidth, about 8 Hz, not its game's timing. Burdea et al. fixed a tracker delay their players noticed without measuring it, and the other seven give at most a sampling rate, or log times with no check. The search has limits: two strings, four databases without Scopus, Web of Science or the grey literature, and abstracts alone for eight papers, so a device reported only under other terms or in a conference talk could be missed. A known-effect check counts as validity evidence, so the screen would have caught any device checked that way.

Table 42: Records of the device searches, PubMed on 15 October, IEEE Xplore on 16 October and OpenAlex on 4 November 2026, by year. Closer: records each finger's pressing force and reports its validity or reliability in people.

First author, year; PMID or source	Device	What it records	Validity or reliability	Closer
Dovat, 2010; 20370489	Cable-driven hand robot with two game-like exercises	Coordination, force modulation and independence	None reported; 2 patients	No
Godfrey, 2010; 21095777	Hand exoskeleton with a virtual reality game	Finger and thumb range of motion	None reported; 4 patients	No
Taheri, 2012; 23366783	FINGER exoskeleton with a Guitar Hero style game	Finger movement and individuation	None reported; 8 patients, 4 controls	No
Ates, 2013; 24187220	Passive wrist and hand orthosis for home games	Rough joint rotation and torque	None reported	No
Lin, 2015; 26409543	Glove with force sensors, vibration and games	Each finger's force	None reported	No
Connan, 2016; 27909406	Forearm band of force and EMG sensors	Muscle signals for prosthesis control	None in numbers; 10 healthy	No
Jaspers, 2018; IEEE Xplore [248]	GriFT, two grip handles with a computer game	Grip force of each hand	Validity in 174 children	No
Sandison, 2020; 33019080	HandMATE exoskeleton with tablet games	Each finger's grip and extension force	None; grip tracking in 10 healthy	No
Burdea, 2021; 31180276	Game controller for therapy at home	Grasp, finger extension and reach	Usability only; 2 healthy	No
Wilhelm, 2022; 35808299	Finger exoskeleton with a game app	Finger motion and force	None reported; 3 patients	No
Băeşu, 2022; OpenAlex W4313375683	Ball with five finger force sensors and games	Each finger's pressure, response time	None reported	No
Gertler, 2023; 36576497	Magnet in a soft finger sheath	Gives vibration; records nothing	Perception tests, 17 users	No
Lapresa, 2023; 36671635 (OpenAlex)	Hand exoskeleton with fingertip sensors and games	Fingertip grip forces	Bench calibration; 10 healthy, 1 patient	No
Valenzuela-López, 2024; 39028607	AMBER, force sensors under each finger	Strength, reaction time and memory	Retest ICC 0.56 to 0.90, 36 healthy adults	Yes
Knill, 2025; 40644104	Force sensors under each finger	Individuation index	Retest ICC 0.48 to 0.80 for flexion	Yes
Khan, 2026; 41721406	3D fingertip force device	Individuation index	Internal consistency R of 0.97 or more; tracks manual ability in cerebral palsy	Yes
Angelini, 2026; IEEE Xplore [247]	Virtual Eggs Test, an instrumented object	Grip force and dexterity	Retest ICC 0.90 to 0.91 fine and 0.63 to 0.64 gross, 79 healthy adults	No
Yang, 2026; IEEE Xplore [249]	Force-sensing glove and grip sensors with two games	Pinch and grip force, joint motion	Said to be reproducible, with no statistic; 39 people	No

First author, year; PMID or source	Device	What it records	Validity or reliability	Closer
Knill, 2026; OpenAlex W7222364440	Doctoral thesis: the individuation device with training games	Each finger's force	Reliability and validity, healthy and patients (abstract)	Yes

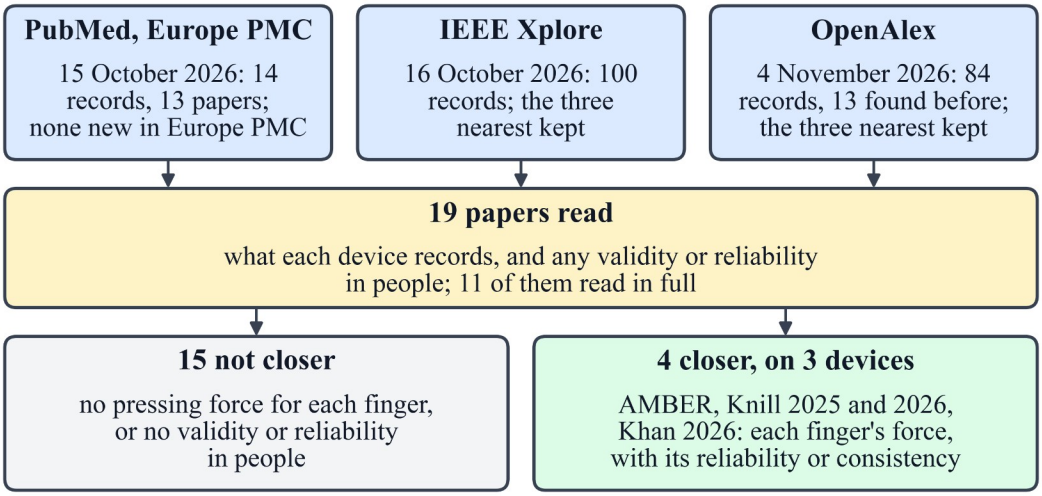


Figure 27: The device search: the records of each database, the papers read and the three closer devices of Table 42.

Appendix K: The study kit's analysis and checks

Table 43 gives, for each mode in the session, the measures the notebook computes, how each is calculated, the reference value beside it and the reliability class predicted before data collection (Section 3.8.1).

Table 43: How each mode is analysed, the healthy value the literature gives and the reliability class expected before data collection.

Mode	Measures	How they are computed	Healthy reference	Expected ICC
Reaction	Median RT, 10th percentile, lapses, false starts, accuracy	RT from the frame that shows the cue to the logged count; RT against the foreperiod	Lapses rare when rested, on a simple reaction test [198]; absolute RT device-specific [116]	Good to excellent
Muscle Memory	Learning score, accuracy rebound	Probe take minus the mean of the trained takes either side, per press	About 10 to 30 ms expected after 108 to 144 trained trials, scaled from [96] (Section 2.6)	Poor for one person [101]
Chords	Enslaving ratio, chord span, clean hit rate, chord cost	Gap-scaled quiet-finger peak over pressed-finger peak; first to last press of a chord; slope of mean onset time on chord size, one to three fingers	8 to 10 percent leak at a quarter of maximum [84]; about 126 ms per extra finger on keys [86]	Moderate for the ratio, good for span
Rhythm	Asynchrony mean and SD, absolute offset, share of presses after +150 ms, lag-1 correlation	Press time minus the scored beat, after the laptop's measured audio delay is removed	Mean about 0 to −20 ms to music [66], SD in the tens of ms [64]	Good for SD, moderate for the mean
Force Pilot	MAE and root mean square error (RMSE), time in corridor, lag, press against release	Offline re-score of the raw stream against the rebuilt target, in percent of the finger's maximum	Release at least as accurate as press [30]; sine gain about 0.83 in young adults [24]	Moderate to good
Buzz Hunt	Accuracy, neighbour errors, catch false alarms, RT, span	Confusion matrix per finger; RT from the buzz command, which precedes the vibration by 74 ms (the four motors' 71 to 80 ms on the bench)	Near ceiling, errors rare [121]	Poor for accuracy and span, good for RT
Echo	Mean span of two games, best span, items before the first error, edit distance	Mean and longest sequence repeated; edit distance [135], exploratory	Corsi 6.2 [128]; Simon about 7 [129]	Moderate for the mean
Adaptive	Pace rise, steady pace, hit rate	Peak minus start; median of trials 21 to 40	Four-choice RT 390 ms at ages 18 to 25 [216]	Not estimated

Each mode's pre-specified checks are in Table 44. Of the eleven that could fail (below), R1, Rh1 and B1 set low bars that a working mode should clear, and B3 rests on the game's design. The seven most demanding are therefore R3, P1, P3, C2, C6, F1 and F2, which the normative study reports apart.

Table 44: Pre-specified known-effect checks, read on pass 1 unless stated.

ID	Mode	Prediction	Pass criterion	Basis
R1	Reaction	The exponential foreperiod removes anticipation	Median rho of reaction time against foreperiod above −0.2; false starts under 10 percent	[19], [225]
R3	Reaction	No practice effect between passes	90 percent CI of pass 2 minus pass 1 inside 20 ms either way	[118] for simple reaction; [223]
P1	Muscle Memory	Sequence learning	Mean learning score above zero	[92], [96]
P3	Muscle Memory	Less accurate on random than on learned order	Mean accuracy rebound above zero	[92]
C1	Chords	Little enslaving at light effort	Median enslaving ratio under 0.15	[84]
C2	Chords	Difficulty rank predicts hit rate	Rank negatively related to hit rate; the enslaving half exploratory	[82]
C3	Chords	Ring finger least independent in movement (reported)	Largest mean leak in the ring finger column	[83]
C6	Chords	Each extra finger delays the chord	Median slope of mean onset time on chord size above zero	[85], [86]
Rh1	Rhythm	Presses anticipate the beat rather than react to the buzz	Median share of presses later than +150 ms under 10 percent	[250], [251]
Rh2	Rhythm	Asynchrony spread in the tens of milliseconds	Median SD between 10 and 100 ms	[64]
F1	Force Pilot	Tracking is worse on irregular waves	Force-target r lower than on repeating waves	[252]
F2	Force Pilot	Irregular waves are followed later	Lag longer than on repeating waves	[154]
F3	Force Pilot	Release error equals press error	90 percent CI of the difference inside 2 percent of maximum	[24], [30]
F4	Force Pilot	The corridor is flyable	More than half of runs at 0.8 time in corridor or above	Game design
B1	Buzz Hunt	Localisation near ceiling	Accuracy above 0.9	[121]
B2	Buzz Hunt	Errors fall on neighbours (exploratory)	Neighbour errors above the buzzed finger's chance: a third for an outer finger, two thirds for an inner one	[120], [121]
B3	Buzz Hunt	Players wait for the buzz rather than guess	Catch-trial false alarms under 10 percent	Game design
B4	Buzz Hunt	Tactile span in a plausible range	Median span between 3 and 5 items	[253]
E1	Echo	Best of two games in the Simon range	Median span between 5 and 9 items	[129]

ID	Mode	Prediction	Pass criterion	Basis
A5	Adaptive	The pace rises for a healthy hand, as the challenge point asks	Median pace rise above zero	[50]
A6	Adaptive	The rising pace does not cause failure	Median hit rate above 0.65	Game design

A check passes when its statistic lies on the predicted side of the reference and its interval clears it; if only the point estimate does, it is reported as direction only. Two are read on the point estimate, as the design registered them: R1 on its median rho and false-start rate, and B3 on the mean false-alarm rate with the exact upper bound beside it. A B3 pass therefore says no false alarm was seen, not that the rate was measured. The equivalence checks, R3 and F3, pass when the 90 percent interval of the difference lies inside the margin, the standard two one-sided tests at the 5 percent level [223]. R3's margin spans the practice reported between short blocks, from 7 ms slower on an immediate four-choice retest [216] to 19 ms faster a week later [224]. A failed R3 is named practice when pass 2 is faster, fatigue when it is slower and inconclusive when only its interval reaches past the margin, as worded on 15 October 2026 before any session's data were read (Appendix B). With catch trials, reaction time rises a little with the wait [225], so only a fall counts against R1. Tapping to music shows less of the lead seen with a metronome [66], so Rh1 asks only whether presses anticipate the beat rather than react to the buzz, since a reaction lands near 200 ms [251]. One-sample and paired checks use the Wilcoxon signed-rank test, reported with the rank-biserial correlation and Cohen's dz. Holm's correction [195] runs across every check with a p value, since the same people contribute to all of them. Eight checks (Rh2, B4, A5, A6, E1, F3, F4 and C1) cannot fail for a healthy hand on a working device, because the game's design or the sensors' own noise meets their band. They are feasibility checks, so validity rests on the eleven that could fail. Three exploratory rows sit outside the correction: neighbour errors (B2), probe errors on the trained next finger (P3i) and late misses in Echo (E2p).

A block is also read for change across its own trials, the progress a participant can see inside a session. Each person's measure is compared on the first and last thirds of the scored trials, with a slope against the logarithm of the trial number, and read only where a later trial is not a harder one. That gives Reaction's median RT (W1), the anchor, which should be no faster in its last third than its first, Chords' second sub-block against its first (W4) and Buzz Hunt's localisation RT (W5), with W4 and W5 read against W1. Muscle Memory's row (W3) takes P1's verdict. Rhythm, Echo,

Adaptive and Force Pilot are described but not tested, because a later trial in them is harder.

Pass 1 and pass 2 are compared on five measures, each with the class predicted before collection. They are Reaction's median reaction time (T1, good to excellent), Force Pilot's mean absolute error and time in corridor (T2 and T3, moderate to good) and Chords' median enslaving ratio (T4, moderate) and chord span (T5, good). Each is read as ICC(2,1) with its approximate F-based interval [190], [191], on the bands of Koo and Li [22]. The SEM and MDC95 follow Equations 7 and 8 [192], with Bland-Altman limits of agreement [193], reported to the GRRAS guideline [221]:

$$\mathrm{SEM} = \mathrm{SD} \times \sqrt{(1 - \mathrm{ICC})} \tag{7}$$

$$\mathrm{MDC}_{95} = 1.96 \times \sqrt{(2)} \times \mathrm{SEM} \tag{8}$$

$$r_{\mathrm{sb}} = 2r / (1 + r) \tag{9}$$

where SD is the pooled standard deviation of the two passes and the split-half correlation r is corrected to the full block by Equation 9. With 5 to 20 participants an ICC interval spans more than one band [194], so each row classes its estimate rather than its interval (Section 2.13), gives the interval beside it and sets both beside the class predicted before data collection. The passes are about 25 minutes apart with the same calibration. The ICC is therefore an upper bound on reliability between days, and the MDC95 a lower bound on a real change. Read at the count rather than the start of the push, Section 4.5's retest differed by −18, +53, −22 and +4 ms (Table 18), for an ICC(2,1) of 0.37, an SEM of 23 ms and an MDC95 of 65 ms. Each median is taken separately over about 40 presses, so the count and push readings can differ by more than the lag itself changed: by 32 ms in Person 4, whose lag fell only 10 ms. Rhythm, played twice in the minutes Syllables gave back, is an exploratory row. Internal consistency comes from the measures that can be recomputed from part of a block, as Spearman-Brown corrected correlations between random halves over 1000 splits. It is never used for an SEM or an MDC, because it leaves out the error between administrations. The rule that combines the checks and this reliability into one verdict per mode was written on 15 October 2026, as the sessions opened and before any session's data were read. Under it, a mode's score is ready to serve as a patient outcome when a check passes and a measure reaches its predicted reliability class. A mode played by fewer than eight people is Not tested (Appendix B).

The five rows were run on the twelve simulated players of Section 3.4, given a second pass, on the 26 October code (Table 45). Every row returned its ICC(2,1), interval, SEM and MDC95; four were classed at or above their prediction, and chord span fell just below its floor, 0.747 against 0.75, with an interval of 0.34 to 0.92 that spans all four bands. T2 and T3 differ only in the fourth decimal, 0.9032 and 0.9026. The classes test the notebook and the rule, not the device, since a simulated player's noise between passes is what the simulator gave it.

Table 45: The five reliability rows on twelve simulated players, two passes each (26 October 2026).

Row	Measure	Predicted class	ICC(2,1) (95 percent CI)	SEM	MDC95	Class of the estimate
T1	Reaction, median reaction time	Good to excellent	0.80 (0.45 to 0.94)	15.2 ms	42.3 ms	Good
T2	Force Pilot, mean absolute error	Moderate to good	0.90 (0.70 to 0.97)	0.09 percent of maximum	0.24 percent of maximum	Excellent
T3	Force Pilot, time in corridor	Moderate to good	0.90 (0.70 to 0.97)	0.005	0.014	Excellent
T4	Chords, median enslaving ratio	Moderate	0.92 (0.75 to 0.98)	0.003	0.007	Excellent
T5	Chords, chord span	Good	0.747 (0.34 to 0.92)	2.2 ms	6.2 ms	Moderate

The model hand of the session rehearsals and the traits given to the twelve players are in Table 46; each simulated player's traits are drawn once from the ranges shown, with seed 2026. The recovered medians run 9 ms below the given reaction times, since the simulated players get about 20 ms faster within a block, and asynchrony reads 7 ms late from the detector's smoothing. Echo's mean span is recovered at r = 0.78, since its simulated players miss by chance rather than at a fixed length.

Normal ranges are reported as the median, interquartile range and full range, with every participant shown as a point. Handedness, taken from the writing hand entered at login, is analysed as an exploratory factor between people, since right-handers' reaction time differs little between hands [254] and the gap between the hands in tapping is smaller in left-handers [255]. Every check is also rerun on the right-handers alone. The sample size, about ten, was set by who could take part, a justification by available resources [256]. In a simulation of exact one-sided signed-rank tests, ten people give 80 percent power for dz 0.91 at the 5 percent level and 1.40 at Holm's strictest step. A check that misses significance is therefore read

with its interval, not as evidence of no effect. An exact signed-rank test on n people cannot give a p below one in 2 to the power n. Holm's first step over the nine checks that carried a test in the twelve-person rehearsal is 0.05/9 = 0.0056, which needs eight people (1/256 = 0.0039). With fewer than eight participants, a check or reliability row is not tested, and each participant's values are described instead. The Parkinson's measures from Force Pilot (Section 3.3.5) are reported as normal values without checks of their own, apart from F3. Syllables is not in the session, so the extension work did not change the study that was specified in advance.

Table 46: The model hand of the session rehearsals, and the traits given to the twelve simulated players.

Part	Setting or range given
Model hand, reaction time	Drawn about a base with 60 ms of noise and a 180 ms floor
Model hand, press	Held 120 ms at a fixed force above rest
Model hand, tracking	Follows the target in percent of maximum with 0.6 percent of noise per sample
Players, reaction time	Base 243 to 346 ms with 45 ms of noise per trial; 25 ms slower on the other hand; 0 to 36 ms faster as a block goes on, most of it over the first eight trials
Players, tapping asynchrony	Mean −65 to −14 ms, SD 16 to 32 ms
Players, tracking lag	120 to 255 ms
Players, enslaving ratio	0.031 to 0.055
Players, chords	Presses spread 10 to 21 ms within a chord; 47 to 151 ms added per extra finger
Players, Echo	50 percent point at 4.7 to 8.5 items; 0 to 30 percent of misses left silent
Players, Buzz Hunt and Syllables	Localisation accuracy 0.93 to 1.00; reading accuracy 0.95 to 1.00
Players, hand and age	Eleven right-handed and one left-handed, Edinburgh quotient −92 to 100; ages 21 to 33

Reaction times are summarised by the median, which outliers affect less [257], and also by the 10th percentile. Ex-Gaussian parameters, which split the distribution into a normal part and a slow tail [258], come from the method of moments, and a response of 500 ms or more counts as a lapse, the standard vigilance-test rule [198]. Force Pilot is scored offline from the raw stream against a target rebuilt from the logged wave. Its lag is the delay at which the correlation between force and target peaks, kept only when that correlation reaches 0.35, Brewer et al.'s threshold [31]. The lag is read on the non-periodic waves, since a periodic wave can be tracked ahead of time [154].

The 15-minute session plays Reaction, Force Pilot, Chords and Adaptive once, and the 30-minute plays all eight games once. The 45-minute adds a 3-minute rest and a

second go at Reaction, Rhythm, Force Pilot and Chords, and the 60-minute plays Echo, Buzz Hunt, Muscle Memory and Adaptive again as well. Over 48 simulated sessions at each length, the medians per simulated player were 14.2 to 14.5, 29.2 to 29.5, 44.4 to 44.9 and 58.5 to 58.9 minutes at the device (Table 47).

Table 47: The 45-minute session, with the trials and checks of each block, listed by pass rather than in play order; minutes are medians of 48 simulated sessions.

Block	Pass	Minutes	Per participant	Checks
Login, seating, quick calibration		5.00	Empty, resting and light-press captures	
Reaction	1 and 2	2.30, 2.38	20 scored trials each	R1, R3, W1, T1
Rhythm	1 and 2	2.05, 2.05	107 notes on the song's attacks, vibration on the beat	Rh1, Rh2; exploratory second go
Echo, four lanes	1	3.06	About 16 sequences	E1
Force Pilot	1 and 2	3.54, 3.54	12 runs each	F1 to F4, T2, T3
Chords	1 and 2	2.64, 2.67	40 trials each	C1 to C3, C6, W4, T4, T5
Buzz Hunt	1	2.78	16 localisation, 3 catch and 4 span trials	B1, B3, B4, W5
Muscle Memory	1	6.27	296 presses: a 12-press warm-up, a 32-trial random take, five trained takes of 36 and two 36-trial probes after 108 and 144 trained trials	P1, P3, W3
Adaptive	1	0.46	40 trials	A5, A6
Rest, transitions and stretch		5.83		
Total at the device		**44.58**		

Order A plays Reaction, Rhythm, Echo, Force Pilot, Chords, Buzz Hunt, Adaptive and Muscle Memory; order B plays Force Pilot, Chords, Buzz Hunt, Adaptive, Reaction, Rhythm, Muscle Memory and Echo. Order A opens with the timing modes and order B with the force and vibration modes. Pass 2 keeps the order of pass 1, and in both orders Muscle Memory never follows a game that ends in sequence recall on the same fingers, Buzz Hunt's span stage or Echo. A login by name takes its order from a hash of the name, which evens the orders only on average; a login by study code plays order A for an odd code and order B for an even one. The screen tells the player what each game asks before it starts, and the 3-minute rest before pass 2 unlocks its start button only after a minute. The 15, 30 and 60-minute sessions play part of the 45-minute battery or add to it, each within 1.5 minutes of its length in simulation. On import the session folders are copied off the laboratory computer and copied again with each name replaced by a code, P01 upwards in the order of each person's first session, leaving out test logins and any block played in Test Mode. The

session check then confirms that every step of the plan finished once, in its passes, with the trial counts its games should play and the rest taken.

In Person 5's Chords block, three of the 40 chords were still rising when their trial closed, so their target peaks read low. Re-read to each chord's release on the raw stream, the block's median enslaving ratio falls rather than rises, so the trial window works against C1, the enslaving check, not for it.

The onset detector of Section 3.8.1 low-passes the force at 20 Hz, then searches forward from the start of the last rise in the 0.4 s before a counted press. Onset is the first sample at which the velocity passes the larger of 25 counts per second and 18 percent of its peak, provided the force then rises 12 counts within 50 ms. A counted press is read as a light touch before the real one when the same finger presses again within 0.3 s of letting go and the first peak is under half the player's usual peak and under the second's. The settings are those of the laboratory's own implementation of the Teasdale method, the one its earlier sessions were processed with, so these onsets compare directly with that work; Table 17 gives how far the lags and scores move under other thresholds. The normative study fixes before its data what this study set after it: SQ1's lag criterion, a block-median lag within one 5 ms sample in the two blocks each learning score compares, and the readings of Table 19 for Muscle Memory's scores.

Appendix L: Information and consent for the patient pilot

The sessions asked for consent only by the act of logging in (Section 3.9.1). The session sheet has since gained the line "Recorded: each press with its force and timing, and the login fields. Your name becomes a code before analysis; the de-identified data go to the supervisor with the thesis." The patient pilot of Section 6.3 will give each person the sheet and form of Table 48 before they play, through a clinician outside the teaching or care relationship.

Table 48: Draft information sheet and consent form for the patient pilot.

Part	What it says
What the study is for	To see whether the device's games measure finger movement after a stroke, against the healthy values in this thesis
What you will do	Sit at the device with your hand on four sensors and play its games for up to 45 minutes at one visit, resting whenever you like
What is recorded	The force and timing of each press, your age, your sex if you give it, which hand you write with and the length of the hand you play with; no video or sound
Who sees it	The research team; your name is replaced by a code before any analysis, and results never name you
How it is kept	De-identified files kept permanently with the thesis and on Curtin storage, beyond the period the Australian Code and Curtin's policy require
Risks	Tiredness, and mild discomfort from pressing or from vibration on sensitive skin; the sensors are cleaned between people
Stopping	Press Esc or tell the researcher at any time, without giving a reason; stopping does not affect your care
Questions	A contact inside the research team, and one outside it for concerns
Consent	I have read this sheet, my questions were answered, I agree to take part and I know I can stop at any time (name, signature, date)

Appendix M: The participants' survey

Two participants answered the project team's survey on the laboratory computer straight after their session: Person 3 after the laboratory's sequence task on 15 October and Person 5 after the self-run session on 20 October 2026. Each answer was linked to its session by the participant's age and by the minutes between the end of the session and the answer. Table 49 gives their ratings from 1, disagree, to 5, agree; four background statements, on confidence with devices and games and use of the hand, were rated 4 or 5 by both and are left out.

Table 49: The two participants' survey ratings, from 1 (disagree) to 5 (agree).

Statement	Person 3	Person 5
On-screen instructions easy to follow	5	2
Visual cues easy to understand	5	4
Knew where I was pressing	4	5
Easy to use overall	4	3
Looks approachable	3	3
Looks comfortable	3	3
Comfortable to use	3	4
Felt secure and stable	4	5
Finger adjustment gave a comfortable position	5	3
No discomfort in palm, wrist or thumb	2	4
Spacing between the pads felt natural	3	3
Sounds helped to time the presses	5	5
Vibrations helped to start the presses	5	5
Could feel which finger was buzzed	4	5
Screen, sound and vibration together helped	5	5
Enjoyable and engaging	5	5
Music made it more engaging	5	5
Stayed focused and motivated	5	4
Each pad responded consistently	3	4
No fatigue that affected performance	2	5

Person 3 liked that the device was set to their own hand. They reported that the middle sensor sometimes stopped responding toward the end of the task, when their wrist and fingers were also uncomfortable, and that it worked when Welber Marinovic checked it afterwards. Person 5 liked the vibration feedback, would have liked sensors that move sideways for wider fingers, and asked for a short pause at the start of the trials, since their accuracy dropped before they found how hard to press.